## Low Level Layered Language

# L-LANGUAGE

**(Draft 1d)**

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# 1 Introduction

This document describes **L-Language**, the Layered Language System Low Level Language.

The L-Language is a system programming language built on the following two main ideas:

**Type Checking Segregation Hypothesis** A strongly typed-checked general-purpose computer-efficient language is impossible. What is possible is to segregate non-type-checkable code into small inline library functions and into macro functions, with code that uses these functions being strongly type-checked.

**Fully Capable Macro Sublanguage Hypothesis** It is better for a programming language to have a builtin macro language that is a general purpose interpreted language than it is for the programming language to build into itself many more limited and specialized type declaration and flow control features.

The author of this document does not plan to implement the L-Language. However, a parser for the L-Language is being built on top of the 'layered' system for lexical analysis and parsing, and is being used to debug the design of the 'layered' system.

# 2 Overview

A typical L-Language statement is:

```
int X = Y - C#"0"
```

This allocates a new variable `X` of type `int` and sets its value to the value of the variable `Y` minus the constant `C#"0"` (which is the character code of the character `0`). The 'variable' `X` is readable, but after it is initialized it is not writable.

The following is another example:

```
av *READ-WRITE* uns8 @bp @= allocate[81]
av uns8 @cp = "Hello!"
int i = 0
while i < cp.upper:
    bp[i] = cp[i]
    next i = i + 1
bp[cp.upper] = 0
```

Here '`allocate[81]`' creates an aligned vector of 81 `uns8` (8-bit unsigned) numbers in the local stack and returns an aligned vector pointer, or `av`, to the vector, marking the vector elements as `*READ-WRITE*`. `"Hello!"` is a constant vector of `uns8` numbers and is similar except that it marks the vector elements `co`, for 'constant', which is the implied default qualifier for `@cp`, and therefore is not explicitly given. Vector pointers can be used with indices to reference elements of their vectors, and have `upper` and `lower` bounds on these

indices. Here the `lower` bounds are their defaults, which are `0`.

Here `@bp` is a variable whose name begins with '`@`' and whose value is therefore a pointer. Such a variable has an associated indirect variable `bp` whose name is missing the initial '`@`'. The expression `@bp[i]` designates a pointer to the `i`+1'st element of the vector pointed at by `@bp`, but the expression `bp[i]` designates the value of the element. Similarly for `@cp[i]` and `cp[i]`.[1]

The qualifier `*READ-WRITE*` says that a value can be read or written, the default qualifier `co`, or 'constant', says a value can be read but will never be written no matter what, the qualifier `ro`, or read-only, says that the value can be read but cannot be written using the variable name given, though it might be written by some other piece of code that accesses the value under another name. The qualifier `*WRITE-ONLY*` says that the value cannot be read but can be written using the variable name given, but might be read by some other piece of code or other device.

There are also two qualifiers that specify the lifetime of the target of a pointer: `*GLOBAL*`, and `*HEAP*`. The default is no lifetime qualifier, which means, roughly, that the pointer is only known to be valid until the code block in which the pointer was first calculated is no longer executing. Typically pointer variables are default-undeclared, as in the example. The `"Hello!"` constant in the example is a `*GLOBAL*` pointer, meaning that the pointer is valid during the entire program execution, but such pointers are implicitly convertible to undeclared pointers, as in the example. `*HEAP*` pointers point at garbage collectible values and obey special rules that we shall not discuss here.

Variables allocated directly to local or global memory have names, like `X`, `Y`, and `Z`, and values that are constants. These values most frequently have a size equal to the natural word size of the computer (typically 32 or 64 bits). Although the value of a variable is constant, the value may point at a memory location that is read-write, and that memory location may be in local or global memory.

An aligned vector pointer `av` is a sequence of 4 integers (`int`) containing:

- A 'base pointer' `int` holding the byte address of an `int` in memory that contains the 'base (byte) address' of the memory block containing the vector. Note that the `av` value does not contain the base address, but contains instead this pointer to where the base address is stored in memory. This scheme allows the base address to be changed without changing the `av` value.
- An 'offset' `int` that is added to the base address to form the byte address of the vector element that has index `0` in the vector (this element does not exist if `0` is not an allowed index).
- A 'step' `int` which is multiplied by the index $I$ to determine the offset of the indexed

[1] '`@`' is analogous to C++ '`&`' used in a variable declaration, but here '`@`' can be used with different types of pointers, can be used without restrictions for structure members, and can be used with mutable pointers.

vector element from the 'offset'. Specifically, element address = base address + offset + $I$ * step.

- An 'upper bound' `int` which is the maximum allowed value of the index $I$ `int` plus 1. Specifically, $I$ must be in the range [0,upper bound).

There are other types of pointer. An `bsv`, or 'bit string vector', is like an `av` aligned vector except that the offset `int` is a bit offset instead of a byte offset and the step `int` is measured in bits instead of bytes. The `ap` ('aligned pointer') and `bsp` ('bit string pointer') types are similar but do not have the bounds and cannot be indexed. Lastly there is the direct pointer, `dp`, that is just a single `int` containing a byte address; this is most useful for calling C language functions. New pointer types may be defined by the user.

Variables whose names begin with '`@`' take pointer values, and the variable's name with the initial '`@`' removed is called the associated target variable and names the value pointed at. Thus `@V` is a pointer valued variable and `V` is the value `@V` points at. For example:

```
int X = 5
ap *READ-WRITE* int @Y   // `@= allocate' is implied
Y = X + 2                // Now Y == 7
Y = Y - 4                // Now Y == 3
X = X + 1                // Illegal!  X is co
ap ro int @Z = @Y        // Copies pointer value @Y.
                         // Pointer conversion from *READ-WRITE*
                         // to ro is legal.
```

Here the implied '`@= allocate`' allocates an `int` to the local stack, zeros it, and returns an '`ap *READ-WRITE* int`' pointer to its location.

Instead of making a variable point at a `*READ-WRITE*` location you can update the constant variable using the `next` construct:

```
int X = 5
int Y = X + 2           // Now Y == 7; Y is co
next Y = Y - 4          // Now Y == 3; Y is co
Y = Y + 1               // Illegal!  Y is co
```

Here '`next Y`' is a new variable, distinct from `Y`, but with the same type, pointer type, qualifiers, and name '`Y`', which hides the previous variable of the same name. The advantage of doing this is that it makes compilation more efficient by keeping variables constant (i.e., `co`), and it improves debuggability by retaining the different values of the variable for inspection by a debugger.

Loops also use the '`next ...`' construct. For example:

```
// Compute sum of 4, 5, and 6.
//
int sum = 0
```

```
int i = 4
next sum, next i = while i <= 6:
    next sum = sum + i
    next i = i + 1
```

which is semantically equal to:

```
int sum = 0
int i = 4
next sum, next i:
    next sum = sum + i
    next i = i + 1
next sum, next i:
    next sum = sum + i
    next i = i + 1
next sum, next i:
    next sum = sum + i
    next i = i + 1
```

The '`next sum`' and '`next i`' before the '`:`', which are the output variables for the block of code containing the two '`+`' statements, can also be implied as they appear as output variables of the '`+`' statements, so the above code can be written as:

```
int sum = 0
int i = 4
while i <= 6:
    next sum = sum + i
    next i = i + 1
```

L-Language has a full set of number types: `int8`, `uns8`, `int16`, `uns16`, `flt16`, ..., `int128`, `uns128`, `flt128`; for signed integer, unsigned integer, and floating point respectively. The types `int`, `uns`, `flt` are just these types for the target machine word size. The `bool` type is a single bit interpreted as true if 1 and false if 0: it is in essence a 1-bit unsigned integer, but it is not considered to be a number type.

User defined types have values that consist of a sequence of bytes containing fields. Fields in turn can contain subfields. An example is:

```
type my type:
    uns32                           // Container for:
    [8]      uns8 op code           //   Operation
    *OFFSET* 0
    [1]      bool has constant      //   Format indicator
    *OFFSET* 8
    [24]     int constant           //   Constant
    *OFFSET* 8
```

```
    [8]     uns8 src1               //   Source Register
    [8]     uns8 src2               //   Source Register
    [8]     uns8 des                //   Destination Register

. . . . . .

my type X:
    X.op code = 5       // This is an initialization block
    X.src1 = 2          // for X in which X is *INIT*.
    X.src2 = 3
    X.des = 3
uns op = X.op code      // Now op == 5
int d = X.des           // Now d == 3
ap *READ-WRITE* my type @Y    // `@= allocate' is implied
Y.op code = 129
bsp *READ-WRITE* int @C = @Y.constant
ap *READ-WRITE* uns8 @OP = @Y.op code
next op = OP            // Now op == 129
bool B = Y.has constant // Now B == 1
C = -1234               // Now Y.constant = -1234
```

In this example the declaration '`my type X`' is approximately the same as '`ap co my type @X`', but the pointer `@X` is hidden. In the subblock of code that initializes `X`, the `co` qualifier of `X` is implicitly changed to the `*INIT*` pseudo-qualifier that has the same effect as `*WRITE-ONLY*` except that it forces fields/subfields of `X` to be `*INIT*` even if they are `co` or `ro`. In code after the subblock, `X` is `co`.

In this example there is one field in a `my type` value, an unlabeled `uns32` integer. Inside this unlabeled field there are 6 subfields, the first of which is an `uns8` integer occupying the highest order 8 bits of the unlabeled field, Bits are numbered 0, 1, 2, ... from high to low order, so the offset of this subfield is 0. The second subfield, also at offset 0, is a 1-bit `bool` value that occupies the high order bit of the unlabeled field. Note that subfields can overlap.

Defined type values are aligned on byte boundaries when they are stored in memory. Therefore the '`op code`' subfield is on a byte boundary, and the location of `OP` is an `ap` aligned pointer. Although the `constant` subfield is on a byte boundary, it is shorter than an `int`, and therefore the location of `C` must be an `bsp` bit string pointer. If '`op code`' were in bits 1 through 8 instead of 0 through 7, it would not be on a byte boundary and the location of `OP` would also have to be an `bsp` bit string pointer.

Note that '`Y.op code`' is a `*READ-WRITE* uns8` while '`@Y.op code`' is a `co` pointer to a `*READ-WRITE* uns8`.

Names in L-Language can have multiple lexemes, as in the type name '`my type`', the subfield name '`op code`', and what L-Language calls the associated member name '`.op code`' which

can be used to access the field.

Another example is:

```
type my type:
    pack
    uns8    kind            // Object Kind
    [1] bool animal         // True if Animal
    [1] bool vegetable      // True if Vegetable
    flt64   weight          // Object Weight
    align   64
    *LABEL*   extension
    ***                     // Enables type extension.

type my type:
    *OFFSET*  extension
    flt64   height          // Object Height
    flt64   width           // Object Width
    ***                     // Enables type extension.

type my type:
    *OFFSET*  extension
    flt64   volume          // Object Volume; overlays height.
                            // No further type extension allowed.

type your type:
    *INCLUDE* my type // Copy sub-declarations of my type
    *OFFSET* *SIZE*   // *SIZE* is max origin seen so far.
    av uns8 @name     // Aligned vector pointer to name
                      // character string

. . . . . .

my type X:
    X.kind = BOX
    X.weight = 55
    X.height = 1023
    X.width = 572

your type Y:
    Y.kind = BEER
    Y.weight = 0.45
    Y.volume = 48
```

```
Y.@name = "John Doe's Lager"
```

Here `my type` and `your type` are defined by statements called *type-declarations*. Each of these *type-declarations* contains a sequence of sub-declarations, e.g., for `my type` the first two sub-declarations are '`pack`' and '`uns8 kind`'. There is a current offset in bits that starts at `0` and is updated by each sub-declaration. A sub-declaration such as '`uns8 kind`' allocates a field (i.e., `kind`) at the current offset and adds the size of the field to the current offset.

In the example the fields are `kind`, `weight`, `height`, etc. Fields can be packed or aligned; aligned is the default. An aligned number has an offset that is a multiple of the length of the number. Here fields are initially packed so that since `kind` has offset 0 bytes and size 1 byte, `weight` has offset 1 byte. Subfields `animal` and `vegetable` are 1-bit values inside `kind`.

The `align 64` sub-declaration moves the current offset forward to a 64-bit boundary and causes fields beyond it to be aligned and not packed. A number is aligned if its offset is a multiple of its length. Alignments must be powers of two. A defined type has an alignment equal to the least common multiple (in this case just the largest) of the alignments of its aligned fields.

A `*LABEL*` is like a zero length field that has no value and is used to associate an origin-label with the current offset. Here `extension` has the offset value of 128 bits (16 bytes). The `*OFFSET*` sub-declaration resets the current offset to the offset of a given origin-label, or to `*SIZE*`, which denotes the current size of the type in bits (which may increase with later sub-declarations).

The '`***`' sub-declaration at the end of a *type-declaration* defining a user defined type indicates that the definition may be continued by a later *type-declaration*, as is done for `my type` above. The sub-declarations of the later *type-declaration* are simply appended to those of previous *type-declarations*.

The `*INCLUDE*` sub-declaration copies all the sub-declarations from another user defined type. If the user defined type is defined by multiple *type-declarations*, only sub-declarations from the *type-declarations* in the current scope (see 13 $^{p116}$) are copied.

Defined types can be extended (as per the example), and fields can overlay each other. A defined type value has a size in bytes just large enough to accommodate all its fields. If a defined type has multiple *type-declarations*, this size may not be known until load time.

Values of defined types are strictly run-time values. In contrast, values of `const` type are compile-time values, and are not available at run-time. Number lexemes consisting of digits and optional signs, decimal points, and exponents, are converted to IEEE 64-bit floating point `const` values, as are special numeric-word lexemes such as `inf`, `+inf`, `-inf`, and `nan`. Other number constants represent rational `const` values with unbounded integral numerators and denominators; for example, `D#"1/3"` represents the precise rational one-third. Number constants can be converted to run-time numbers during compilation. However it is a compile error if the result will not fit into a run-time integer. This happens, for example, if `1.1`, `1e20`, or `D#"1/3"` is converted to an `int32`.

Quoted strings denote string `const` values that can be converted during compilation to run-time vectors with `co` unsigned integer elements that encode the string in UTF-8, UTF-16, or UTF-32.

Lastly there are map `const` values that can hold lists and dictionaries. Map values can be mutable at compile-time, but cannot be converted to run-time values.

Expressions, statements, and functions that use only `const` values execute at compile-time and can be used to compute compile-time `const` values including maps that represent code.

By default, functions in L-Language are inline. For example,

```
function int r = max ( int x, int y ):
    if x < y:
        r = y
    else:
        r = x

int x = ...
int y = ...
int z = max ( x, y )
```

L-Language does not support implicit conversions of run-time function results[2], but does support implicit conversion of variables[3] and constants. Any number constant or rational constant may be converted implicitly to any run-time number type as long as the constant value can be stored exactly in a variable of the run-time type or the run-time type is floating point (in which case there may be loss of precision or conversion to an infinity). Any run-time numeric variable value may be implicitly converted at run-time to a number type that will hold all the possible values of the variable, or to any floating point type (in which case the run-time conversion result may be less precise or an infinity).

Language expressions have **target types**. For a function call, the function result cannot be implicitly converted to the target type. However a function call that returns a `const` result is replaced by its value at compile time, and this value can be implicitly converted to the target type. Also, variable values may be implicitly converted to a target type.

Most builtin operators, such as '`+`', have operands of the same type as their result.

An example of all this, using the declaration of `max` above, is:

```
flt w = 1.1                 // flt is target type of 1.1
int x = 123                 // int is target type of 123
int y = 2e5                 // int is target type of 2e5
int z = 1e100               // illegal; int is target type of 1e100
                            // which is too large to fit
```

[2]In this matter L-Language follows ADA.
[3]Unlike ADA.

```
flt r1 = x                  // implicit conversion is legal as x is a
                            // variable name and not a function call
int r2 = 5 + max ( x, y )   // int is target type of +, 5, max, x, y
int r3 = max ( y, w )       // illegal; int is target type of w and
                            // flt cannot implicitly convert to int
int r4 = max ( x, 123 )     // int is target type of max, x, 123
int r5 = max ( x, 123.4 )   // illegal; int is target type of 123.4
                            // which cannot be stored in a int
const c1 = 100
const c2 = 1000
int r6 = max ( x, c1 + c2 ) // legal; c1 + c2 is replaced by 1100
                            // which has int target type
```

For each type `T`, a function:

```
function T r = T ( T v ):
    r = v
```

is provided. Such an 'identity' function might seem useless, but in fact it can be used in an expression '`T(`$e$`)`' to force the target type of $e$ to be `T`. As an example consider the following, where there are builtin functions:

```
function N r = N ( N v1 )
function N r = ( N v1 ) "+" ( N v2 )
function bool r = ( N v1 ) "<" ( N v2 )
```

for every number type `N`:

```
int x = ...
bool b1 = ( x + 3 < 5 )
    // Illegal: any target type N to which int is implicitly
    // convertible can be used as the target type of "+", so
    // this is ambiguous.
bool b2 = ( flt ( x + 3 ) < 5 )
    // Legal: the argument of flt must have type flt, so that
    // becomes the target type of "+".  Because flt (...) is
    // an explicit type conversion, flt is also the target
    // type of 5.
bool b3 = ( x + 3 < int ( 5 ) )
    // Legal: the argument of int must have type int, so that
    // becomes the target type of "5".  Because int (...) is
    // an explicit type conversion, int is also the target
    // type of x + 5.
```

Integer arithmetic ignores overflows (as in C and C++); for example, if integer + produces a value too large for its target, the result is undefined and may or may not cause program termination.

There are conversion functions from a floating point type `F` to any integer type `I` with prototypes:

```
function I r = floor ( F v )
function I r = ceiling ( F v )
function I r = truncate ( F v )
function I r = round ( F v )
```

that take a floating point value and round it toward negative infinity (`floor`), positive infinity (`ceiling`), zero (`truncate`), or nearest (`round`). If the value is too large to be stored in `I`, a floating point exception flag is set, and the result is an indefinite integer, which is a particular value of type `I` analogous to `NaN` for floating point.

Although function results cannot in general be implicitly converted, implicit conversions of qualifiers are allowed. For example:

```
function ap *READ-WRITE* int r = foo ( ... )
ap ro int @p = foo ( ... )   // Implicitly converts *READ-WRITE* to ro.
```

In addition to using target types to select which overloaded function is being called, the types of implicitly convertible arguments, that is, variables and constants, can be used. Specifically, a function definition at least one of whose arguments does not require an implicit conversion is preferred.

Finding a target type for comparison operands is more complicated. If one of the operands is a function call to an explicit type conversion function (a function whose name is a type name), then the type of that function is used. Othersize if some operands are reference expressions, their types are tried. An example is:

```
int x = ...
bool b1 = ( x < 6 )
    // There is a separate "<" operator for every number type,
    // X is a reference expression, so its type int is tried and
    // succeeds.
bool b2 = ( x < 6.5 )
    // Again int is tried and succeeds, but this time 6.5 produces
    // a compiler error when converted to an int.  Inconveribility of
    // particular constant values is NOT considered in selecting
    // function definitions.
bool b3 = ( x <= floor ( 6.5 ) )
    // Again int is tried and succeeds, but this time 6.5 is replaced
    // by 6.0 by the compile-time execution of the floor function,
```

```
    // and 6.0 converts to an int without error.
bool b3 = ( x <= flt ( 6.5 ) )
    // As flt is here an explicit conversion function with type flt,
    // that type must be used, and it succeeds.
```

It is possible to define compile-time functions:

```
constant function const r = max ( const x, const y ):
    if x < y: r = y
    else:     r = x

const x = 2e5
const y = 3e6
const z = 9e4
const w = max ( x, max( y, z ) )
```

A '`constant`' function is just an inline function that is guaranteed to not generate any run-time code. Thus it can be executed at compile time to compute with `const` values.

Inline function definitions may make use of type wildcards. A name that is a single word beginning with `T$` is a type wildcard that denotes an arbitrary type. Thus the example:

```
function T$r r = max ( T$r x, T$r y ):
    if x < y: r = y
    else:     r = x

const x = 2e5
int y = 27e4
int z = max ( x, y )      // T$r is int, x converts to int.
flt u = 3.1415
flt v = max ( x, u )      // T$r is flt, x converts to flt.
flt v = max ( x, y )      // T$r is flt, x and y convert to flt.
int v = max ( x, u )      // Illegal, T$r is int, u cannot
                          // implicitly convert to int.
```

A wildcard type of a result variable gets its value from the target type of a function call. A wildcard type of an argument can get its value from the argument type, but only if the later is a variable, or more generally, a reference expression (e.g., `x[i]`). Typing is mostly done top-down using target types, but reference expressions get types bottom up from the variable explicitly named in the reference expression.

Pointer types can be wildcards which must have names that are single words beginning with `P$`. A list of qualifiers can also be a wild card named by a single word beginning with `Q$`. An example is:

```
out-of-line function
```

```
        uns r = strlen ( dp Q$s uns8 @s ) : *DEFERRED* "strlen"
        // External function in C library.
    function uns r = strlen ( P$s Q$s uns8 @s ):
        // Inline wrapper.
        dp ro uns8 @sdp = *UNCHECKED* ( @s )
        r = strlen ( @sdp )
```

which converts the pointer of type `P$s` to a pointer of type `dp` (direct pointer) and calls the 'foreign' C programming language subroutine `strlen` with the direct pointer. The `*UNCHECKED*` function is needed to produce a direct pointer from other pointer types, though this function is undefined for some argument pointer types (e.g., bit string pointer types). Calling `strlen` with a `dp` pointer calls the external function directly, bypassing the inline wrapper, because the external call has fewer wildcards (only 1 instead of 2).

A pointer type has two places where a qualifier may appear, as in

```
type my control block:
    *READ-WRITE* ap *GLOBAL* ro uns32 @cr
    ....
```

in which `@cr` is a read-write pointer variable pointing at a global read-only `uns32` location `cr`.

Pointer types can be cascades: e.g., one can have a pointer to a pointer to a pointer. If the cascade has $N$ pointer type names the pointer variable name typically begins with exactly $N$ '`@`'s. However it is possible that the name begins with fewer that $N$ '`@`'s, including none at all. For example:

```
ap ap *READ-WRITE* ap int @@@x = ...
    // Now @@@x is a pointer with 3 cascaded pointer type names.
    // Now @@x is @@@x dereferenced once.
    // Now @x is @@x dereferenced once or @@@x dereferenced twice.
    // Now x is @x dereferenced once or @@@x dereferenced three times.
ap ap *READ-WRITE* ap int @y = ...
    // Now @y is a pointer with 3 cascaded pointer type names.
    // But it can only be dereference once using y.
ap ap *READ-WRITE* ap int z = ...
    // Now z is a pointer with 3 cascaded pointer type names.
    // But it cannot be dereferenced at all.  However,
    // it could be copied to a variable with a different
    // name beginning with 3 @'s, and that variable could be
    // dereferenced up to three times.
```

An `inclusion` function can create new code that replaces the statement calling the `inclusion` function. For example:

```
inclusion function const r = max add ( macro V, macro I, macro LIMIT ):
```

```
    r = *INCLUDE* (V, I, LIMIT):
        next V += I
        next V = max ( V, LIMIT )
max add ( my sum, my increment, z + 100 )
    // This statement is replaced by
    //     next my sum += my increment
    //     next my sum = max ( my sum, z + 100 )
    // or more specifically:
    //     V = { "my", "sum" }
    //     I = { "my", "increment" }
    //     LIMIT = { { "z" }, "+", { 100 } }
    //     r = { { { "next", "my", "sum" }, "+=",
    //             { "my", "increment" } },
    //           { { "next", "my", "sum" }, "="
    //             { "max", { { "my", "sum" },
    //                        { { "z" }, "+", { 100 } },
    //                        ".initiator" => "(",
    //                        ".terminator" => ")",
    //                        ".separator" => "," } } } }
```

# 3 Lexemes

An L-Language source file is a sequence of bytes that is a UTF-8 encoding of a sequence of UNICODE characters. This is scanned into a sequence of **lexemes**.[4]

Unless otherwise specified, the term '**character**' in this document means a 32-bit UNICODE character.

Lexemes are defined in terms of the following character classes:

***horizontal-space-character*** ::= characters in UNICODE category `Zs` (includes *ASCII-single-space*)
| *horizontal-tab-character*

***vertical-space-character*** ::= *line-feed* | *carriage-return*
| *form-feed* | *vertical-tab*

***space-character*** ::= *horizontal-space-character* | *vertical-space-character*

***graphic-character*** ::= characters in UNICODE categories `L`, `M`, `N`, `P`, and `S`

***control-character*** ::= characters in UNICODE categories `C` and `Z`

***isolated-separating-character*** ::=

[4] L-Language lexemes are layered system standard lexemes, which are a compromise between the needs of programming languages and the needs of natural languages.

characters in UNICODE categories `Ps`, `Pi`, `Pe`, and `Pf`;
includes `{ ( [ « » ] ) }`

***separating-character*** ::= `|` | *isolated-separating-character*

***leading-separator-character*** ::= `‘` | `¡` | `¿`
***trailing-separator-character*** ::= `’` | `!` | `?` | `.` | `:` | `,` | `;`

***quoting-character*** ::= `"`

***letter*** ::= characters in UNICODE category `L`
***ASCII-digit*** ::= `0` | `1` | `2` | `3` | `4` | `5` | `6` | `7` | `8` | `9`
***digit*** ::= characters in UNICODE category `Nd` (includes *ASCII-digits*)
***lexical-item-character*** ::= *graphic-character* other than *separating-character* or `"`

Comments may be placed at the ends of lines:

***comment*** ::= `//` *comment-character*$^\star$

***comment-character*** ::= *graphic-character* | *horizontal-space-character*

Lexemes may be separated by *white-space*, which is a sequence of *space-characters*, but, with some exceptions mentioned just below, is not itself a lexeme:

***white-space*** ::= *space-character*$^+$
***horizontal-space*** ::= *horizontal-space-character*$^+$
***vertical-space*** ::= *vertical-space-character*$^+$

The following is a special virtual lexeme:

***indent*** ::= virtual lexeme inserted just before the first *graphic-character* on a physical line

**Indent lexemes** have no characters, but do have an **indent**, which is the indent of the graphic character after the indent lexeme. The **indent** of a character is the number of columns that precede the character in the character's physical line. *Control-characters* other than *horizontal-space-characters* take zero columns, as do characters of classes `Mn` (combining-marks) and `Me` (ending marks). All other characters take one column, except for tabs, that are set every 8 columns. Indent lexemes are used to form logical lines and blocks ($4\,^{p21}$).

One kind of *vertical-space* is given special distinction:

***line-break*** ::= *vertical-space* containing exactly one *line-feed*

This is the *line-break* lexeme.

Non-*indent*, non-*line-break white-space*, such as occurs in the middle of text or code outside comments, is discarded and not treated as a lexeme. Such *white-space* may be used to separate lexemes.

*Horizontal-space-characters* other than single space are illegal inside *quoted-string* lexemes (defined below). *Vertical-space* that has no *line-feeds* is illegal (see below). *Control-characters*

not in *white-space* are illegal. Characters that have no UNICODE category are *unrecognized-characters* and are illegal:

***misplaced-horizontal-space-character*** :::=
*horizontal-space-character*, other than ASCII-single-space

***misplaced-vertical-space-character*** :::=
*vertical-space-character* other than *line-feed*

***illegal-control-character*** :::=
*control-character*, but not a *horizontal-space-character* or *vertical-space-character*

***unrecognized-character*** :::=
character with no UNICODE category or
with a category other than `L`, `M`, `N`, `P`, `S`, `C`, or `Z`

Sequences of these characters generate warning messages, but are otherwise like *horizontal-space*:

***misplaced-horizontal*** :::= *misplaced-horizontal-space-character*$^+$

***misplaced-vertical*** :::= *misplaced-vertical-space-character*$^+$

***illegal-control*** :::= *illegal-control-character*$^+$

***unrecognized*** :::= *unrecognized-character*$^+$

*Misplaced-horizontal* only exists inside a *quoted-string*, but the other three sequences can appear anywhere. When they occur, these sequences generate warning messages, but otherwise they behave like *horizontal-space*. Specifically, outside *quoted-strings* and *comments* these sequences can be used to separate other lexemes, just as *horizontal-space* can be used, whereas inside *quoted-strings* and *comments* these sequences do nothing aside from generating warning messages.

The lexemes in a L-Language program are specified in Figure 1 $^{p19}$. This specification assumes there are no illegal characters in the input; see text above to account for such characters.

The symbol '`:::=`' is used in syntax equations that define lexemes or parts of lexemes whose syntactic elements are character sequences that must not be separated by *white-space*. The symbol '`::=`' is used in syntax equations that define sequences of lexemes that may and sometimes must be separated by *white-space*.

There is a special ***end-of-file*** lexeme that occurs only at the end of a file.

Files are scanned into sequences of lexemes which are then divided into logical lines as per 4 $^{p21}$. After each logical line is formed, *indent*, *comment*, *line-break*, and *end-of-file* lexemes are deleted from the logical line.

A ***special-character-representative*** can consist of a UNICODE character name surrounded by angle brackets. Examples are `<NUL>`, `<LF>`, `<SP>`, `<NBSP>`. There are three other cases: `<Q>` represents the doublequote `"`, `<NL>` (new line) represents a line feed (same as `<LF>`), and `<UUC>` represents the '**unknown UNICODE character**' which in turn is

***lexeme*** ::= *numeric-word* | *word* | *natural* | *number* | *numeric*
| *mark* | *separator* | *quoted-string*
| *indent* | *line-break* | *comment* | *end-of-file*

***strict-separator*** ::= *isolated-separating-character* | `|`$^+$

***leading-separator*** ::= `‘`$^+$ | `¡`$^+$ | `¿`$^+$

***trailing-separator*** ::= `’`$^+$ | `!`$^+$ | `?`$^+$ | `.`$^+$ | `:`$^+$ | `;` | `,`

***separator*** ::= *strict-separator* | *leading-separator* | *trailing-separator*

***quoted-string*** ::= `"` *character-representative*$^\star$ `"`

***character-representative*** ::= *graphic-character* other than `"`
| *ASCII-single-space-character*
| *special-character-representative*

***special-character-representative*** ::= `<` { *upper-case-letter* | *digit* }$^+$ `>`

***lexical-item*** ::= *lexical-item-character*$^+$ not beginning with `//`

***lexical-item*** ::= *leading-separator*$^\star$ *middle-lexeme*$^?$ *trailing-separator*$^\star$

***middle-lexeme*** ::= *lexical-item* not beginning with a *leading-separator-character*
or ending with a *trailing-separator-character*

***numeric-word*** ::= *sign*$^?$ `nan` | *sign*$^?$ `inf` [where *letters* are case insensitive]

***word*** ::= *middle-lexeme* that contains a *letter* before any *digit*
and is not a *numeric-word*

***natural*** ::= *ASCII-digit*$^+$ not beginning with `0` | `0`
[but lexical type may be changed; see p21]

***number*** ::= *sign*$^?$ *integer-part exponent-part*$^?$ that is not a *natural*
| *sign*$^?$ *integer-part*$^?$ *fraction-part exponent-part*$^?$
[but lexical type may be changed; see p21]

***integer-part*** ::= *ASCII-digit*$^+$ ***fraction-part*** ::= `.` *ASCII-digit*$^+$

***exponent-part*** ::= *exponent-indicator sign*$^?$ *ASCII-digit*$^+$

***sign*** ::= `+` | `-` ***exponent-indicator*** ::= `e` | `E`

***numeric*** ::= *middle-lexeme* that contains a *digit* before any *letter*
and is not a *natural* or *number*

***mark*** ::= *middle-lexeme* not containing a *letter* or a *digit*

*indent* ::= see p17 *line-break* ::= see p17

*comment* ::= see p17 *end-of-file* ::= see p18

Figure 1: L Language Program Lexemes

used to represent illegal UTF-8 character encodings.

A *special-character-representative* can also consist of a hexadecimal UNICODE character code, which must begin with a digit. Thus **<0FF>** represents ÿ whereas **<FF>** represents a form feed.

**Quoted string lexemes** separated by the '**#**' mark are glued together if they are in the same logical line. Thus

```
"This is a longer sentence" #
    " than we would like."
"And this is a second sentence."
```

is equivalent to

```
"This is a longer sentence than we would like."
"And this is a second sentence."
```

This is useful for breaking long quoted string lexemes across line continuations. But there is an important case where there is not an exact equivalence between the glued and unglued versions. **"<" # "LF" # ">"** is not equivalent to **"<LF>"**. The former is a 4-character quoted string, the characters being **<**, **L**, **F**, and **>**. The latter is a 1-character quoted string, the character being a line feed.

The definition of a **middle-lexeme** is unusual: it is what is left over after removing *leading-separators* and *trailing-separators* from a *lexical-item*. The lexical scan first scans a *lexical-item*, and then removes *leading-separators* and *trailing-separators* from it. Also *trailing-separators* are removed from the end of a *lexical-item* by a right-to-left scan, and not the usual left-to-right scan which is used for everything else. Thus the *lexical-item* '**¿4,987?,,::**' yields the *leading-separator* '**¿**', the *middle-lexeme* '**4,987**', and the four *trailing-separators* '**?**', '**,**' '**,**' and '**::**'.

*Words*, *numerics*, and *marks* in the same logical line are glued together if the first ends with '**#**' and the second begins with '**#**'. Thus

```
This is a continued-#
    #middle# #-lexeme.
```

is equivalent to

```
This is a continued-middle-lexeme.
```

For compatibility, two consecutive '**#**' marks may be used to glue together two quoted strings, as in

```
"This is a continued-"#
    #"quoted"# #"-string".
```

which is equivalent to

```
"This is a continued-quoted-string".
```

A *numeric-word*, *natural*, or *number* lexeme is a C/C++ constant, and conversely a C/C++ decimal constant that does not end in a *decimal-point* is a *numeric-word*, *natural*, or *number* lexeme. All these lexemes are given an IEEE double precision number value after the manner of C/C++, and then their lexical type is changed as follows:

- If the value is not a finite number, the new type is *numeric-word*. For example, this applies to `1e500` which converts to the same value as `+inf`.
- If the value is an integer in the range $[0, 10^{15})$ the new type is *natural*. For example, this applies to `1e3` which converts to the same value as `1000`.
- Otherwise the new type is *number*. For example, this applies to `1e20` or `1.1`.

In contrast, a *numeric*, like `02/28/2022`, represents a character string and in this is like a *word*. The lexeme `2/3` is also a *numeric* and is not used to represent a number; instead the lexeme pair `D# "2/3"` is used to represent a rational number constant (value of `const` type).

# 4 Logical Lines, Blocks, and Statements

Each non-blank physical line begins with an *indent* lexeme that is followed by a lexeme that cannot be an *indent*, *line-break*, or *end-of-file*.

Lexemes are organized into **logical lines**.[5] A logical line begins immediately after an *indent* lexeme, and the **indent** of the logical line is the indent of this *indent* lexeme (i.e., the indent of the first graphic character of the logical line).

A logical line ends with the next *indent* lexeme whose indent is not greater than the indent of the logical line, or with an *end-of-file*. Thus physical lines with indent greater than that of the current logical line are **continuation lines** for that logical line.

A code file is a sequence of '**top level**' logical lines that are required to have indent `0`.

A logical line may end with a **block** that is itself a sequence of logical lines that have indents greater than the indent of the logical line containing the block. The block is introduced by a '`:`' at the end of a physical line, provided the '`:`' is not inside brackets or quotes (e.g., not inside `( )` or `‘ ’`). If the first *indent* lexeme after the '`:`' has an indent that is not greater than the indent of the logical line containing the '`:`', the block is empty. Otherwise the indent of this *indent* lexeme becomes the **indent** of the block and the indent of all the logical lines in the block. The first logical line of the block starts immediately after this *indent* lexeme. The block ends just before the first logical line with lesser indent than the block indent, or the end of file. More specifically, the last logical line of the block ends with an *indent* whose indent is less than the block indent, or with an *end-of-file*.

[5] L-Language logical lines and indented paragraphs are layered system bracketed subexpression recognition pass logical lines and paragraphs.

Examples are:

```
this is a top level logical line ending with a block:
    this is the first line of the block
    this is the
         second line of the block
    this is the third line of the block:
        this is the first line of a subblock
        this is the second line
                of the subblock:
            this is the only line of a sub-subblock
        this is the third line of the subblock
    this is the fourth line
            of the block:
        this is the only line of the second subblock
    this is the fifth line of the block
         and it ends with an empty subblock:
this is the second top level
     logical line
```

A warning message is output if two indents that are being compared differ by more than `0` and less than `2` columns, in order to better detect indentation mistakes.

*Line-break* lexemes are effectively ignored. A sequence of *line-break* lexemes is followed by an *indent* or *end-of-file* which is not ignored. Blank physical lines are represented by sequences of more than one *line-break* lexeme, and are effectively ignored.

A logical line that contains *comments*, but no lexemes other than *comments*, *line-breaks*, *indents* and a possible *end-of-file*, is a '**comment line**'.

It is an error to begin non-comment logical lines with a *comment*. *Comments* can be used freely in the middle of or at the end of any logical line, or at the beginning of a comment line.

It is an error for the first logical line of a file to have an indent that is greater than `0`, the top level indent.

It is an error for a block to be in the middle of a logical line. This means that the first *indent* following the block must have an indent no greater than that of the logical line containing the block.

Examples are:

```
// this is a logical line that is a single comment

// this is a logical line that has two
    // comments
```

```
this is a logical line // with a comment
     // and another comment
     with three comments // and a last comment

this is a logical line ending with a block:
     First line of the block
     Second line of the block
// Comment that ends block
// Comment that is in error because
    it begins a logical line that this continues

this is a logical line with a block:
     First line of the block
     Second line of the block
  but the block is in error because it is before
  this continuation of the logical line that contains
  the block

this is a logical line ending with a block:
        First line of the block
        Second line of the block
  // comments that end the block, but are in error,
  // because they continue the logical line
  // containing the block
```

After a logical line has been formed, any *indent*, *comment*, *line-break*, and *end-of-file* lexemes in the logical line are removed from the logical line. If the result is empty, e.g., the logical line is a comment line, it is discarded. Otherwise the modified logical line becomes a L-Language '***statement***'.

Therefore a file is a sequence of top-level statements.

Since a logical line can end with a block that itself consists of a sequence of logical lines, a statement can end with a block that itself consists of a sequence of statements.

# 5 Data Structure

Run-time data is built from numbers, memory blocks, and pointers into memory blocks. A **memory block** is a continguous sequence of bytes that may be moved in memory. Each memory block has a **memory block root** that points at the memory block and is fixed in memory. A pointer contains a pointer to the memory block root and an offset from the beginning of the memory block the memory block root points at.

The bytes of a memory block are grouped into numbers, and the numbers are grouped into larger data structures such as character strings, vectors of numbers, pointers, and user defined structures.

## 5.1 Builtin Run-Time Number Types

| Type | Length in Bits | Format |
|---|---|---|
| **int8** | 8 | two's complement integer |
| **int16** | 16 | |
| **int32** | 32 | |
| **int64** | 64 | |
| **int128** | 128 | |
| **int** | integer word length | |
| **intptr** | address length | |
| **uns8** | 8 | unsigned binary integer |
| **uns16** | 16 | |
| **uns32** | 32 | |
| **uns64** | 64 | |
| **uns128** | 128 | |
| **uns** | integer word length | |
| **unsptr** | address length | |
| **ptr** | address length | `intptr` or `unsptr` |
| **flt8** | 8 | IEEE floating point number |
| **flt16** | 16 | |
| **flt32** | 32 | |
| **flt64** | 64 | |
| **flt128** | 128 | |
| **flt** | floating point word length | |

- Some computers use `intptr` for addresses, and some use `unsptr`. `ptr` is which ever of these the target computer uses.
- Addresses are assumed to be byte addresses. So the difference between two addresses is a number of bytes, and adding an integer I to an address moves the address by I bytes (I may be negative).

- Integer word length in the length of the integer registers, floating point word length is the length of the floating point registers, address length is the length of the address registers. On many computers the same registers are used for integers, floating point numbers, and addresses.
- `bool` [p26] is not a number type.

## 5.2 Builtin Run-Time Pointer Types

A pointer type has an associated data type which is a structure whose components are the parts of the pointer. For example, the pointer type `dp` has the associated data type `data for dp`.

**dp** A **direct pointer** is just a byte-address. Direct pointers are mostly used to communicate with subroutines written in languages such as C and C++.

```
pointer type std dp is type std data for dp
type std data for dp:
    ptr address
```

Here `address` is the byte address.

**ap** An **aligned pointer** points at a byte in a memory block.

```
pointer type std ap is type std data for ap
type std data for ap:
    dp ro ptr @base
    intptr offset
```

Here `@base` is the address of `base`, the address of the first byte of the memory block, and `base + offset` is the address of the byte pointed to in the block.

**av** An **aligned vector pointer** points at a vector of elements in a memory block. The elements must begin on byte boundaries and the difference between consecutive elements, known as the **step**, must be an integral number of bytes. The index $I$ used to access an element must be in the range [0,**limit**).

```
pointer type std av is type std data for av
type std data for av:
    dp ro ptr @base
    intptr offset
    intptr step
    intptr limit
```

Element $I$ has address `base + offset +` $I$ `* step`, and to be legal $I$ must satisfy $0 \leq I <$ limit.

**bsp** A **bit string pointer** points at a **bit string** which is a sequence of consecutive bits in a memory block.

```
pointer type std bsp is type std data for bsp
type std data for bsp:
    dp ro ptr @base
    intptr offset
```

Here `@base` is the address of `base`, the address of the first byte of the memory block, and `8 * base + offset` is the bit-offset of the first bit of the bit string from the beginning of the containing memory block.

**bsv** A **field vector pointer** points at a vector of field elements in a memory block. The difference between consecutive elements, known as the **step**, is an integral number of bits. The index $I$ used to access an element must be in the range [0,**limit**).

```
pointer type std bsv is type std data for bsv
type std data for bsv:
    dp ro ptr @base
    intptr offset
    intptr step
    intptr limit
```

Bit string element $I$ has bit-offset `8*base + offset +` $I$ `* step` from the beginning of the containing memory block, and to be legal $I$ must satisfy $0 \leq I <$ limit.

## 5.3 Other Builtin Types

**bool** A `bool` datum stores one of two values: **true** or **false**. When used as the type of a structure field, a `bool` is stored as if it were `int8`, `true` is `-1`, and `false` is `0`. When used as a subfield type the subfield must be 1 bit long, `true` is `1` and `false` is `0`.

**const** `const` is the type of all compile time values, which are also known as constants. See Constants p39.

# 6 Expressions

Expressions[6] are built from operators, such as `+` and `*`, and primaries, such as variable names and function calls.

Operators are characterized by fixity, precedence, and format. The L-Language operators are listed in Figures 2 p29, 3 p30, 4 p30, and 5 p31.

[6] L-Language expressions are layered system standard operator parsing pass expressions with the addition of some operators.

Given this, expressions have the following syntax, where an *P-expression* is an expression all of whose operators that are outside brackets have precedence equal to or greater than P:

***expression*** ::= *(L-1)-expression*
***P-expression*** ::= *P-initial-operators*$^?$
{ *(P+1)-expression* | *P-middle-operator)* }$^+$
*P-final-operators*$^?$
***P-basic-expression*** ::= *P-initial-operator*$^?$
{ *(P+1)-expression* | *P-middle-operator)* }$^+$
*P-final-operator*$^?$
***(H+1)-expression*** ::= *primary* [see p34]
***P-initial-operators*** ::= *P-prefix-operator*$^\star$ *P-initial-operator*
***P-final-operators*** ::= *P-final-operator P-postfix-operator*$^\star$
***P-operator*** ::= operator of precedence P
***P-initial-operator*** ::= *P-operator* with **`initial`** flag
***P-final-operator*** ::= *P-operator* with **`final`** flag
***P-middle-operator*** ::= *P-operator* with neither **`initial`** nor **`final`** flag
***P-prefix-operator*** ::= *P-operator* with both the **`initial`** and **`right`** flags
***P-postfix-operator*** ::= *P-operator* with both the **`final`** and **`left`** flags

where in a *P-expression*:
P is any precedence in the range [L-1,H];
no two *(P+1)-expressions* may be adjacent;
each *P-operator* with a **`left`** flag must be preceded by a *(P+1)-expression*;
each *P-operator* with a **`right`** flag must be followed by a *(P+1)-expression*;
any operator with an **`afix`** flag must not be the first *P-operator*
in a *P-expression*;
no operator may have both an **`initial`** and a **`left`** flag;
no operator may have both a **`right`** and a **`final`** flag;
no operator may have both an **`initial`** and an **`afix`** flag;
if there is more than one operator in a *P-expression-initial-operators*,
implicit '**`(`**' left parentheses are inserted between operators;
if there is more than one operator in a *P-expression-final-operators*,
implicit '**`)`**' right parentheses are inserted between operators;
if the implicit parentheses are not balanced, implicit parentheses
are inserted at the beginning or end to balance the implicit parentheses;

Essentially the expression being parsed is organized into *P-expressions* where *P* is the precedence of the *P-expression*. Generally a *P-expression* consists of a sequence of *(P+1)-expressions* and operators of precedence *P*. If a *P-expression* is not a *P-basic-expression*, implicit parentheses are inserted to make the *P-expression* into a *P-basic-expression* that

contains a subexpression that is also a *P-basic-expression*. As an example of this, if '`-`' is a prefix operator, '`- - x`' is converted to the equivalent of '`- (- x)`'.

The highest precedence expression, the *(H+1)-expression*, contains no operators outside brackets, and is thus a primary like '`x`' and '`f(y+1)`'.

The operators can have any combination of the following **base fixities**:

| | |
|---|---|
| **`initial`** | *P-operator* must be the first thing in its *P-expression*. |
| **`final`** | *P-operator* must be the last thing in its *P-expression*. |
| **`left`** | *P-operator* must be immediately preceded by a *(P+1)-expression* in its *P-expression*. |
| **`right`** | *P-operator* must be immediately followed by a *(P+1)-expression* in its *P-expression*. |
| **`afix`** | *P-operator* must be after a (not necessarily immediately) preceding *P-operator* in its *P-expression*. |

The following **combination fixities** are defined:

| | |
|---|---|
| **`prefix`** | `initial` + `right` |
| **`infix`** | `left` + `right` |
| **`postfix`** | `left` + `final` |
| **`nofix`** | none of `initial`, `final`, `left`, or `right` |

All of these but `initial` and `prefix` can be combined with `afix`.

Line Level Operators: Part I
Must Occur Outside Parentheses and Brackets
At Top Level or Inside {* ... *}

| Operator | Meaning | Fixity | Format | Precedence |
|---|---|---|---|---|
| **if** | conditional | initial | conditional | 0000 |
| **else if** | | | | |
| **else** | terminating conditional | | terminating conditional | |
| **:** | conditional completion | afix right | (none) | |
| **subblock** | conditional or declaration completion | afix | | |
| | assignment or loop | postfix | postfix | |
| **type** | declaration | initial | declaration | |
| **pointer type** | | | | |
| **out-of-line function** | | | | |
| **function** | | right | | |
| **reference function** | | infix | | |
| **is type** | | afix infix | (none) | |
| **is function** | | | | |
| **--->** | abbreviate | infix | binary | |
| **=** | assignment | left | assignment | 1000 |
| **+=** | increment | infix | binary | |
| **-=** | decrement | | | |
| ***=** | multiply by | | | |
| **/=** | divide by | | | |
| **\|=** | include | | | |
| **&=** | mask | | | |
| **^=** | flip | | | |
| **<<=** | shift left | | | |
| **>>=** | shift right | | | |
| **@=** | pointer assignment | infix | binary | 1100 |

Figure 2: L-Language Line Operators: Part I

Line Level Operators: Part II
Must Occur Outside Parentheses and Brackets
At Top Level or Inside {* ... *}

| Operator | Meaning | Fixity | Format | Precedence |
|---|---|---|---|---|
| **do** | iterator | initial | iterator | 2000 |
| **while** | | | | |
| **until** | | | | |
| **exactly** | | | | |
| **at most** | | | | |
| **do** | | afix | (none) | |
| **while** | | | | |
| **until** | | | | |
| **exactly** | | | | |
| **at most** | | | | |
| **times** | iteration modifier | afix | (none) | |

Figure 3: L-Language Line Operators: Part II

Non-Line Level Operators: Part I
May Occur Inside or Outside Parentheses and Brackets

| Operator | Meaning | Fixity | Format | Precedence |
|---|---|---|---|---|
| **,** | separator | nofix | separator | 3000 |
| **if** | selector | infix | selector | 10000 |
| **else** | | infix afix | (none) | |
| **BUT NOT** | logical and not | infix | binary | 11000 |
| **AND** | logical and | infix | n-ary | 11100 |
| **OR** | logical or | | | |
| **NOT** | logical not | prefix | unary | 11200 |
| **==** | is equal | infix | infix | 12000 |
| **!=** | is not equal | | | |
| **<** | is less than | | | |
| **<=** | is less than or equal | | | |
| **>** | is greater than | | | |
| **>=** | is greater than or equal | | | |

Figure 4: L-Language Non-Line Operators

Non-Line Level Operators: Part II
May Occur Inside or Outside Parentheses and Brackets

| Operator | Meaning | Fixity | Format | Precedence |
|---|---|---|---|---|
| `+` | addition | infix | sum | 13000 |
| `-` | subtraction | | | |
| `\|` | bitwise or | | n-ary | |
| `&` | bitwise and | | | |
| `^` | bitwise xor | | | |
| `/` | division | infix | binary | 13100 |
| `*` | multiplication | | n-ary | 13200 |
| `**` | exponentiation | | binary | 13300 |
| `<<` | left shift | | | |
| `>>` | right shift | | | |
| `+` | no-op | prefix | unary | H-1 |
| `-` | negation | | | |
| `~` | bitwise complement | | | |
| `#` | length | | | |
| `D#` | decimal rational | | | |
| `B#` | binary rational | | | |
| `X#` | hexadecimal rational | | | |
| `C#` | character rational | | | |

Figure 5: L-Language Non-Line Operators

The operators in Figures $2^{p29}$, $3^{p30}$, $4^{p30}$, and $5^{p31}$ have precedences in the range *[L,H]*. Precedence *(L-1)* is reserved for the 'error operator' which is a nofix operator inserted by the parser to 'fix up' parsing errors so parsing can continue. Precedence *(H-1)* is reserved for prefix operators and precedence *H* is reserved for postfix operators (although not all prefix and postfix operators have these precedences).

The first *P-operator* in a *P-expression* determines the *P-expression*'s **format**, which is one of the following, where in describing expression formats we use:

'expression' to mean *P-expression*,
'operator' to mean *P-operator*,
and 'operand' to mean *(P+1)-expression*:

**conditional** The expression must consist of the operator followed by an operand followed by either a **:** operator and an operand or by just a subblock operator (**:** indented paragraph, which can be an operator).

**terminating conditional** The expression must consist of the operator followed by either a **:** operator and an operand or by just a subblock operator.

**postfix** The expression must consist of an operand followed by the operator.

**declaration** The expression must consist of an operator followed by an operand (that may contain higher precedence operators such as **=** and **,**) followed optionally by a subblock operator.

**binary** The expression must consist of an operand followed by the operator followed by an operand. There must be only one operator in the expression.

**assignment** The expression must consist of an operand followed by the operator followed by an optional operand (in 'x =:' the : subblock has lower precedence than = so 'x =' becomes a subexpression in which = has no right operand).

**selector** The expression operators must all be either **if** or **else**. The expression must consist of alternating operands and operators and begin and end with an operand. The two possible operators alternate, with **if** first and **else** last.

**separator** All operators in the expression must be identical. There are no other constraints on the expression. An implied empty operand is inserted between two consecutive operators, at the beginning if the expression begins with an operator, and at the end if the expression ends with an operator. Then the operators are deleted from the expression and the expression operator is attached to the expression as its **.separator** attribute.

**infix** The expression must consist of alternating operands and operators and begin and end with an operand.

**n-ary** All operators in the expression must be identical. The expression must consist of alternating operands and operators and begin and end with an operand.

**unary** The expression must consist of the operator followed by an operand.

**iteration** The expression must conform to the syntax of an *iteration-control*[p67], where operands are assumed to be *loop-labels*, *int-expressions*, or *bool-expressions*.

**sum** The expression operators must all be either **+** or **-**. The expression must consist of alternating operands and operators and begin and end with an operand.

There is an additional special syntactic rule:

1. Non-line bitwise operators (`|`, `&`, `^`, `<<`, `>>`, and `~`) cannot be mixed with non-line arithmetic operators (`+`, `-`, `/`, `*`, and `**`) outside parentheses in a subexpression. E.g., '`x + (y * ~ z)`' is illegal but '`x + (y * (~ z))`' is legal.

Full semantics of operators and expressions is described later, but the following examples give an idea of some of this semantics:

`T v = x + y * z`
: Here `T` is the **target type** of the expression '`x + y * z`' and thus must be the result type of the prototype of the '`+`' function, since function results cannot be implicitly converted. Because it is the result type of `+` and arithmetic operators (with a few exceptions) have operands that are of the same type as their result, `T` is also the target type of `x` and `*`, and since it is the target type of `*` it will be the target type of `y` and `z`. Implicit conversions of variables are allowed, so `x`, `y`, and `z` will all be converted to type `T` before any computation is done.

`T v = x if y else z`
: If `y` is not a `const`, it is evaluated with target type `bool`. If that value is `true`, `x` is evaluated and returned; otherwise `z` is evaluated and returned. Both `x` and `z`, have target type `T`.

  However if `y` is a `const` value, the right-side of the statement is replaced by `x` or `z`, whichever is discarded is also not compiled, and if it would be in error were it compiled, the error is not detected (unless it is a parsing error).

`bool v = x AND y`
: If either operand evaluates at compile-time to `FALSE`, compile-time evaluation stops and the statement is replaced by '`bool v = FALSE`'.

  Otherwise same as '`bool v = y if x else FALSE`'.

  The `const` values `TRUE` and `FALSE` are implicitly convertible to run-time `bool` `true` (`1`) and `false` (`0`), respectively.

`x < y < z`
: This is logically equivalent to '`x < y` AND `y < z`', except that `y` is evaluated at most once.

  If any comparison evaluates to `FALSE` at compile-time, compile-time evaluation stops and the entire expression is replaced by '`FALSE`'.

  If a single comparison evaluates to `TRUE` at compile-time, that comparison is removed from the `AND`-containing version of the expression.

  If run-time evaluation is necessary, some operands need to be evaluated at run-time, and a target type `T` needs to be found for these operands. A single target type `T` must work as the target type of all the run-time operands. If one of the operands is

a call to an explicit conversion function, the type of that conversion is the only one considered (i.e., if one operand has the form `T (...)` then `T` is the target type for all the operands). Otherwise if some of the operands are reference expressions, their types are tried. If neither of these methods works, it is a compile error.

Thus changing the expression to '`T(x) < y < z`' will force the target type to be `T` if that works, or a compile error otherwise.

`v[x+5] = y`
The target type of subscript expressions such as '`x + 5`' is `int`.

`~ x`
The '`~`' operator evaluates on signed integers as if they were represented in two's complement by binary values of unbounded size, and similarly for other bitwise operators.

`x ** y`
Requires that `y` be a `const` integer; `x ** 0 == 1`, `x ** 1 == x`, `x ** -1 == 1/x` for all `x`.

`x += y`
Means '`x = x + y`' if `x` is `*READ-WRITE*` and '`next x = x + y`' if `x` is `co` and '`next x`' is defined.

`next x += y`
Means '`next x = x + y`'. '`next x`' must be defined.

# 7 Primaries

A **primary** is an *expression* that has no operators outside parentheses or brackets:

***primary*** ::= *constant-primary*
| *reference-expression* [p48]
| *function-call* [p57]
| *bracketed-expression* [p58]

***constant-primary*** ::= *constant* other than *rational-constant*

*constant* ::= see p39

*rational-constant* ::= see p40

Note that a *rational-constant* is an operator (e.g. `D#`) followed by a *string-constant*, and therefore contains an operator and is not a *primary*.

## 7.1 Names

A **name** is a sequence of lexemes used to name things like variables and functions. Names are building blocks of primaries.

***name*** ::= *initial-name-item continuing-name-item*$^\star$
***initial-name-item*** ::= *name-item* other than *natural*
***continuing-name-item*** ::= *name-item* not containing '`.`'
***name-item*** ::= *word* containing no '`.`' that follows a character that is not a '`.`'
[see text about splitting words with embedded '`.`'s]
| *natural* [see p21]
| *quoted-word* containing no '`.`' that follows a character
that is not the beginning `"` or another '`.`'
| *quoted-mark* containing no '`.`' that follows a character
that is not the beginning `"` or another '`.`'
| *quoted-separator* not containing '`.`'s
***quoted-word*** ::= `"` *word* `"`
***quoted-mark*** ::= `"` *mark* `"`
***quoted-separator*** ::= `"` *separator* `"`

- '`.`'s must appear at the beginning of *words* and `marks` (examples: '`..size`' and '`.*`'). See text about splitting *words* and *marks* with embedded '`.`'s.

*Words* and *marks* containing embedded '`.`'s are split into parts which contain '`.`'s only at their beginning. Thus:

| | | |
|---|---|---|
| `bill.1.weight..size` | splits into | `bill .1 .weight ..size` |
| `p.@.*` | splits into | `p .@ .*` |
| `.@.*` | splits into | `.@ .*` |
| `.,.*` | splits into | `., .*` |
| `.*.p` | splits into | `.* .p` |

However it is a compile error if one of the parts is not a *word* or *mark*, as in the examples where `.1` is a *number* and `.,` is not a legal lexeme.

*Quoted-words* and *quoted-marks* are not split.

A function may be defined with a *name* that is a *quoted-word* or *quoted-mark*, such as `"AND"` or `"+"`. It may then be called by an expression in which the quotes are omitted, such as `x AND y` or `x + y`. Similarly a function may be defined with a function-term `".*"` and called by the expression `p.*` which is split into `p .*`.

Name items beginning with more than one '`.`' are reserved for use by systems and compilers (e.g., `..size` in the examples). Name items that are words containing '`$`' or that both begin and end with '`*`' are similarly reserved. For example, words of the form '`T$`...' are reserved for use as type wildcards, and the word `*READ-WRITE*` is reserved for use as a qualifier.

A name may begin with a *word* that is a *module-abbreviation* that designates a code module: see 11.7 $^{p113}$. For example `std` abbreviates the builtin standard module.

L-Language uses many kinds of names (the numbers in brackets refer to rules below):

***simple-name*** ::= *word* not containing any '`.`'s or '`@`'s [2, 3, 6]
***module-abbreviation*** ::= *simple-name*
***ma*** ::= *module-abbreviation*
***pointer-type-name*** ::= *ma*? *simple-name* [4,5]

***basic-name*** ::= *name* not containing a '`.`', *quoted-mark*, or *quoted-separator* [1, 2, 3, 6]
***type-name*** ::= *ma*? *basic-name* not containing '`@`'s [4,5]
***variable-name*** ::= *ma*? *basic-name*
***pointer-variable*** ::= *variable-name* whose *basic-name* may, but need not, begin with '`@`'s [see text below]
***non-pointer-variable*** ::= *variable-name* whose *basic-name* does not begin with an `@`
***target-variable*** ::= *variable-name* [see text below]
***primary-target-variable*** ::= *target-variable* [see text below]
***function-variable-name*** ::= *variable-name*
***function-constant-name*** ::= *variable-name*
***statement-label*** ::= *basic-name* not containing '`@`'s

***member-name*** ::= *name* beginning with a *word* or *quoted-mark* containing a '`.`', but not containing a *quoted-separator* [2, 3, 6]
(note: all '`.`'s in a *name* must be at the beginning of the *name*)
***pointer-member-name*** ::= *member-name* that may, but need not, have '`@`'s immediately following the initial '`.`'s
***target-member-name*** ::= *member-name* that is not a *pointer-member-name*

***data-label*** ::= *basic-name* | *member-name* not containing a *quoted-mark*

***function-member-name*** ::= *member-name*
***function-term-name*** ::= *name* not containing a '`.`' [1, 2, 3, 6]

***qualifier-name*** ::= *access-qualifier* | *lifetime-qualifier*

A list of qualifiers may have at most one *access-qualifier* and one *lifetime-qualifier*, and if both are present their order never matters (e.g., if two lists are compared, both lists are effectively sorted before the comparison).

***access-qualifier*** ::= `co` | `ro` | `*READ-WRITE*` | `*WRITE-ONLY*` | `*INIT*`

`co` abbreviates 'constant' meaning 'never changes'
`ro` abbreviates 'read-only' meaning 'other code may change'
`*INIT*` forces write-only during initialization

***lifetime-qualifier*** ::= `*GLOBAL*` | `*HEAP*`

`*GLOBAL*` has global (forever) lifetime
`*HEAP*` has lifetime managed by a garbage collector

Absence of a lifetime qualifier indicates the lifetime of a pointer target may only be as long as the lifetime of the pointer.

***initial-operator*** ::= `if` | `else` | `type` | `pointer type`
| `out-of-line function` | `function` | `do` | `repeat`
| `while` | `until` | `exactly` | `at most` | `NOT`
| `+` | `-` | `-` | `~` | `#` | `D#` | `B#` | `X#` | `C#`

***middle-operator*** ::= `:` | `function` | `is type` | `do` | `while` | `until` | `exactly`
| `at most` | `times` | `if` | `else` | `BUT NOT` | `AND` | `OR`
| `--->` | `=` | `+=` | `-=` | `*=` | `/=` | `|=` | `&=` | `~=` | `<<=` | `>>=`
| `@=` | `==` | `!=` | `<` | `<=` | `>` | `>=` | `+` | `-` | `|` | `&` | | `^` | `/` | `*`
| `<<` | `>>`

***final-operator*** ::= `times`

***function-keyword*** ::= `no` | `not` | `function`
| `"="` | `","` | `"("` | `")"` | `"["` | `"]"`

***wild-card*** ::= *simple-name* beginning with *wild-card-prefix*

***wild-card-prefix*** :::= one of:

| | |
|---|---|
| `T$` | name is assigned a *type-name* |
| `P$` | name is assigned a *pointer-type-name* |
| `PST$` | name is assigned a *type-name* preceeded by a possibly empty sequence of *pointer-type-names* and *qualifiers* beginning with a *pointer-type-name* |
| `Q...$` | *qualifier-wild-card*; name is assigned a list of *qualifier-names* subject to *qualifier-wild-card-flags* '...' |

***qualifier-wild-card-flag*** :::= one of:

| | |
|---|---|
| `R` | readable, excludes `*WRITE-ONLY*` and `*INIT*` |
| `W` | writable, excludes `ro` and `co`<br>(e.g., `QRW$1` requires `*READ-WRITE*`) |
| `U` | excludes `*GLOBAL*` and `*HEAP*` (a.k.a., undeclared)<br>(e.g., `QU$1` requires undeclared - a.k.a., local)<br>(e.g., `QU$1 *GLOBAL*` requires `*GLOBAL*`) |
| `L` | excludes *access-qualifiers*<br>(allows only *lifetime-qualifiers*)<br>(e.g., `QL$1 ro` requires `ro`) |

1. Pointer types can be cascades: e.g., one can have a pointer to a pointer to a pointer. If the cascade has $N$ pointer type names, a *pointer-variable* with this type typically begins with exactly $N$ '`@`'s. It may have fewer '`@`'s, including none, but may not have more. See the example on p15.

   The usual reason for having fewer then $N$ '`@`'s is that the pointer type contains `PST$...` and therefore the total number of *pointer-type-names* in the pointer type

is unknown.

2. A *pointer-variable* beginning with '`@`'s has associated ***target-variables*** made by removing one or more '`@`'s from the beginning of the *pointer-variable*. A *target-variable* names the value pointed at by dereferencing the associated *pointer-variable* value as many times as there are removed '`@`'s. The ***primary-target-variable*** associated with a *pointer-variable* beginning with '`@`'s is the associated *target-variable* made by removing a single '`@`',

The name resolver treats a sequence of names in certain contexts as having the form:

$$\{\ \textit{pointer-type-name}\quad \textit{qualifier-name}^{\star}\ \}^{\star}\ \textit{type-name}\quad \textit{variable-name}$$

where any *pointer-type-name*, *type-name*, or *variable-name* may begin with a *module-abbreviation*. While scanning this sequence from left to right, the name resolver does not back up after identifying a *qualifier-name*, *module-abbreviation*, *pointer-type-name* or *type-name* in the sequence. As a consequence, the following rules should be followed, least there be various confusing syntax or semantic errors (some violations of these rules will be detected as compilation errors, but some will not be):

1. *Basic-names* and *function-term-names* should not begin with a *module-abbreviation*.
2. *Names* should not contain *function-keywords*.
3. *Names* that are not themselves operator names should not begin with *initial-operators*, end with *final-operators*, or contain *middle-operators*.
4. A *type-name* should not begin with a *pointer-type-name*.
5. A *pointer-type-name* should not begin with a *type-name*.
6. *Names* that are not *qualifier-names* should not contain *qualifier-names*, with the exception that a *qualifier-name* by itself can be a *function-term-name*.

*Variable-names*, *type-names*, and *pointer-type-names* that begin with a *module-abbreviation* are called **external**. Non-external names are called **internal**.

Names can abbreviate other names, using the statement:

***abbreviation-statement*** ::= *abbreviating-name* `--->` *abbreviated-name*

For example:

```
bool ---> std bool
```

Note that it is whole names that are abbreviated, and not parts of names.

The `--->` operator executes at compile time. The *abbreviation-statement* must be within the scope[p116] of a definition of the *abbreviated-name*, which must be one of the following kinds:

*pointer-type-name*
*type-name*
*qualifier-name*
*variable-name*
*statement-label*
*member-name*

The *abbreviating-name* will be of the same kind as the *abbreviated-name*, and must follow the syntax rules of that kind. For example, if the *abbreviated-name* is a *variable-name* for a non-pointer variable value, the *abbreviating-name* must be a *non-pointer-variable*, and if the *abbreviated-name* names a variable whose type has $N > 0$ cascaded pointer types, the *abbreviating-name* must be a *pointer-variable* that may begin with up to $N$ '`@`'s, but not more.

Also see *field-abbreviations*[p73] and *subfield-abbreviations*[p73].

Note that *function-term-names* used in *function-calls* cannot be abbreviated.

## 7.2 Constants

A **constant** is a value of type `const` computed at compile-time. One type of constant, the map constant, is not actually constant and can be changed.

There are five of types of constants:

***constant*** ::= *special-constant*
| *string-constant*
| *number-constant*
| *rational-constant*
| *map-constant*

***special-constant*** ::= `TRUE` | `FALSE` | `UNDEF` | `NONE`
| `*LOGICAL-LINE*` | `*INDENTED-PARAGRAPH*`

***string-constant*** ::= *quoted-string*

The meanings of the *special-constants* are:

`TRUE` The boolean value true. Convertible to `bool ( 1 )`.

`FALSE` The boolean value false. Convertible to `bool ( 0 )`.

`UNDEF` The value exists but is undefined (unknown).

`NONE` The value does not exist.

`*LOGICAL-LINE*` see Section 12[p115]

`*INDENTED-PARAGRAPH*` see Section 12[p115]

A special constant is not equal to any other constant. The constant `TRUE` can be implicitly converted to the run-time `bool` value `1`. The constant `FALSE` can be implicitly converted to the run-time `bool` value `0`.

A ***string-constant*** is just a *quoted-string* lexeme that denotes a character string: see p19 and p20.

String constants can be used to load run-time vectors with `uns8`, `uns16`, or `uns32` type elements. UTF-8, UTF-16, or UTF-32 encodings are used according to element size.

A ***number-constant*** is an *natural*, *number*, or *numeric-word* lexeme converted to an IEEE 64-bit floating point number.

A number constant may be converted to a run-time number type such as `int32` or `flt64`. It is a compile error to convert to an integer type that cannot hold the exact value of the number. Conversion to a run-time floating type is however never a compile error. If necessary the converted value is `+Inf` or `-Inf` or loses precision.

*Rational-constants* and *map-constants* are described in the following sections.

### 7.2.1 Rational Constants

A **rational constant** is a rational number with unbounded numerators and denominators, where the denominator is at least 1 and the numerator and denominator have no common factors (other than 1). If the denominator is 1, the rational is called a **rational integer**.

Non-negative rational constants may be computed at compile-time by the prefix operators:

| **Operator** | Argument String |
|---|---|
| `D#` | *decimal-constant-string* |
| `B#` | *binary-constant-string* |
| `X#` | *hexadecimal-constant-string* |
| `C#` | *character-constant-string* |

Each of these operators takes a `const` string as its sole argument. The syntax of the argument strings is:

***sign*** ::= `+` | `-`
***exponent*** ::= { `e` | `E` } *sign*$^{?}$ *dit*$^{+}$
***decimal-constant-string***
::= `"` *decimal-natural* *decimal-fraction*$^{?}$ *exponent*$^{?}$ `"`
| `"` *decimal-natural* `/` *decimal-natural* `"`
***decimal-natural*** ::= *dit*$^{+}$ { `,` *dit dit dit* }$^{\star}$
***decimal-fraction*** ::= `.` { *dit dit dit* `,` }$^{\star}$ *dit*$^{+}$
***dit*** ::= `0` | `1` | `2` | `3` | `4` | `5` | `6` | `7` | `8` | `9`

***binary-constant-string***
::= " *binary-natural* *binary-fraction*? *exponent*? "
| " *binary-natural* / *binary-natural* "

***binary-natural*** ::= *bit*+ { , *bit bit bit bit* }*

***binary-fraction*** ::= . { *bit bit bit bit* , }* *bit*+

***bit*** ::= 0 | 1

***hexadecimal-constant-string***
::= " *hexadecimal-natural* *hexadecimal-fraction*? *exponent*? "
| " *hexadecimal-natural* / *hexadecimal-natural* "

***hexadecimal-natural*** ::= *hit*+ { , *hit hit* }*

***hexadecimal-fraction*** ::= . { *hit hit* , }* *hit*+

***hit*** ::= 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | a | b | c | d | e | f | A | B | C | D | E | F

***character-constant-string*** ::= " *character-representative* "

*character-representative* ::= see p19

where

1. Denominators in fractions must not be zero.

Decimal naturals may have commas every 3 digits from the end and decimal fractions may have commas every 3 digits from the decimal point. Similarly for binary naturals and fractions with commas every 4 binary digits, and with hexa-decimal naturals and fractions with commas every 2 hexa-decimal digits. If there is a decimal point, there must be at least one integer digit and one fraction digit.

For decimal constants without / the denominator is specified by the exponent and is a power of 10; for binary constants, a power of 2; and for hexadecimal constants, a power of 16.

The value of a character constant is the integral UNICODE code point of the *character-representative*.

Negative rational constants can be computed at compile-time by applying the minus operator '-' to a non-negative rational constant.

A rational constant may be converted to a run-time number type such as int32 or flt64. It is a compile error to convert to an integer type that cannot hold the exact value of the rational constant. Conversion of a rational constant to a run-time floating type is however never a compile error. If necessary the converted value is +Inf or -Inf or precision is lost.

D#, B#, and X# are compile-time operators and allow the *constant-string* may be replace by a *constant-expression*$^{p93}$ (which, if it contains an operator, will likely have to be placed in parentheses).

D#, B#, and X# are also run-time operators converting strings to run-time number values (which cannot be rationals): see p158.

### 7.2.2 Map Constants

A **map constant** has two parts, a list (a.k.a, a vector) and a dictionary. Either or both can be empty.

A map constant is computed by an *expression* whose syntax is:

***map-constant*** ::= `{ }`
| `{` *map-list* `}`
| `{` *map-dictionary* `}`
| `{` *map-list*`,` *map-dictionary* `}`
| *phrase-constant*
| *expression-constant*
| *type-constant*
| *pointer-type-constant*

***map-list*** ::= *list-element* { `,` *list-element* }$^\star$

***map-dictionary*** ::= *dictionary-entry* { `,` *dictionary-entry* }$^\star$

***dictionary-entry*** ::= *dictionary-label* `=>` *dictionary-value*

***list-element*** ::= *constant-expression*

***dictionary-label*** ::= *constant-expression* evaluating to a string

***dictionary-value*** ::= *constant-expression*

*constant-expression* ::= a `const` valued *expression* as defined on p93

*expression* ::= see p27

***phrase-constant*** ::= `‘` *expression* `’`

***expression-constant*** ::= `{*` *expression* `*}`

***type-constant*** ::= *type-name*

***pointer-type-constant*** ::= *pointer-type-name*

*type-name* ::= see p36

*pointer-type-name* ::= see p36

Maps cannot be represented at run-time.

By abuse of language, **list** is used to refer to a map whose dominant mode of usage is to go through the map list elements sequentially. Similarly **vector** is used to refer to a map whose dominant mode of usage is to access the map list elements randomly using subscripts. And **dictionary** is used to refer to a map whose dominant mode of usage is to access the map's dictionary elements.

Dictionary entries are also called **attributes**. For lists and vectors, they are also called **annotations**.

Each *map-constant* creates a distinct map: no two such maps are `==`. A map created by a *map-constant* is initially set so that it and all its dictionary entries are read-only. This can

be changed: see p142.

An ***expression-constant*** is shorthand for the *map-constant* produced when the *expression* is parsed: see p115. Generally, parsing an expression groups expression elements into sublists and moves bracket and separator punctuation to annotations (dictionary entries). Some examples are:

| The expression: | Is equivalent to: |
|---|---|
| `{* X = ( Y + 1 ) *}` | `{ { "X" }, "=",`<br>`  { { "Y" }, "+", { 1 }, ".initiator" =>"(",`<br>`  ".terminator" =>")" } }` |
| `{* X, Y = Y, X *}` | `{ { "X", "Y", ".separator" => "," },`<br>`  "=",`<br>`  { "Y", "X", ".separator" => "," } }` |
| `{* X + Y * Z *}` | `{ { "X" }, "+", { { "Y" }, "*", { "Z" } } }` |
| `{* X 3 = Y Z + 1 *}` | `{ { "X", 3 }, "=", { { "Y", "Z" }, "+", { 1 } } }` |

In an **{*** *expression* ****}*** constant, line level operators (those in Figures 2 [p29] and 3 [p30]) are recognized if and only if they are outside parentheses in the *expression*.

***Phrase-constants*** are like *expression-constants* except that operators (including separators, e.g. ‘`,`’) are not recognized. Brackets are recognized and create sublists. Some examples contrasting with *expression-constants* are:

| The Expression: | Is Equivalent To: |
|---|---|
| `‘X = Y + 1’` | `{ "X", "=", "Y", "+", 1 }` |
| `{* X = Y + 1 *}` | `{ { "X" }, "=", { { "Y" }, "+", { 1 } } }` |
| `‘X = ( Y + 1 )’` | `{ "X", "=", { "Y", "+", 1,`<br>`  ".initiator" => "(",`<br>`  ".terminator" => ")" } }` |
| `{* X = ( Y + 1 ) *}` | `{ { "X" }, "=", { { "Y" }, "+", { 1 },`<br>`  ".initiator" => "(",`<br>`  ".terminator" => ")" } }` |

Map constants containing parsed code can be computed by *include-assignment-statements*: see p109.

*Type-names* and *pointer-type-names* can be used at compile-time as if they were variables of type `const` with map values. These map values are partly read-only, with the read-only part including elements with labels like `.size` for the size in bits of run-time values of the type. Users can add their own elements if these do not conflict with the names of the read-only elements. See p145.

## 7.3 Reference Pointers

A reference pointer points at a location in memory where a value is stored. At run-time, it can point into the local or global stack or into the heap. At compile-time, it can point into the local or global stack or it can point at a map element.

Reference pointers are computed by *variable-declarations* and *reference-expressions*. A reference pointer consists of:

**Reference Pointer Components**

- A **pointer part**. At run-time this may be a regular run-time pointer, or may be a similar pointer that points into a local or global stack. At compile-time this may be a pointer into the compiler's local or global stack, or it may be a map plus an index of an element in the map.

  The pointer part may be missing provided the call part is not missing.

- A **call part**. The call part may be missing (usually is) provided the pointer part is not missing. The call part provides a function to load or store values pointed at by the (virtual) pointer.

  The pointer part and the call part cannot both be missing.

  The call part is the declaration of a reference function plus values of any indices used by that function. This reference function must be called with the pointer part (if not missing) and indices to load a value from or store a value into the target of the reference pointer.

  A call part plus pointer part define a kind of virtual location. The reference function called must be either a `load` reference function to load a value or a `store` reference function to store a value. Thus a reference pointer with a call part can be use to either load or store a value, but not both.

## 7.4 Variable Declarations

A *variable-declaration* creates a reference pointer that points into either the local stack or the global stack, and gives it a *variable-name*. If a `const` type variable is declared, the stack is in compile-time memory; otherwise the stack is in run-time memory.

The syntax is:

***variable-declaration*** ::= *initial-variable-declaration*
| *next-variable-declaration*

***initial-variable-declaration*** ::=

*type-name non-pointer-variable*
| *pointer-type-name qualifier-name*$^{\star}$
*pointer-target-type pointer-variable* { `@=` *allocation-call* }$^{?}$

***next-variable-declaration***
::= **`next`** *non-pointer-variable*
| **`next`** *pointer-variable* { `@=` *allocation-call* }$^{?}$

***allocation-call*** ::= *function-call* to an `allocator` function$^{p90}$

***pointer-target-type*** ::= *type-name* | *bit-string-type*
| *pointer-type-name qualifier-name*$^{\star}$ *pointer-target-type*

***bit-string-type*** ::= `[` *bit-string-size* `]` *bit-string-base-type*

***bit-string-size*** ::= `constant-expression` with integer value in range [1,64]

***bit-string-base-type*** ::= *number-type-name* | **`bool`**

***number-type-name*** ::= **`int`** | **`int8`** | **`int16`** | **`int32`** | **`int64`** | **`int128`**
| **`uns`** | **`uns8`** | **`uns16`** | **`uns32`** | **`uns64`** | **`uns128`**
| **`flt`** | **`flt16`** | **`flt32`** | **`flt64`** | **`flt128`**

***isolated-declaration-statement*** ::= *initial-variable-declaration*

*type-name* ::= see p36

*non-pointer-variable* ::= see p36

*pointer-type-name* ::= see p36

*qualifier-name* ::= see p36

*pointer-variable* ::= see p36

*constant-expression* ::= see p93

where

1. As there are no compile-time (non-reference) pointers, there are no compile-time *pointer-variables*. A compile-time *non-pointer-variable* has type (i.e., *type-name*) **`const`**.
2. In a *variable-declaration*, a *module-abbreviation* may appear as part of a *type-name* or *variable-name*. A type or variable name with a *module-abbreviation* names a different type or variable than does the same name without the *module-abbreviation*.
3. An *initial-variable-declaration* of a variable V cannot be within the scope of another *initial-variable-declaration* of V: see Scope$^{p116}$. An *initial-variable-declaration* of a *pointer-variable* also declares all the *target variables*$^{p36}$ of the *pointer-variable*. For example, declaration of `@@p` also declares `@p` and `p`.
4. An *initial-variable-declaration* creates a reference pointer and assigns a variable name to it.
5. The smallest block containing a *next-variable-declaration* '`next V`' must also con-

tain an *initial-variable-declaration* of V. The *initial-variable-declaration* of V must preceed the '`next V`' declaration in this block.

A '`next V`' expression that does not conform to this rule is a *reference-expression* and not a *variable-declaration*.

6. Given a '`next V`' expression that is a *reference-expression*, it must be contained in a smallest block $B$ that has an *initial-variable-declaration* for `V`. Let $B'$ be the immediate subblock of $B$ that contains the *reference-expression* '`next V`'. Then the *next-variable-declaration* '`next V`' should be a *variable-declaration* of $B'$. If it is not, it is automatically added, and thus implied.

   Example:

   ```
   int x = 5
   do:
       next x = x + 1
   ```

   is equivalent to:

   ```
   int x = 5
   next x = do:
       next x = x + 1
   ```

7. A *next-variable-declaration* '`next V`' creates a new reference pointer and assigns it to `V`. At this point `V` has both a new and an old reference pointer; the reference expresssion '`next V`' names the new reference pointer; and the reference expression '`V`' with no '`next`' names the old reference pointer. When the *statement* containing the *next-variable-declaration* completes, the old reference pointer is discarded and the new reference pointer becomes the only reference pointer named by '`V`'.

   Examples:

   ```
   int V = 5      // V names reference pointer RP1.
   next V = V + 5 // V has a new reference pointer RP2 and
                  // the old reference pointer RP1.
                  // V in `V+5' names RP1.
   // Now V names only RP2.
   next V:        // V has a new reference pointer RP3 and
                  // the old reference pointer RP2.
       next V = V + 5   // `next V' names RP3.  V in `V+5'
                        // names RP2.
   // Now V names only RP3.
   ```

8. If an *initial-variable-declaration* has $N > 0$ *pointer-type-names*, the *pointer-variable* it declares may begin with up to $N$ '`@`'s. This variable with exactly one of its '`@`'s removed is the associated *primary-target-variable*[p36], and names the value the *pointer-variable* value points at.

9. For a *pointer-variable* with an *allocation-call*, the *allocation-call* allocates and zeros memory for the value of the associated *primary-target-variable*, and returns a

pointer to the allocated memory that is stored in the *pointer-variable*.

10. A *bit-string-type* describes a number or `bool` value that is stored in a bit string with *bit-string-size* bits. A *type-name* descibes a datum that is a sequence of memory bytes, and is not a bit string.
11. *Bit-string-types* are only for targets of bit string pointers (e.g., `bsp`) and *type-names* are only for targets of byte aligned pointers (e.g., `ap`).
12. An *isolated-declaration-statement* is a statement that consists of nothing but an *initial-variable-declaration*.
13. For an *isolated-declaration-statement* declaring a *pointer-variable* with `*READ-WRITE*` target and with no *allocation-call*, '`@= allocate`' is implied. If the pointer has the `*GLOBAL*` target qualifier, this will allocate to the global stack. If the pointer has the `*HEAP*` target qualifier, this will allocate to the heap. Otherwise this will allocate to the local stack. It is a compile error to allocate to the local stack when the *isolated-declaration-statement* is top level: see Lifetimes [p121].
14. For an *isolated-declaration-statement* '`= *DEFERRED*`' [p66] is implied unless the declaration is of a *pointer-variable* with `*READ-WRITE*` target.

Examples:

```
// Here the `stack' is global if this code is top-level,
// and local otherwise.

int x = 5
    // Allocates an int variable to the run-time stack and creates
    // a reference pointer named x to that variable.  Stores 5 into
    // variable pointed at by x.
int y = x
    // Allocates an int variable to the run-time stack and creates
    // a reference pointer named y to that variable.  Loads the value
    // pointed at by x and stores it into the variable pointed at by y.
next x = 6
    // Allocates an int variable to the run-time stack and creates
    // a reference pointer with name x to that variable (int is the
    // type of x before this statement is compiled).  The reference
    // pointer previously named x no longer has a name and cannot
    // be accessed. Stores 6 into the variable now pointed at by x.
ap *READ-WRITE* flt @z
    // Allocates an `ap *READ-WRITE* flt' pointer variable to the
    // run-time stack and creates a reference pointer named @z to
    // that variable.  As `@= allocate' is implicit, allocates a
    // flt variable to the stack, zeroes that variable's value,
    // and stores a pointer to that variable into the variable
```

```
        // named by @z.  After this, `z' is a reference expression
        // computing a reference pointer whose pointer-part is the
        // value pointed at by @z and whose call part is missing.
    z = x
        // Load from the reference pointer named x a value and then
        // store that value using the reference pointer computed by
        // the reference expression z.  The loaded value is an int
        // and before it is stored it is converted to a flt.
    const d = 23
        // Allocates an const variable to the compiler stack and creates
        // a reference pointer named d to that variable.  Stores 23 into
        // variable pointed at by d.
    next d = d + 1
        // Allocates an const variable to the compiler stack and creates
        // a reference pointer with name d to that variable (const is the
        // type of d before this statement is compiled).  On the right
        // side of the =, d names the old reference pointer named d, and
        // the value pointed at by this, 23, is loaded.  Stores 23 + 1 = 24
        // into the variable newly pointed at by d.
```

Most *variable-names* are declared by *variable-declarations* in *assignment-statements*[p59], but the following other declarations create `co` variables:

| Declaration | Variable Name | Type | |
|---|---|---|---|
| *type-declaration* | *type-name* | `const` | p71 |
| *pointer-type-declaration* | *pointer-type-name* | `const` | p82 |
| *function-type-declaration* | *function-type-name* | `const` | p112 |
| *function-constant-declaration* | *function-constant-name* | *function-type-name* | p112 |

## 7.5 Reference Expressions

A ***reference-expression*** computes a reference pointer. Syntactically *reference-expressions* are simple or compound. A **simple** *reference-expression* is a *variable-name*, '**`next`** *variable-name*', or *function-variable-name*. A **compound** *reference-expression* consists of two parts: a **base** that is a smaller *reference-expression*, and an **offset** that follows the base with a `[` or *member-name* (which begins with '`.`': see p36).

Every *reference-expression* begins with a simple *reference-expression* called the **root** of the *reference-expression*. The root is then followed by zero or more offsets.

The full syntax of a *reference-expression* is:

***reference-expression*** ::= *simple-reference-expression*
| *compound-reference-expression*

***simple-reference-expression*** ::= *variable-name*
| **`next`** *variable-name*
| *function-variable-name* *function-index-list*$^{?}$

***compound-reference-expression*** ::= *reference-base* *reference-offset*

***reference-base*** ::= *reference-expression*

***reference-offset*** ::= *member-name* *offset-index-list*$^{?}$
| *1D-index-list*
| *function-member-name* *function-index-list*$^{?}$

***index-list*** ::= **`[`** *index* { **`,`** *index* }$^{\star}$ **`]`**

***index*** ::= *expression* | **`*`**

*function-index-list* ::= *index-list* with no **`*`** indices

*offset-index-list* ::= *index-list* with at most one **`*`** index

*1D-index-list* ::= **`[`** *expression* **`]`**

*variable-name* ::= see p36

*function-variable-name* ::= see p36

*member-name* ::= see p36

*function-member-name* ::= see p36

1. A primary (*expression* with no operators outside brackets) that begins with a *variable-name*, '**`next`** *variable-name*', or *function-variable-name*, not followed by an explicitly or implicitly parenthesized subexpression, is a *reference-expression*, with the exception that '**`next`** *variable-name*' without any following *reference-offset* can be either a *reference-expression* or a *variable-declaration*.

   Note: If a reference function has argument lists, the first such must be **`[ ]`** bracketed. If a non-reference function has argument lists, the first such must be **`( )`** or implicitly bracketed.

2. An expression of the form '`next V`' for variable name V is a reference expression if and only if the smallest block containing the expression does not contain an *initial-variable-declaration*$^{p44}$ of V.

   Example:

```
int v = 5
next v:                 // `next v' is variable-declaration
    next v = 6          // `next v' is reference-expression
    do:
        next v = v + 2 // `next v' is reference-expression
```

3. A *reference-expression* is used to store a value if and only if it is also an *assignment-result*$^{p60}$ (i.e., is on the left side of an = sign in an *expression-assignment-statement*

or a *call-assignment-statement*). In this case the *reference-expression* is said to be **storing**.

Otherwise it is used to load a value. In this case the *reference-expression* is said to be **loading**. Note that in *compound-reference-expressions* the *reference-base* is always loading.

4. A *function-variable-name* acts like a *variable-name* of a virtual variable. It can, for example, name a pair of *reference-function-declarations*[p102], one of which is used to store a value in the virtual variable, and one of which is used to load a value from the virtual variable.
5. In a *reference-expression*, a *module-abbreviation* may appear as part of a *variable-name* or *function-variable-name*, but may not appear elsewhere. A *variable-name* or *function-variable-name* with a *module-abbreviation* names a different variable or reference function than does the same name without the *module-abbreviation*.
6. An *function-index-list* immediately after a *function-variable-name* or *function-member-name* is either the argument list for the named function, or, if the named function takes no arguments, the *function-index-list* itself must be a separate *1D-index-list reference-offset* independent of the function name *reference-offset*.
7. *1D-index-lists* can be used with `av` and `bsv` reference pointers, and with user defined pointer types that convert to `av` or `bsv` pointers.
8. *Offset-index-lists* can be used with a *field-label* or *subfield-label*, but the total dimensions[p80] of this name must equal the length of the *offset-index-list*. An *offset-index-list* of the form '`[*]`' may be omitted, and will be implied if it is omitted and the total dimensions is 1.

Run-time and compile-time *reference-expressions* are described separately in the following sections.

### 7.5.1 Run-Time Reference Expressions

The pointer part of a run-time reference pointer is just a normal run-time pointer value, except that `ap` and `bsp` pointers that are the pointer part of a reference pointer may point into the global or local stacks.

The **type** of a reference pointer is defined to be the type of the value that is loaded or stored by the reference pointer. More specifically:

**Reference Pointer Type**

- If there is no call part, the type of the reference pointer is the *pointer-target-type* of its pointer type.

- If there is a call part, the type of the reference pointer is the result type of a `load` reference function declared in the call part or the type of a `store` function value input argument declared in the call part.

The **type** of a *reference-expression* is the type of the reference pointer it computes. This is not affected by the within-*statement* context in which the *reference-expression* occurs. Thus the types of the *reference-expressions* in a *statement* can be computed before *function-calls* in the *statement* are matched to *function-declarations*.

Examples of run-time reference pointers are:

```
type my type:
    int X;
    ap my type @Y

my type Z:        // The Z reference pointer is an `ap my type' pointer
                  // to Z in the stack.
    Z.X = 5       // The Z.X reference pointer is an `ap int' pointer
                  // to Z.X in the stack.

ap my type @W:    // The @W reference pointer is an `ap ap my type'
                  // pointer to @W in the stack.
    W.X = 5       // The W reference pointer is a `ap my type' pointer,
                  // the value of @W.
                  // The W.X reference pointer is an `ap int' pointer,
                  // computed from @W and the offset of X in my type.
    W.@Y = @W     // The W.@Y reference pointer is an `ap ap my type'
                  // pointer to the member @Y of W.  It is used
                  // to store the value of @W.
    W.Y.X = 6     // The W.Y reference pointer is an `ap my type'
                  // pointer, the target value of W.@Y.
                  // The W.Y.X reference pointer is an `ap int' pointer,
                  // computed from W.@Y and the offset of X in my type.
                  // It is used to store the value 6.
```

The special member name '`.*`' may be used to dereference a run-time reference pointer. Specifically if '*reference-base*`.*`' is a *reference-expression*, $P$ is the pointer part of the reference pointer computed by the *reference-base*, $C$ is the call part, and $T$ the type, then the reference pointer computed by the entire *reference-expression* has pointer part $P'$, and call part $C'$ where:

- If $T$ is a run-time pointer type, $P'$ =the pointer value loaded by P/C, $C'$ = missing.
- If $T$ is not a run-time pointer type, it is a compile error.

Dropping an '`@`' from the beginning of a pointer variable name is equivalent to appending '`.*`' to the name. Cases taken from the above example are:

```
W.X   is equivalent to  @W.*.X
W.@Y  is equivalent to  @W.*.@Y
W.Y.X is equivalent to  @W.*.@Y.*.X
```

The number of '`@`'s at the beginning of a *pointer-variable* must be equal to or less than the number of *pointer-type-names* in the type of the variable. Given the *pointer-variable* `@@@p` one has:

```
@@p  is equivalent to  @@@p.*
 @p  is equivalent to  @@@p.*.*
  p  is equivalent to  @@@p.*.*.*
```

We proceed to specify how the reference pointer of a run-time *reference-expression* is computed. This computation proceeds partly at compile-time and partly at run-time. Specifically, for a reference pointer the pointer part value is run-time but its type is compile-time, the declaration in any call part is compile time, the number of indices is compile-time, the target types of the indices are compile-time, and the values of the indices are run-time.

We begin with the run-time root that computes a reference pointer with pointer part $P$ and call part $C$. There are the following cases:

1. If the root is a *variable-name* or '**`next`** *variable-name*', the reference pointer is that associated with the *variable-name* by the variable's *variable-declaration*: see notes 4 [p45] and 7 [p46]. Then $P$ is set to an `ap` pointer to the variable in the local or global stack and $C$ is set missing.

2. Otherwise assume the *reference-expression* begins with a *function-variable-name* and search for *reference-function-declarations* [p102] that have the *function-variable-name* and no *reference-base* argument. The declaration may have zero indices, in which case any *index-list* following the *function-variable-name* in the *reference-expression* will be split off as a separate *reference-offset*. Otherwise the declaration must have the same number of indices as the *index-list* following the *function-variable-name* in the *reference-expression*. If the *reference-expression* is storing exclude `load` declarations, and if loading, exclude `store` declarations. See p105 for details.

   (a) If a `load` or `store` declaration is found set $P$ = missing and $C$ to the declaration and any index value arguments. The declaration specifies the target types of any indices.

   (b) If a `point` declaration is found set $P$ = the value returned by a call to the reference function with any index value arguments, and set $C$ = missing.

3. It is a compile-time error if none of the above cases applies.

In a compound run-time *reference-expression*, the *reference-offset* operates on the reference pointer computed by the *reference-base*. This reference pointer must have missing call part and non-missing pointer part $P$. Let $PT$ be the type of $P$. Then the computation has the following cases (the first case that succeeds is used):

1. Search for a *reference-function-declaration*[p102] whose *reference-pattern-base* has type $PT$ and whose *reference-pattern-offset* matches the *reference-offset*. If the *reference-offset* has a *member-name* and an *index-list*, the declaration may have zero indices, in which case the *index-list* will be split off as a separate *reference-offset*. Otherwise the declaration must have the same number of indices as the *reference-offset*. If the *reference-expression* is storing exclude `load` declarations, and if loading, exclude `store` declarations. See p105 for details.

   If a declaration is found:

   (a) If the declaration is of a `load` or `store` reference function, set $P' = P$ and $C'$ to the declaration and index values.

   (b) If the declaration is of a `point` reference function, set $P'$ = the value returned by a call to the reference function with $P$ as reference base argument and the index values. Set $C'$ = missing.

   In either case, the declaration specifies the target types of any indices.

2. If $PT$ is not a builtin pointer type (not `dp`, `ap`, `av`, `bsp`, or `bsv`), search for a *reference-function-declaration* whose *reference-pattern-base* has type $PT$, whose *member-name* is '`..point`', and which has no indices. If a declaration is found, set $P'$ = the value returned by a call to the reference function with $P$ as reference base argument. Set $C'$ = missing. Then go back to case 1 (this current case does not consume a *reference-offset*, but changes $PT$).

3. If $PT$ is a `dp` (or `ap`) pointer type pointing at a user defined type and the *reference-offset* has the form '`.`*field-label*' of '`.`*subfield-label*' possibly followed by an *offset-index-list* whose number of *indices* must equal the total dimensions[p80] of the *field-label* or *subfield-label* (or if the *offset-index-list* would be '`[*]`' it may be omitted and implied):

   (a) For $PT$ a `dp` pointer type, a *field-label* with 0 total dimensions (so there is no *offset-index-list*), $P'$ = a `dp` pointer pointing at the field value.

   (b) For $PT$ a `dp` pointer type, a *field-label* with non-zero total dimensions, and an *offset-index-list* with no `*` index, $P'$ = a `dp` pointer pointing at the field array element value designated by the indices. The target type of the indices is `int`.

   (c) For $PT$ an `ap` pointer type, a *field-label* (or *subfield-label*) with 0 total dimensions (so there is no *offset-index-list*), $P'$ = an `ap` (or `bsp`) pointer pointing at the field (or subfield) value.

(d) For $PT$ an `ap` pointer type, a *field-label* (or *subfield-label*) with non-zero total dimensions, and an *offset-index-list* with no `*` index, $P'$ = an `ap` (or **`bsp`**) pointer pointing at the field (or subfield) array element value designated by the indices. The target type of the indices is `int`.

(e) For $PT$ an `ap` pointer type, a *field-label* (or *subfield-label*) with non-zero total dimensions, and an *offset-index-list* with one `*` index (including the case of an implied '**`[*]`**'), $P'$ = an `av` (or **`bsv`**) pointer pointing at a vector whose $I$'th index element is the the field (or subfield) array element value designated by the indices with `*` replaced by $I$. The target type of the indices is `int`.

In all cases $C'$ = missing.

It is a compile-time error if the user defined type has no field with the *field-label* or *subfield-label* in the *reference-offset*, or if the total dimensions of the field or subfield does not equal the size of the *offset-index-list*.

4. If $PT$ is an `av` (or `bsv`) pointer type and the *reference-offset* is nothing but a *1D-index-list*, $P'$ = an `ap` (or **`bsp`**) pointer pointing at the vector element designated by the index. $C'$ = missing, and the target type of the index is `int`.

5. All other cases are compile errors.

A run-time *reference-expression* of the form '*reference-base*`[`*constant-expression*`]`', where the *constant-expression* evaluates to the `const` string $S$, is equivalent to '*reference-base*`.`$S$'. An *offset-index-list* for `.`$S$ may appear after these expressions.

When a *reference-offset* specifying a field in a user defined type produces a new reference pointer with a pointer part and no call part (as in Case 3), pointing at a field, the access and lifetime qualifiers of the new pointer part are determined according to Figure 6 from the effective qualifiers of user defined type (those in $PT$), which is the container of the field, and the coded qualifiers of the field.

When instead the *reference-offset* specifies a subfield, the effective qualifiers of the new pointer part are determined by Figure 6 from the effective qualifiers of the field containing the subfield (after these have been determined according to the last paragraph), which is the container of the subfield, and the coded qualifiers of the subfield.

Note that if the qualifiers of the container are `co` or `*INIT*`, these remain the qualifiers of the contained field or subfield.

In Cases 1 and 2 above, the qualifiers of the computed reference pointer are determined by the *reference-function-declaration* involved.

### 7.5.2 Compile-Time Reference Expressions

The pointer part of a compile-time reference pointer is one of:

Effective Field or Subfield Access Qualifier
Given Container Access Qualifier
and Coded Field or Subfield Access Qualifier

| Container Access Qualifier | Coded Field or Subfield Access Qualifier | | | | |
|---|---|---|---|---|---|
| | (none) | co | ro | *READ-WRITE* | *WRITE-ONLY* |
| co | co | co | co | co | (illegal) |
| ro | ro | co | ro | ro | (illegal) |
| *READ-WRITE* | *READ-WRITE* | co | ro | *READ-WRITE* | *WRITE-ONLY* |
| *WRITE-ONLY* | *WRITE-ONLY* | (illegal) | (illegal) | *WRITE-ONLY* | *WRITE-ONLY* |
| *INIT* | *INIT* | *INIT* | *INIT* | *INIT* | *INIT* |

1. Here *INIT* is a pseudo-qualifier used for containers that are being initialized. It has the same effect as *WRITE-ONLY* except that it forces fields/subfields of the container to be *INIT* even if they are co or ro.

2. A pointer part pointing at a field or subfield is given the same lifetime qualifier (*GLOBAL*, *HEAP*, or undeclared ) and lifetime depth as the pointer pointing at the field or subfield's container (see 14 [p121]).

Figure 6: Reference Expression Qualifier Computation

- A compiler global stack pointer, pointing at a const variable in the compiler global stack.
- A compiler local stack pointer, pointing at a const variable in the compiler local stack.
- A pair of const values, the first a map, and the second an index that is either an integer or a string.
- A single const value, a map. In this case the call part must exist, and this pointer part is an argument to the call part function.

There are no point compile-time reference functions. All *reference-expression* indices must be *constant-expressions* evaluating to const value. The reference pointer of a compile-time *reference-expression* is completely computed at compile-time.

The compile-time root has the following cases:

1. If the root is a *variable-name* or '**next** *variable-name*', then the pointer part of the root's reference pointer points into the compile-time local or global stack and the call part is missing.

2. Otherwise assume the *reference-expression* begins with a *function-variable-name* and search for *reference-function-declarations* that have the *function-variable-name* and no *reference-base* argument. The declaration may have zero indices, in which case any *index-list* following the *function-variable-name* in the *reference-expression* will be split off as a separate *reference-offset*. Otherwise the declaration must have the same number of indices as the *index-list* following the *function-variable-name* in the *reference-expression*. If the *reference-expression* is storing exclude `load` declarations, and if loading, exclude `store` declarations. See p105 for details.

   Any reference function declaration found must be a `load` or `store` declaration with `const` results and arguments. Set the pointer part of the reference pointer to missing and the call part to the declaration and any index value arguments.

3. Other cases are compile errors. For example, a *function-variable-name* in a storing *reference-expression* that has no `store` *reference-function-declaration*.

A *reference-offset* of a compile-time compound *reference-expression* computes the reference pointer of the *reference-expression* by first reading the target of the reference pointer of the *reference-base*, which must be a map. Then either an index is computed and added to the map to make the pointer part of a new reference pointer with no call part, or a call part is computed and the new reference pointer is given the map by itself as its pointer part.

The computation has the following cases (the first case that succeeds is used):

1. If the *reference-offset* has a *member-name*, search for *reference-function-declarations* with a `const` *reference-base*, and the given *member-name*. The declaration may have zero indices, in which case any *index-list* following the *member-name* in the *reference-expression* will be split off as a separate *reference-offset*. Otherwise the declaration must have the same number of indices as the *index-list* following the *member-name* in the *reference-expression*. For a storing *reference-expression*, exclude `load` declarations, and for a loading *reference-expression*, exclude `store` declarations. See p105 for details.

   If a declaration is found its declaration becomes the call part of the computed reference pointer along with the values of any indices in the *reference-offset*

2. If the *reference-offset* is a *member-name* '`.`*M*', set the pointer part index to the `const` string *M*.

3. If the *reference-offset* is an *index-list* with more than one index, split the *reference-offset* into two *reference-offsets*, one with just the first index and one with the remaining indices. Then proceed as in the next case.

4. If the *reference-offset* is an *index-list* with one index, and the evaluated index *I* is an integer or string `const` value, set the pointer part index to *I*.

5. All other cases are compile errors.

Map `const` values are represented internally by pointers to where the map is stored, so that if `X` is a variable equal to, i.e., pointing at, a map, then `Y = X` copies the pointer to the map to the variable `Y`. By default map constants are read-only and cannot be changed, but it is possible to mark a whole map as read-write, and to independently mark dictionary members as either read-only or read-write. Dictionary members are marked read-only when they are initially created.

The following example illustrates compile-time *reference-expressions*:

```
const U = {"A", "B"}[0]  // Illegal: {"A", "B"} is not a simple
                         // reference expression.
const X = {"A", "B"}
const V = X[0]           // Legal, == "A".
X[0] = "C"               // Illegal, X is read-only.
read-write ( X )
X[0] = "C"               // Now X is {"C", "B"}.
const Y = X              // Now X and Y are both {"C", "B"}.
Y[1] = "D"               // Now X and Y are both {"C", "D"}.
const Z = { Y, "M" }     // Now Z is {{"C", "D"}, "M"}.
Z[0].W = "N"             // Now Z is {{"C", "D", "W" => "N"}, "M"},
                         // X and Y are both {"C", "D", "W" => "N"}.
X.W = "P"                // Illegal, X.W is read-only.
read-write ( X, "W" )
X.W = "P"                // X and Y are both {"C", "D", "W" => "P"}.
                         // Now Z is {{"C", "D", "W" => "P"}, "M"}.
```

## 7.6 Function Calls

The syntax of function calls is:

***function-call*** ::=
*module-abbreviation*$^{?}$ *parenthesized-call-argument-list*$^{\star}$ *call-term*$^{+}$

***call-term*** ::= *call-term-name call-argument-list*$^{\star}$
| **no** *call-term-name*
| **not** *call-term-name*

***call-term-name*** ::= *function-term-name* with quotes optionally removed from *quoted-words*, *quoted-marks*, and *quoted-separators*

*function-term-name* ::= see p36

***call-argument-list*** ::= `(` *actual-argument* { `,` *actual-argument* }$^{\star}$ `)`
| `[` *actual-argument* { `,` *actual-argument* }$^{\star}$ `]`
| `(` `)` | `[` `]`

***parenthesized-call-argument-list*** ::=

*call-argument-list* with parentheses `( )` (and not square brackets `[ ]`)

***actual-argument*** ::= *expression*

*expression* ::= see p27

*module-abbreviation* ::= see p36

1. *Call-term-names* cannot be abbreviated.
2. *Call-terms* of the form '`no x`' and '`not x`' are equivalent to '`x(FALSE)`'.
3. *Parenthesized-call-argument-lists* may have implicit parentheses.
4. There must be at least one explicit or implicit *parenthesized-call-argument-list* just before or just after the first *call-term-name* (to distinguish *function-calls* from *reference-expressions*).

Thus a *function-call* is an optional *module-abbreviation* followed by a sequence of *function-term-names* and *call-argument-lists*. Note that *function-term-names* cannot contain '`.`'s and therefore cannot be *member-names*, which are reserved for *reference-calls*[p102]. Also any `[ ]` bracketed *call-argument-lists* must be preceded by a *parenthesized-call-argument-list*, a syntactic distinction to their use in *reference-calls*.

*Function-calls* are matched to *function-prototypes* in *function-declarations*. The *call-term-names* in a match are identical to the *function-term-names* taken from the prototype being matched, except that quotes (`"`) in a prototype *quoted-word*, *quoted-mark*, or *quoted-separator* may (or may not) be omitted in the *function-call*. The first step in matching is to scan the *function-call* to identify the *call-term-names*. There is no parser backing up after this is done: if the results of this initial scan do not lead to a satisfying match, the entire call-declaration match fails.

## 7.7 Bracketed Expressions

The syntax of a *bracketed-expression* is:

***bracketed-expression*** ::= *ma*? `(` *expression* `)`
| `[` *expression* `]`
| `{` *expression* `}`
| `‘` *expression* `’`
| `{*` *expression* `*}`

*ma* ::= *module-abbreviation* [see p36]

Arithmetic subexpressions and some function argument lists are bracketed with `()` brackets. Reference expression index lists and some function argument lists are bracketed with `[]` brackets. Expressions that compute map constants are bracketed with `{ }`, `‘ ’`, or `{* *}`, brackets (see 7.2.2[p42]).

An expression of the form '*ma* `(` *expression* `)`' is just syntactic sugar for '`(` *ma expression* `)`', except that the *expression* is parsed before the *ma* is moved inside the `()`'s. Thus if `mom` is a *module-abbreviation*, '`mom ( x + y * z )`' is syntactic sugar for

```
( mom x "+" ( y "*" z ) )
```

in which the parenthesis pair surrounding '`y "*" z`' is implied. This allows the *module-abbreviation* to be applied to the outermost operator in the *expression*.

# 8 Assignment Statements

*Assignment-statements* have a list of variables on the left side of an `"="` operator which receive values from a list of expressions or a block of code on the right side of the operator. The left-side variables and the `"="` may be omitted if the right side produces no values, or if all left-side variables have the form of '`next` *variable-name*' and are implied by the right side.

The forms of an *assignment-statement* are:

***assignment-statement*** ::= *expression-assignment-statement*
| *call-assignment-statement*
| *block-assignment-statement*
| *deferred-assignment-statement*
| *loop-assignment-statement*
| *isolated-declaration-statement*

***expression-assignment-statement*** ::=
*assignment-result* { `,` *assignment-result* }$^\star$ `=` *expression-list*

*expression-list* ::= see p62

***call-assignment-statement***
::= *assignment-result* { `,` *assignment-result* }$^\star$ `=` *assignment-call*
| *assignment-call*

***assignment-call*** ::= *function-call* | *call-expression*

*function-call* ::= see p57

*call-expression* ::= see p113

***block-assignment-statement*** ::=
*variable-declaration* { `,` *variable-declaration* }$^\star$
{ `=` { `do` *block-label* }$^?$ }$^?$ `:`
*statement*$^\star$
*exit-subblock*$^\star$
| `do` *block-label*$^?$ `:`
*statement*$^\star$
*exit-subblock*$^\star$

***deferred-assignment-statement*** ::=
*initial-variable-declaration* { **,** *initial-variable-declaration* }$^\star$ **= *DEFERRED***

***loop-assignment-statement*** ::=
*variable-declaration* { **,** *variable-declaration* }$^\star$ **=**
*iteration-control* **:**
*statement*$^\star$
*exit-subblock*$^\star$
| *iteration-control* **:**
*statement*$^\star$
*exit-subblock*$^\star$

***isolated-declaration-statement*** ::= see p45

***exit-subblock*** ::= *exit-label* **exit:**
*statement*$^\star$

*iteration-control* ::= see p67

***block-label*** ::= *statement-label*

***exit-label*** ::= *statement-label*

*statement-label* ::= see p36

***assignment-result*** ::= *variable-declaration*
| *reference-expression*

*variable-declaration* ::= see p44

*reference-expression* ::= see p48

Associated with *block-assignment-statements* and *loop-assignment-statements* there are *control-statements* to control the flow of execution within the more complex *assignment-statement*:

***control-statement*** ::= *block-control-statement* [p65]
| *loop-control-statement* [p67]

A *variable-declaration* allocates memory for its variables in the global or local stack. The sizes of these variables must be known at compile time. For a *pointer-variable*, it is the pointer itself that is allocated to the local or global stack; the memory pointed at can be allocated elsewhere. For sizes not known at compile time, a pointer to the variable can be allocated by an *allocation-call* that is given the variable size at run-time.

Allocations to the local stack are more specifically allocations to the frame of the current out-of-line function within the local stack, or to the portion of the local stack that is allocated to a subblock of a top level statement.

*Expression-assignment-statements* set the values of their result variables to the values of the *expressions* in the *expression-list*. *Call-assignment-statements* set the values of their result variables to the values returned by the *function-call*.

A *variable-declaration* initializes its variables according to the declaration syntax as follows:

Non-`next` *variable-declaration* with no *allocation-call*:
: The memory is zeroed.

`Next` *variable-declaration* with no *allocation-call*:
: Previous value of the named variable.

*variable-declaration* with *allocation-call*:
: Value returned by the *allocation-call* (the allocated memory is zeroed).

Zeroed numbers are zero, while zeroed pointers typically cause segmentation faults when de-referenced.

When a declaration has an *allocation-call*, its variable must have pointer type, and the *allocation-call* is executed to set the pointer before the rest of the *assignment-statement* is executed. The *allocation-call* is executed with a pre-pended argument list consisting of two `uns` values in `()` parentheses. The first value is the number of bytes to be allocated, and the second value is the byte alignment of the memory to be allocated. The *prototype-pattern* of the called function's prototype [p88] must begin with '`( uns length, uns alignment )`', although the argument names may be different. The called function must allocate a block of memory with the required number of bytes and alignment and zero that block. The prototype must have exactly one result variable whose type is identical to the type of the *pointer-variable* being set (but the prototype result type may contain wildcards).

As a general rule, allocator functions that return a value of type `av` or `bsv` or a user defined vector pointer have a `[]` argument list with a single argument giving a vector size `N`. Then the allocator allocates not a single block of the given length and alignment, but instead a vector of `N` such blocks, with zero padding between the blocks if necessary to obtain proper alignment for each block. However, this is by convention and is not a builtin requirement of the L-Language. The convention is followed by the builtin allocators (e.g., `allocate`).

In a *block-assignment-statement*, variables declared by *variable-declarations* without *allocation-calls* are given the qualifier `*INIT*` in *statements* of the *block-assignment-statement* and `co` after the *block-assignment-statement*. The `*INIT*` qualifier is equivalent to `*WRITE-ONLY*` except that it forces fields and elements of the value to also be `*INIT*` [p55].

In a *block-assignment-statement*, *pointer-variables* declared by *variable-declarations* with *allocation-calls* are initialized by the *allocation-call* (see p61) and are thereafter `co`, but the memory pointed at is zeroed initially, `*INIT*` in *statements* of the *block-assignment-statement*, and subject to the *pointer-variable qualifiers* after the *block-assignment-statement*.

A *loop-assignment-statement* is similar to a *block-assignment-statement* except that variables declared by *next-variable-declarations* are reallocated for each iteration of the loop, and the value given by the previous iteration can be read by the current iteration using the *variable-name* (without a preceding '`next`'). Non-`next` variables are allocated only at the start of the *loop-assignment-statement*, and are `*INIT*` (and not readable) throughout all iterations.

Note that a variable declaration does not allow qualifiers on anything but the target of a pointer. The implicit qualifier of a declared variable is `co` after its *assignment-statement* executes meaning that the value of the variable once initially set is never changed. The qualifiers

of a *primary-target-variable* associated with a *pointer-variable* allocated by an *allocation-call* are `*INIT*` in the subblock *statements* of a *block/loop-assignment-statement* and those of the target of its associated pointer after the *block/loop-assignment-statement* executes.

A *next-variable-declaration* for a variable `v` enables '`next v`' to be used as an *assignment-result* within the subblock *statements* of a *block/loop assignment-statement*. Use of `v` within these statements without the '`next`' refers to the value of `v` just before a *block-assignment-statement* or current *loop-assignment-statement* iteration was executed.

Under some circumstances '`next v`' will be implicitly added to the *assignment-result* list of a *call-assignment-statement* (see p64), the *variable-declaration* list of a *block-assignment-statement* (see p46), or the *variable-declaration* list of a *loop-assignment-statement* (see p69).

A *deferred-assignment-statement* behaves like a *block-assignment-statement* with an empty block, but in addition allows each of its variables to be redeclared in a subsequent 'companion' *variable-declaration* which does not reallocate the variable (there may be separate companions for different variables).

For loops, 4 copies of the variables allocated by an iteration are allocated to the local stack, and the loop cycles among these copies. A call to '`allocate`' to a local pointer (i.e., a pointer with no lifetime qualifiers) inside a loop stores a pointer value in one of these 4 copies. If a pointer in one of the copies is already allocated by a previous loop iteration, its memory is reused, or doubled in size and reallocated. This means that the total memory allocated to a pointer inside a loop copy by '`allocate`' will never be more than 4 times the size of the maximum memory needed for any single loop iteration. As there can be 4 copies of the iteration variables, a call to '`allocate`' inside a loop may allocate to the stack at most 4x4 = 16 times as much memory as any single iteration call to '`allocate`'.

## 8.1 Expression Assignment Statements

The syntax of an *expression-assignment-statement* is:

***expression-assignment-statement*** ::=
*assignment-result* { **,** *assignment-result* }$^\star$ **=** *expression-list*

***expression-list*** ::= *expression* { **,** *expression* }$^\star$

*assignment-result* ::= see p60

*expression* ::= see p27

where

1. The number of *expressions* must equal the number of *assignment-results*.
2. The *expression-list* must not consist of a single *expression* which is a *function-call* (else the *statement* is a *call-assignment-statement* as described in 8.2 $^{p63}$). However the *expression-list* may consist of a single *reference-call*.

Sub-*expressions* computable at compile-time are evaluated in bottom-to-top left-to-right order and replaced by their `const` values before the *statement* is compiled into run-time code.[p93] At run-time *expressions* are evaluated in bottom-to-top left-to-right order and then the *expression* values are stored in the *assignment-results*.

Variable names declared by *initial-variable-declarations* that are *assignment-results* of an *expression-assignment-statement* are not visible in the *expressions*. But variable names declared by *next-variable-declarations* (e.g., '`next V`') are visible (e.g., as '`V`') and refer to the variable that exists before the *expression-assignment-statement* is executed.

The type of an *assignment-result* becomes the **target type** of its corresponding *expression*. If the *expression* is a *function-call*, the type of the first result of the *function-prototype* must exactly match the target type of the *expression*, the remaining prototype results are discarded, and the types of the prototype arguments become the target types of the *actual-argument* sub-expressions.

If an *expression* is a *reference-expression* or a *constant-expression* (which is replaced by a constant), it will be implicitly converted to its target type if possible.

An example is:

```
int x = 5              // target type of 5 is int
flt y = 6              // target type of 6 is flt
flt z = 7              // target type of 7 is flt
flt r1 = x + y         // target type of +, x, y is flt
flt r2 = x + y * z     // target type of +, x, *, y, z is flt
next x = z             // illegal; target type of z is int
```

Here the `+` and `*` operator functions are only defined for cases where their operand types are the same as their result type, and `int` variables may be implicitly converted to `flt` but not vice-versa.

## 8.2 Call Assignment Statements

The syntax of a *call-assignment-statement* is:

***call-assignment-statement***
::= *assignment-result* { **,** *assignment-result* }$^\star$ **=** *assignment-call*
| *assignment-call*

***assignment-call*** ::= *function-call* | *call-expression*

*assignment-result* ::= see p60

*function-call* ::= see p57

*call-expression* ::= see p113

A *call-assignment-statement* with *assignment-results* follows the same general rules as *expression-assignment-statements* except that its right side is a single *function-call* or the function-

ally similar *call-expression* and the result values of the call, excluding any results with '**next** *v*' *prototype-next-variable-declarations*[p88], are stored in the *assignment-results* proceeding left-to-right (see item 8, p100). Excess call results are discarded. There must be an least as many call results as there are *assignment-results*.

The types of the function prototype results cannot be implicitly converted to the types of the *assignment-results*.

The right side of a *call-assignment-statement* will have an implied *assignment-result* for every *prototype-next-variable-declaration*[p88]of the *function-prototype*. Every such *prototype-next-variable-declaration* '**next** *w*' must have a corresponding a *prototype-argument-declaration* '**...** *w*', and the call *actual-argument* matched to this must be either a **co** *variable v* for which '**next** *v*' is legal, or a readable and writable *reference-expression e* (see item 7, p89, and also item 13, p101). The implied right side result is then either '**next** *v*' or *e*. An example is:

```
function next w = inc ( int w ):
    next w = w + 1
int v = 5
inc ( v )    // Equivalent to `next v = v + 1'.
ap *READ-WRITE* int @e:
    e = 5
inc ( e )    // Equivalent to `e = e + 1'.
```

## 8.3 Block Assignment Statements

The syntax of *block-assignment-statements* is:

***block-assignment-statement*** ::=
*variable-declaration* { **,** *variable-declaration* }$^\star$
{ **=** { **do** *block-label* }$^?$ }$^?$ **:**
*statement*$^\star$
*exit-subblock*$^\star$
| **do** *block-label*$^?$ **:**
*statement*$^\star$
*exit-subblock*$^\star$

*variable-declaration* ::= see p44

***exit-subblock*** ::= *exit-label* **exit:**
*statement*$^\star$

***exit-label*** ::= *statement-label*

*statement-label* ::= see p36

*statement* ::= see p23

***exception-subblock*** ::= *exit-subblock* whose *exit-label* is ***EXCEPTION***

***block-control-statement*** ::= *goto-exit-statement* | *throw-statement*
*go-to-exit-statement* ::= see p65
*throw-statement* ::= see p65

where

1. The *exit-label* ***EXCEPTION*** is special and any *exception-subblock* must be the last *exit-subblock* in the *block-assignment-statement*.
2. *Next-variable-declarations* may be implied: see note 6 [p46].

The *block-assignment-statement* first allocates and initializes memory in the local or global stack for the variables declared by the *variable-declarations*. [p60]

Then any *statements* and *exit-subblocks* are executed. During this execution block variables not set by an *allocation-call* are `*INIT*` [p55], and after this execution, these variables become `co`. During this execution pointer variables set by an *allocation-call* are `co`, but their target type is changed to `*INIT*`, and after execution the target type qualifiers become whatever the pointer variable declarations specify.

A *go-to-exit-statement* within a block may exit the block or enter an *exit-subblock* of the block:

***go-to-exit-statement*** ::= **`go to`** *go-to-label* **`exit`**
***go-to-label*** ::= *block-label* | *exit-label*

Unless a *go-to-exit-statement* is executed, a block exits after the last *statement* in the block, and an *exit-subblock* exits its containing block after the last *statement* in the *exit-subblock*.

A *go-to-exit-statement* in an *exit-subblock* may only enter a subsequent *exit-subblock* or exit any of its containing *block-assignment-statements* by using that block's *block-label*.

An *exception-subblock* is entered if a *throw-statement* in the *block-assignment-statement* but not in the *exception-subblock* is executed:

***throw-statement*** ::= **`throw`** *throw-ID*?
***throw-ID*** ::= *expression* evaluating to an **`uns`** integer

If a *throw-statement* is executed in an *exception-subblock*, it behaves as if the containing *block-assignment-statement* were replaced by the *throw-statement*. If a *block-assignment-statement* has no *exception-subblock*, it behaves as if it did have an *exception-subblock* whose only statement was a '**`throw`**' statement (without *throw-ID*).

A *throw-statement* with *throw-ID* stores the *throw-ID* and statement program counter in the `std current exception data`: see section 19.7 [p164]. A *throw-statement* without a *throw-ID* leaves the `std current exception data` untouched.

A statement that has a fault, such as a memory fault or an integer divide-by-zero fault, executes as a *throw-statement* with one of the builtin *throw-IDs* of section 19.7 [p164].

*Go-to-exit-statements* define various possible execution paths through a *block-assignment-*

*statement* (these are paths in an acyclic graph). It is a compile error if a *statement* within the *block-assignment-statement* uses a declaration and the statement can be reached by a path that does not contain the declaration. Declarations not in *exit-subblocks* have scope that includes the *exit-subblocks*, but declarations within an *exit-subblock* have scope that ends with the end of the *exit-subblock*. A *function declaration* is 'used' if and only if its prototype matches a *function-call*.[7]

There is an exception to the last paragraph for *exception-subblocks*. In these any `co` variable allocated in the *block-assignment-statement* (but not in one of its *exit-subblocks*) may be read, but the value will be zero or `null` if the variable's declaration has not been executed. The '`is set`' function p164 should be used to test a variable in this case.

## 8.4 Deferred Assignment Statements

The syntax of *deferred-assignment-statements* is:

***deferred-assignment-statement*** ::=
*initial-variable-declaration* { `,` *initial-variable-declaration* }$^{\star}$ `= *DEFERRED*`
*initial-variable-declaration* ::= see p44

Each variable declared by a *initial-variable-declaration* of a *deferred-assignment-statement* must be declared identically, except for addition or subtraction of an *allocation-call*, as a *initial-variable-declaration* of an *assignment-statement* that is within the scope p116 of the *deferred-assignment-statement*. The *assignment-statement*, known as the **companion** of the *initial-variable-declaration*, computes the value of the declared variable, except in the case of a pointer variable whose value is set by an *allocation-call* in the *deferred-assignment-statement*. If a *deferred-assignment-statement* is in a module (11.7 p113), companions of its *initial-variable-declarations* must be in that module or its bodies.

A pointer variable may have an *allocation-call* in either its *deferred-assignment-statement* or in its companion, but not both.

*Deferred-assignment-statements* and their companions must be top-level.

*Deferred-assignment-statement* variable initialization is the same as *block-assignment-statement* variable initialization, except that after the statement the variables are made *ro* and not *co* if they are not set by an *allocation-call* in the *deferred-assignment-statement*. Code that reads such `ro` variables before companions compute their values will read zero. For pointers this will typically reference undefined memory which will cause a memory fault if accessed. For code following a companion the variable qualifier will be changed to `co`. This paragraph applies to pointer variables not allocated by the *deferred-assignment-statement*.

[7] Stack space for every stack variable that might be used by an out-of-line function is allocated when the out-of-line function is called. Space allocated to the local stack by the `allocate` function is treated differently: see p133.

For pointers allocated by a *deferred-assignment-statement*, the pointer will be made `co`, the pointer target value will be zeroed, and the pointer target will be given the `ro` qualifier. For code following a companion the pointer target qualifiers will be changed to the qualifiers of the pointer type.

The variables declared in a *deferred-assignment-statement* are treated as normal *variable-declaration* variables inside their companions. In particular, inside their companions the `ro` variables are `*INIT*` [p55], and the `co` pointer variables set by an *allocation-call* in their *deferred-assignment-statement* have their target type changed to `*INIT*` inside their companions.

## 8.5 Loop Assignment Statements

A *loop-assignment-statement* has the syntax:

***loop-assignment-statement*** ::=
*variable-declaration* { **,** *variable-declaration* }$^\star$ **=**
*iteration-control* **:**
*statement*$^\star$
*exit-subblock*$^\star$
| *iteration-control* **:**
*statement*$^\star$
*exit-subblock*$^\star$

***iteration-control*** { **do** | **repeat** } *loop-label*$^?$ *iteration-item*$^+$
| { **do** | **repeat** } *int-expression* **times**
| **repeat** *loop-label*$^?$

***iteration-item*** ::= **exactly** *int-expression* **times**
| **at most** *int-expression* **times**
| **while** *bool-expression*
| **until** *bool-expression*

***loop-label*** ::= *statement-label*

*statement-label* ::= see p36

***int-expression*** ::= *expression* evaluating to an `int`

***bool-expression*** ::= *expression* evaluating to a `bool`

***loop-control-statement*** ::= *break-statement* | *continue-statement*

***break-statement*** ::= **break** *loop-label*$^?$

***continue-statement*** ::= **continue** *loop-label*$^?$

where

1. The *exit-label* ***EXCEPTION*** is special and any *exception-subblock* must be the last *exit-subblock* in the *loop-assignment-statement*.
2. *Next-variable-declarations* may be implied: see note 6 [p46].

A *loop-assignment-statement* that has no *initial-variable-declarations* (as opposed to *next-variable-declarations*) is the semantic equivalent of a sequence of zero or more copies of the statement with its *iteration-control* deleted, making these copies into *block-assignment-statements*. Each copy is called an **iteration** of the *loop-assignment-statement*. The number of iterations is determined at run-time by the *iteration-control* and *loop-control-statements*.

A simple example is:

```
int sum = 0
int i = 1
next sum, next i = while i < 4:
    next sum = sum + i
    next i = i + 1
```

which is semantically equivalent to:

```
int sum = 0
int i = 1
next sum, next i =:
    next sum = sum + i
    next i = i + 1
next sum, next i =:
    next sum = sum + i
    next i = i + 1
next sum, next i =:
    next sum = sum + i
    next i = i + 1
// Now sum == 6 and i == 4
```

However at run-time the variable values of all but the last 4 iterations of the *loop-assignment-statement* are discarded, which would not be the case if the compiler actually inserted iterations in the source code. This only affects debugging.

If there are *initial-variable-declarations* (i.e., without '**next**'), these are treated as if they had been moved to an immediately enclosing *block-assignment-statement*. For example:

```
int sum = 0
int i = 1
int result, next sum, next i = loop:
    next sum = sum + i
    next i = i + 1
    if i + 1 >= 4:
        result = sum
```

```
            break // terminate loop
```

is semantically equivalent to:

```
int sum = 0
int i = 1
int result:
    next sum, next i = loop:
        next sum = sum + i
        next i = i + 1
        if i + 1 >= 4:
            result = sum
            break // terminate loop
```

Another way of putting this is that the variables of *next-variable-declarations* get allocated anew each iteration, but the variables of *initial-variable-declarations* are only allocated once before the first iteration. As they have the `*INIT*` qualifier throughout the *loop-assignment-statement*, they can only be written inside that *statement*.

The *iteration-items* are independent of each other: `exactly x times` and `at most x times` both terminate the loop after `x` iterations (both do the same thing), `while x` terminates the loop if `x` is `false` at the start of an iteration, and `until x` terminates the loop if `x` is `true` at the start of an iteration.

The *break-statement* exits the current iteration of the *loop-assignment-statement* and prevents further iterations. A *continue-statement* exits the current iteration of the *loop-assignment-statement* but lets the *iteration-control* determine whether there will be any more iterations. If there are nested loops, a *loop-label* may be used with these statements to designate which nested iteration is being exited or continued.

As in *block-assignment-statements*, if '`next V`' occurs as a *reference-expression* within the loop *statements*, '`next V`' will be added to the *next-variable-declaration* list of the *loop-assignment-statement*. Therefore the above example could be written as:

```
int sum = 0
int i = 1
next sum = while i < 4:
    next sum = sum + i
    next i = i + 1
```

or

```
int sum = 0
int i = 1
while i < 4:
    next sum = sum + i
    next i = i + 1
```

*Go-to-exit-statements*, *throw-statements*, and *exception-subblocks* are treated within a loop iteration just as they are for *block-assignment-statements*. A *throw-statement* in an *exception-subblock* behaves as if the *loop-assignment-statement* were replaced by the *throw-statement*.

# 9 Conditional Statements

A *conditional-statement* executes another *statement* or block of *statements* according to what a *bool-expression* evaluates to. *Conditional-statements* have the syntax:

***conditional-statement*** ::=
`if` *bool-expression* `:`
*statement*$^\star$
| `else if` *bool-expression* `:`
*statement*$^\star$
| `else :`
*statement*$^\star$
| `if` *bool-expression* `:` *restricted-statement*
| `else if` *bool-expression* `:` *restricted-statement*
| `else :` *restricted-statement*

*bool-expression* ::= see p67

*restricted-statement* ::=
*expression* with no operator of precedence equal to or less than 0
(the precedence of the '`:`' operator)

where

1. An '`else if`' or '`else`' *statement* must be immediately preceded by an '`if`' or '`else if`' *statement*.

An example is:

```
int x = 5
int y = 6
int z:
    int sum = x + y          // Sets sum = 11
    int product = x * y      // Sets product = 30
    if sum < product:
        z = sum              // Sets z = 11
    else: z = product        // Is NOT executed
// Now z = 11
```

# 10 Trace Statements

A *trace-statement* optionally writes an information record into a trace file:

***trace-statement*** ::= `trace` *trace-ID* `,` *trace-description* { `,` *trace-value* }$^{\star}$
***trace-ID*** ::= *expression* with an `uns` integer value in the range [0,63]
***trace-description*** ::= *expression* with a `const` string value
***trace-value*** ::= *reference-expression* with a value of any run time type

A *trace-statement* with *trace-ID* X is executed if bit `1 << X` of the `std current trace mask` [p164] is on. Otherwise the *trace-statement* is a no-operation.

When a *trace-statement* is executed it outputs a trace record into a trace file (not specified by this document) that contains the program counter (as a `dp uns8 @pc` value) followed by the *trace-values*. The program counter, when used with debugging information files output by the compiler and loader (not specified by this document), identifies the particular *trace-statement*, its *trace-description*, and the data types of the *trace-values*. This information allows the trace record to be printed in a human readable form (in a manner not specified by this document).

# 11 Declarations

The following is a complete list of declarations:

***declaration*** ::= *variable-declaration* [p44]
| *prototype-result-declaration* [p88]
| *prototype-argument-declaration* [p88]
| *declaration-statement*

***declaration-statement*** ::= *type-declaration* [p71]
| *field-declaration* [p72]
| *subfield-declaration* [p73]
| *pointer-type-declaration* [p82]
| *function-declaration* [p87]
| *reference-function-declaration* [p102]
| *out-of-line-function-declaration* [p111]
| *function-type-declaration* [p112]
| *function-constant-declaration* [p112]
| *module-declaration* [p113]
| *body-declaration* [p114]

## 11.1 Type Declarations

The syntax of a type declaration is:

***type-declaration*** ::= `type` *defined-type-name* `:`
*type-subdeclaration*$^\star$
| `type` *defined-type-name* `: *DEFERRED*`
***defined-type-name*** ::= *type-name*
*type-name* ::= see p36

***type-subdeclaration*** ::=
::= *field-declaration*
| *field-abbreviation*
| `align` *alignment*$^?$
| `pack`
| `*EXTERNAL*`
| `*PRIVATE*`
| `*INCLUDE*` *defined-type-name*
| `*LABEL*` *origin-label*
| `*OFFSET*` *offset-specification*
| `***`

***alignment*** ::= *constant-expression* with power of 2 integer value
***origin-label*** ::= *basic-name* not containing `@`
***basic-name*** ::= see p36

***offset-specification*** ::=
| *origin-label* { {`+`|`-`} *offset-increment* }$^?$
| {`+`|`-`} *offset-increment*
| *offset*
| `*SIZE*`
*offset-increment* ::= integer valued *constant-expression*
*offset* ::= non-negative integer valued *constant-expression*
*constant-expression* ::= a `const` valued expression as defined on p93

***field-declaration*** ::= *field-without-subfields-declaration*
| *field-with-subfields-declaration*

***field-without-subfields-declaration*** ::=
*qualifier-name*$^\star$ *field-type-name* *non-pointer-label* *dimensions*$^?$
| *qualifier-name*$^\star$ *pointer-type-name qualifier-name*$^\star$
*field-pointer-target-type* *pointer-label* *dimensions*$^?$

***field-pointer-target-type*** ::=
*field-type-name*
| *pointer-type-name qualifier-name*$^\star$ *field-pointer-target-type*

***field-type-name*** ::= *type-name* that is not `const`

*type-name* ::= see p36
*pointer-type-name* ::= see p36
*qualifier-name* ::= see p36

***field-label*** ::= *data-label*
***pointer-label*** ::= *field-label* beginning with zero or more '`.`'s
followed by zero or more '`@`'s [see text below]
***non-pointer-label*** ::= *field-label* that has no '`@`'s after any beginning '`.`'s
***target-label*** ::= *field-label* [see text below]
***data-label*** ::= see p36

***dimensions*** ::= `[` *dimension-size* { `,` *dimension-size* }$^\star$ `]`
***dimension-size*** ::= *constant-expression* with strictly positive integer value

***field-with-subfields-declaration*** ::=
*qualifier-name*$^\star$ `std`$^?$ *number-type-name* *non-pointer-label*$^?$ *dimensions*$^?$ `:`
*subfield-declaration*$^+$
*number-type-name* ::= see p45

***subfield-declaration*** ::=
*bit-string-type subfield-label dimensions*$^?$
| *subfield-abbreviation*
| `*EXTERNAL*`
| `*PRIVATE*`
| `*LABEL*` *origin-label*
| `*OFFSET*` *offset-specification*
*bit-string-type* ::= see p45
***subfield-label*** ::= *non-pointer-label*

***field-abbreviation*** ::= *abbreviating-field-label* `--->` *abbreviated-field-label*
***abbreviating-field-label*** ::= *field-label*
***abbreviated-field-label*** ::= *field-label*
***subfield-abbreviation*** ::=
*abbreviating-subfield-label* `--->` *abbreviated-subfield-label*
***abbreviating-subfield-label*** ::= *subfield-label*
***abbreviated-subfield-label*** ::= *subfield-label*

where

1. If a *field-without-subfields-declaration* has $N > 0$ *pointer-type-names*, the *pointer-label* it declares typically has exactly $N$ '`@`'s after any initial '`.`'s. It may have fewer '`@`'s, including none, but may not have more.
2. A *pointer-label* has associated ***target-labels*** made by removing one or more '`@`'s from the beginning of the *pointer-label* (after any '`.`'s). A *target-label* names the

value pointed at by dereferencing the associated *pointer-label* field value as many times as there are removed '`@`'s.

3. The labels of fields and subfields in one *type-declaration* must all be distinct. For example, a *type-declaration* may not have distince fields respectively labelled `@p` and `p`, and may not have a field and a subfield both labeled `x`.
4. A *field-abbreviation* may be used to provide an alternative label for a field, and a *subfield-abbreviation* may be used to provide an alternative label for a subfield. These abbreviations can only be used when labelling fields or subfields of the type in whose *type-declaration* they appear.

Values of user defined types are byte strings. Fields of a user defined type are byte substrings. The number of bytes in a value that are before a field is the **byte offset** of the field. *Type-subdeclarations* are used to define fields and the the offsets of these fields.

A number-valued field may be treated as a bit string into which bit substrings called subfields may be allocated. The order of these bits is from the high order end of the number to the low order end. The number of bits in the field that are before a subfield is the **bit offset** of the subfield. *Subfield-declarations* are used to define subfields and the offsets of these fields.

The *type-subdeclarations* of a *type-declaration* are processed in order. At the beginning of each, there is a align/pack switch value, an byte offset value (the '**current offset**'), and an external/private switch value.

A *field-declaration* defines a field of its containing user defined type. If the align switch is '`pack`', the offset of the field is the current offset. If the align switch is '`align`', the current offset is first incremented until it is an integer multiple of the alignment of the new field, and then the current offset is used as the byte offset of the new field.

The alignment of a field is the alignment of its values, as determined by the type of the field, as follows:

1. The alignment of a number is the length of the number. E.g., the alignment of a 4 byte integer is 4.
2. The alignment of a user defined type is the least common multiple of the alignment of all its fields. As alignments are always powers or 2, this is the same as the maximum alignment of any of its fields.

   E.g., the alignment of builtin pointers (see 5.2 $^{p25}$) is the length of a `ptr` integer (see 5.1 $^{p24}$).

The current offset is set by a *field-declaration* to be the sum of the byte offset of the field and the length in bytes of the field.

An '**`align`** $N$' sub-declaration behaves like an unnamed field of alignment $N$ bytes and zero length, and in addition sets the align/pack switch to 'align'. Here $N$ must be a power of 2. An '**`align`**' sub-declaration just sets the align/pack switch to 'align'.

A **pack** sub-declaration sets the align/pack switch to 'pack'.

A ***LABEL*** sub-declaration assigns the current offset to the given *origin-label* and provides the *origin-label* for use by subsequent *OFFSET* sub-declarations. Each *origin-label* may be defined only once for a given defined type.

An ***OFFSET*** sub-declaration changes the current offset to the value of a given *origin-label*, or to that value plus or minus an integer, or to the current value plus or minus an integer, or to a given non-negative integer.

'***OFFSET* *SIZE***' sets the current offset to the maximum current offset computed for the defined type so far.

The processing of *subfield-declarations* within a *field-with-subfields-declaration* is similar to the processing of *field-declarations* within a *type-declaration* except that offsets are in bits and not bytes, and the align/pack switch is ignored. The current offset is saved and reset to 0 before the first *subfield-declaration* processed, and is restored after the last *subfield-declaration* is processed.

A *field-label* or *subfield-label* will be either external or private according to the setting of the external/private switch when the *type-subdeclaration* or *subfield-declaration* defining the label is executes. If the defining declaration is in a body (modules have bodies, and bodies cannot be imported by other modules or bodies: see 11.7 [p113]), the label must be private. If the defining declaration is in a module (which can be imported by other modules or their bodies), the label may be external, in which case it is visible throughout the scope [p116] of its declaration, even where this scope includes other modules or their bodies. A private label, on the other hand, is not visible outside the module in which it is defined and the bodies of that module.

The external/private switch is set to private before the first *type-subdeclaration* of a *type-declaration*. It is set to external by the *EXTERNAL* *type-subdeclaration*, and to private by the *PRIVATE* *type-subdeclaration*.

The external/private switch may only be set to external in a *type-declaration* of an external *defined-type-name* (this name must begin with a *module-abbreviation*).

Before the first *subfield-declaration* of a *field-with-subfields-declaration* the external/private switch is saved, and after the last *subfield-declaration* it is restored. It may be set by *EXTERNAL* and *PRIVATE* *subfield-declarations*, but is not otherwise changed at the beginning or during the *field-with-subfields-declaration*.

There may be several *type-declarations* for the same defined type. One of these is the **root**, or first, of the set of these *type-declarations*, and all the others are called **expansions**, and are in the scope [p116] of the root. At the beginning of each *type-declaration* the align/pack switch is set to align, and the external/private switch is set to private. At the beginning of the root the current offset is set to zero, but at the beginning of all others the current offset is undefined, and must be defined by an *OFFSET* sub-declaration before it is used.

If a *type-declaration* does not end with a ******* sub-declaration, expansions of the *type-declaration* are forbidden within the scope of the *type-declaration*. If this *type-declaration* is in the same module or body as the root of its defined type, then the final size of the defined type's values, and the fields and subfields of the defined type, are fixed and known within the scope of this *type-declaration*. It is common for the root to not have a *** sub-declaration.

If a *type-declaration* ends with a ******* sub-declaration, expansions of the *type-declaration* are permitted within its scope. If the root and all expansions of the root within the module or body containing the root have a *** sub-declaration, then the size of the defined type's values and the complete set of fields and subfields of the defined type are not known at any point during compilation. The value size is computed at load time, after all modules and bodies containing the defined type's root and its expansions have been loaded, and is implicitly used by *allocate-calls* that allocate values of the defined type.

The ***INCLUDE*** sub-declaration copies all the sub-declarations from another user defined type. The text of the sub-declarations is copied, not the effects which may vary according to context. If the user defined type is defined by multiple *type-declarations*, only sub-declarations visible in the current scope [p116] are copied.

A ***DEFERRED*** *type-declaration* declares a *defined-type-name* without declaring the definition of the named type. It is the equivalent of a *type-declaration* whose only sub-declaration is ***. Such a *type-declaration* is typically used to allow the *defined-type-name* to be used as the target type of a pointer before its fields are declared.

The *defined-type-name* in a *type-declaration* is declared before the *type-declaration*'s sub-declarations are processed, so these sub-declarations may use the *defined-type-name* as a pointer target type.

### 11.1.1 Type Fields

A field of a value of a user defined type is accessed by prepending '**.**' to the *field-label* to form a *member-name* in a *reference-expression* (see p53). An example is:

```
type my type:
    uns8    kind         // Object Kind
    flt     weight

my type X:
    // Within this block X is *INIT*.
    X.kind = HIPPOPOTAMUS
    X.weight = 152.34
uns8 kind = X.kind
flt weight = X.weight
```

If a *field-label* is a *pointer-label*, associated ***target-labels*** are declared consisting of the

*pointer-label* minus some of its first '`@`'s. A *target-label* references a virtual field consisting of the target value stored at the location pointed at by the *pointer-label* field's pointer value. Note that as '`next`' is only recognized as a keyword at the beginning of a *next-variable-declaration*, it can be used as a *field-label*.

An example is:

```
type list element:
    int value
    ap list element @next

// Make circular list.
//
ap list element @Y      // `@= allocate = *DEFERRED*' is implied
ap list element @X:     // `@= allocate' is implied
    X.value = 1
    X.@next = @Y
ap list element @Y:
    Y.value = 2
    Y.@next = @X
//
// Now X.value == 1 and X.next.value == 2 and
// similarly Y.value == 2 and Y.next.value == 1,
// while X.@next == @Y and Y.@next == @X are
// pointers to local memory.
```

A field of the defined type can only be accessed by code in the scope of a *type-declaration* declaring the field.

An example is:

```
type my type: *DEFERRED*
ap *READ-WRITE* my type @X    // `@= allocate' is implied
                    // Legal, my type members need not be declared.
                    // Size and alignment of my type values is
                    // computed at load time.  The allocated value
                    // will be zeroed.
ap ro my type @Y = @X  // Legal, only ap copied.
ap my type @Z:      // Legal, `@= allocate' is implied
    Z = X           // Legal, the value at @X is copied to
                    // the value at @Z.  However in this
                    // case the value is completely zero.

type my type:       // Definition of my type that was *DEFERRED*.
    *LABEL* origin  // `origin' is set to offset 0
```

```
    int32 I          // Offset of I is 0.
    ***              // Allows expansion.

X.I = 55             // Legal, .I has been declared.  X is
                     // *READ-WRITE*.

type my type:        // Expansion of my type
    *OFFSET* *SIZE* // offset set to 4.
    int32 J          // Now X.J == 0
    ***              // Allows expansion.

X.J = 66             // Legal, .J has been declared.

type my type:        // Expansion of my type
    *OFFSET* origin // Offset set to 0.
    int32 K1         // Overlays I at offset 0.
    int32 K2         // Overlays J at offset 4.
        // No ***, so this expansion is a linear leaf.

// Now X.K1 == X.I == 55; X.K2 == X.J == 66; but you must know
// how offsets are assigned to believe this.
```

A value of `bool` type may be stored as a 1-byte field value or a 1-byte pointer target value. When this is done, the value is encoded as an unsigned 8-bit integer equal to 1 for `true` and 0 for `false`.

### 11.1.2 Type Subfields

Subfields are parts of the previously declared number type field. The bits occupied by subfields are allocated from high order to low order, with the bits being numbered 0, 1, ... from the high order end of numbers. The bit number of the next subfield can be reset by the `*OFFSET*` subdeclaration. Subfield types must themselves be number types or `bool`.

A subfield value may have fewer bits than values of the number-type of the subfield. For integer types, the value is the low order bits of the integer, with the high order bits added when the value is read by adding 0 bits for unsigned integers or copies of the highest order bit for signed integers. For floating types, the value is missing low order mantissa bits, which are added as zeros. If a value outside the representable range is stored, it is not a detected error unless the value is a `const` value (in which case it is a compile-time error). Unsigned and two's complement integer values are truncated, and IEEE floating values have low order mantissa bits dropped (there is no rounding). However, it is a compile error to have a floating type whose sign and exponent parts plus 1 mantissa bit cannot be stored in the subfield value.

*Subfield-labels* and *field-labels* have the same standing within *reference-expressions*. Both have associated *member-names* made by adding a single '`.`' to the beginning of the *field-label* or *subfield-label*. For example:

```
type my type:
    uns8     kind:       // Object Kind
        [1] bool animal      // True if Animal (bit 0 of kind)
        [1] bool vegetable   // True if Vegetable (bit 1 of kind)
    flt      weight

my type X:
    // Within this block X is *INIT*.
    X.kind = HIPPOPOTAMUS
        // Also sets animal bit and clears vegetable bit.
    X.weight = 152.34
uns8 kind = X.kind
bool animal = X.animal         // == true
bool vegetable = X.vegetable   // == false
flt weight = X.weight
```

A value of `bool` type may be stored as a 1-bit subfield value or a 1-bit bit string pointer target value. When this is done, the value is encoded as 1 for `true` and 0 for `false`.

### 11.1.3 Type Dimensions

If a *field-declaration* with *field-label* `F` contains a single *dimension-size* `[`$n$`]` then $n$ fields are allocated to ascending offsets. If the align/pack switch is set to '`align`', zero padding is used as necessary before each of the $n$ fields to align them. The labels of these fields are `F[`$i$`]` for $0 \leq i < n$, where $i$ is the index. If there is a subfield labeled `S` of the field, `S[`$i$`]` refers to the subfield in `F[`$i$`]`. For example:

```
type character attributes:
    uns8 [128]:
        [0] bool is graphic

character attributes X:
    int i = 0
    while i < 128:
        X.is graphic[i] = 32 < i && i < 127
        next i = i + 1
bool line feed is graphic = X.is graphic [C#"<LF>"]  // == false
bool A is graphic = X.is graphic [C#"A"]             // == true
```

A *field-declaration* with two *dimension-sizes* `[`$n1$`, `$n2$`]` allocates $n1$*$n2$ fields and `F[`$i1$`, `$i2$`]`

references the $i1$*$n2$+$i2$+1'nth of these.[8] Similarly `[`$n1$`, `$n2$`, `$n3$`]` allocates $n1$*$n2$*$n3$ fields with `F[`$i1$`, `$i2$`, `$i3$`]` referencing the $i1$*$n2$*$n3$*+$i2$*$n2$+$i3$+1'nth. And so forth for any number of *field-dimensions*.

If a *subfield-declaration* with *subfield-label* `S` contains a single *dimension-size* `[`$n$`]` then $n$ subfields are allocated to the containing field in order of increasing bit numbers. There is no padding. For example:

```
type hex digits:
    uns32 container:
        [4] uns hex digit [8]
        // hex digit[i] is stored in bits 4*i through 4*i+3

hex digits X:
    int i = 0
    while i < 8:
        X.hex digit[i] = i
        next i = i + 1
// Now X.container == X#"01234567"
```

A *subfield-declaration* with *subfield-label* S and more than one *dimension-size* is treated in the same manner as a *field-declaration* with more than one *dimension-size*. For example, `[`$n1$`, `$n2$`, `$n3$`]` allocates $n1$*$n2$*$n3$ subfields with `S[`$i1$`, `$i2$`, `$i3$`]` referencing the $i1$*$n2$*$n3$*+ $i2$*$n2$+$i3$+1'nth of these.

If a *field-declaration* with *field-label* `F` has one *dimension-size* and also a subfield with *subfield-label* `S` and no *dimension-sizes*, then `S[`$i$`]` references the subfield in the field value `F[`$i$`]`. If in addition the subfield has one *dimension-size*, `S[`$i$`, `$j$`]` references the subfield selected by `[`$j$`]` in the field value `F[`$i$`]`. In general, if the *field-declaration* has $f$ *dimension-sizes* and the *subfield-declaration* has $s$ dimension sizes, then $S$ subfields are referenced using $f + s$ indices, with the field indices first. In this caseand $f + s$ is called the **total dimensions** of the subfield and $f$ is also called the total dimensions of the field.

### 11.1.4 Type Conversions

When the compiler is confronted with code such as:

```
T1 v1 = ...
T2 v2 = v1
```

where `T1` and `T2` are different types, the compiler just rewrites the code to:

```
T1 v1 = ...
T2 v2 = implicit-conversions ( v1 )
```

[8] Thus arrays are stored in row major order: last subscript varies fastest, as in C/C++.

where *implicit-conversions* denotes one or more applications of `*IMPLICIT* *CONVERSION*` functions. If you define a function with the prototype

```
function T2 r = ma? *IMPLICIT* *CONVERSION* ( T1 v )
```

the compiler will used this function. Otherwise the compiler will try to chain implicit conversions together to get to a successful compile. Of course such chaining will only work if there is at least one `*IMPLICIT* *CONVERSION*` function with target type `T2`. If there is more than one such function, and none have argument type `T1`, ambiguity may lead to a compile error.

You can define `*IMPLICIT* *CONVERSION*` functions provided at least one of the two types `T1` and `T2` is user defined, and not builtin, and `T2` is not `const`. There are builtin `*IMPLICIT* *CONVERSION*` functions in which both types are builtin: see 19.2 p155.

The set of `*IMPLICIT* *CONVERSION*` functions defines a graph, the **implicit conversion graph** in which types are nodes and `*IMPLICIT* *CONVERSION*` functions are directed edges. A *function-declaration* $d$ of the form

```
function T2 r = ma? *IMPLICIT* *CONVERSION* ( T1 v )
```

creates an edge labeled $d$ in this graph from node `T2` to node `T1`. This graph must be acyclic. A path $d_1, d_2, \ldots, d_n$ in this graph denotes a chain of *IMPLICIT* *CONVERSION* *function-declarations* that together convert the result type of $d_1$ to the argument type of $d_n$.

When implicitly converting a value of type `T1` to a value of type `T2`, the compiler will choose a shortest path in the implicit conversion graph.

When defining an implicit conversion from type `T1` to type `T2`, each value of type `T1` should be exactly representable by a value of type `T2`. This rule should be followed, but is not checked by the compiler. This rule is also violated by the builtin implicit conversions for any integer type to any floating point type. p155

It is important that a shortest path in the implicit conversion graph be chosen by the compiler, so that, for example, `int32 --> flt64` will be chosen instead of `int32 --> flt32 --> flt64`, as the latter would lose precision.

And there is one notable exception: conversions from `const` which are done at compile-time may fail (the failure will be at compile-time and be a compiler error). For example, conversions from `const` to any integer type will fail if the `const` value cannot be encoded exactly in the integer type.

For types `T1` and `T2` you can also define an explicit conversion:

```
function T2 r = ma? T2 ( T1 v )
```

If the result may not properly represent the value `v`, you may wish to define instead an unchecked conversion:

```
function T2 r = ma? *UNCHECKED* ( T1 v )
```

The compiler will not allow you to define such functions if both `T1` and `T2` are builtin or if

`T2` is `const`, but some such functions are builtin: see 19.3 p157.

The types `T1` and `T2` may also be pointer types: see p86. Or one may be a pointer-type and one a non-pointer type. However, `T2` cannot be `const`, and you cannot define your own `*IMPLICIT* *CONVERSION*` function if both types are builtin.

## 11.2 Pointer Type Declarations

A pointer type has an **associated data type** specified by a pointer type declaration. The syntax is:

***pointer-type-declaration*** ::=
    **`pointer type`** *defined-pointer-type-name* **`is type`** *type-name*

*pointer-type-name* ::= see p36

*type-name* ::= see p36

An `*UNCHECKED*` conversion is implicitly defined from the associated data type to the given pointer type, and an explicit conversion is implicitly defined in the other direction.

It is important that there be a 1-1 correspondence between pointer types and their associated data types. In particular, `int` must not be used as the associated data type of more than one pointer type. This is why associated data types are generally user defined types.

A *pointer-type-declaration* '`pointer type` *P* `is type` *D*' implicitly declares the functions:

```
function P Q$1 PST$1 @p = ma? *UNCHECKED* ( D d )
function D d = D ( P Q$1 PST$1 @p )
```

where *ma*? denotes the *module-abbreviation* of *P*, if any. These just copy values changing type.

For example, the following are builtin:

```
pointer type std dp is type std data for dp
pointer type std ap is type std data for ap
type std data for dp:
    ptr address
type std data for ap:
    dp ro ptr @base
    int offset

function dp Q$1 PST$1 @r = std *UNCHECKED* ( std data for dp ddp )
function ap Q$1 PST$1 @r = std *UNCHECKED* ( std data for ap dap )
function std data for dp r = std data for dp ( dp Q$1 PST$1 @ptr )
function std data for ap r = std data for ap ( ap Q$1 PST$1 @ptr )
    // These functions just copy the argument value to the
    // result value changing the type of the value.  Here
```

```
    // Q$1 is a wild-card that matches any list of qualifier-names,
    // and PST$1 is a wild-card that matches any pointer target
    // type.
```

Reading and writing values using a pointer of type $P$ can be accomplished by the functions:

```
load reference function PST$1 r = ( P QR$1 PST$1 @p ) ..load
store reference function ( P QW$1 PST$1 @p ) ..store = PST$1 r
```

These functions can be defined after $P$ is defined. They allow the pointer to be used to read a copy of the value pointed at, or write the value, but do not allow members or elements of the value to be accessed (members and elements of the copy may be accessed).

These functions are implicitly called when the target of a pointer variable is read or written. For example:

```
// Assume that ..load and ..store are defined for P = ap
//
ap int @p:
    p = 5
        // This translates first to:
        //   @p.* = 5
        // and then translates further to:
        //   @p..store = 5
int x = p
  // This translates first to:
  //   int x = @p.*
  // and then translates further to:
  //   int x = @p..load
```

An alternative strategy is to convert a pointer of type $P1$ to a pointer of type $P2$ that allows members and elements to be accessed. Suppose we are given:

```
pointer type P1 is type D1
pointer type P2 is type D2
```

Then we can define a reference function with the prototype:

```
point reference function P2 Q$1 PST$1 @p2 =
            ( P1 Q$1 PST$1 @p1 ) ..point
```

which converts a pointer of type $P1$ to a pointer of type $P2$. This conversion function is automatically called if a value pointed at by a pointer of type $P1$ is to be accessed, in preference to directly calling the `..load` and `..store` functions above. Then if $P2$ is `dp`, `ap`, `bsp`, `av`, or `bsv`, the $P2$ pointer can be used to not only read or write the value, but to also read or write members or elements of the value.

For example:

```
// Assume that ..point is defined for P1 = ap and P2 = dp and
```

```
// ..load and ..store are defined for P = dp.
//
ap int @p:
    p = 5
        // This first translates to:
        //   @p.* = 5
        // and then translates further to:
        //   @p..point.* = 5
        // and then translates to:
        //   @p..point..store = 5
int x = p
    // This first translates to:
    //   int x = @p.*
    // and then translates further to:
    //   int x = @p..point.*
    // and then translates to:
    //   int x = @p..point..load
```

Since `@p..point` is syntactically a *reference-call*, this *expression* can only match *reference-function-prototypes*, and such matches ignore the prototype result types, and use only the prototype argument types. [p104] Generally an inserted call to `..point` can be matched only if there is at most one `"..point"` function in the current context with suitable qualifiers and type (e.g., `Q$1` and `PST$1`) for a given $P1$).

Because there can be at most one matching `..point` reference function declaration, `..point` calls can be chained by the compiler. In particular, a `..point` reference function that produces an `ap` result will be chained to the builtin:

```
point reference function dp Q$1 PST$1 @r =
        std ( ap Q$1 PST$1 @p ) ..point:
    std data for ap dap = std data for ap ( @p )
    std data for dp ddp:
        ddp.address = dap.base + dap.offset
    @r = *UNCHECKED* ( ddp )
```

You can provide reference function definitions of `..point` with different actual types in place of `PST$1`, and these will be used in preference to a definition that has the wildcard `PST$1`.

Similar considerations apply to `..load` and `.store`. However if `..point` is defined, its use will be preferred.

As a convenience feature, instead of defining `..point`, you can define a (non-reference) function with prototype:

```
function D2 d2 =
        ma? *POINTER* *ACCESS* *CONVERSION* ( D1 d1 )
```

where *ma* refers to any module in which both $D1$ and $D2$ are defined. This implicitly defines the reference function:

```
point reference function P2 Q$1 PST$1 @p2 =
            ( P1 Q$1 PST$1 @p1 ) "..point":
    D1 d1 = D1 ( @p1 )
    D2 d2 = *POINTER* *ACCESS* *CONVERSION* ( d1 )
    @p2 = *UNCHECKED* ( d2 )
```

An example usage is the builtin:

```
// This function enables conversion of `ap ...' to `dp ...'
// when an ap pointer is being used to access a value or
// member or element of a value.  The dp is the sum of the
// base and offset of the ap.
//
function std data for dp r = std *POINTER* *ACCESS* *CONVERSION*
        ( std data for ap dap ):
    std data for dp ddp:
        ddp.address = dap.base + dap.offset
    r = ddp
```

An example implementing a new pointer type is:

```
type file:
    *READ-WRITE* av uns8 @name
    . . . . .
av *GLOBAL* *READ-WRITE* file @files @= allocate [1000]
ap *GLOBAL* *READ-WRITE* int @number of files @= allocate

// Implement a file descriptor (fd) that addresses a file
// in files.  The fd contains an index and addresses
// files[index].

type data for fd:
    int index

pointer type fd is type data for fd

point reference function ap Q$1 file @r = ( fd Q$1 file @p ) ..point:
    data for fd d = data for fd ( @p )
    ap file *READ_WRITE* @f = @files[d.index]
    @r = @f  // Implicitly converts *READ-WRITE* to Q$1

allocator function fd *READ-WRITE* file @r = allocate fd:
```

```
        data for fd d:
            d.index = number of files
        @r = *UNCHECKED* ( d )
        number of files = number of files + 1

    fd *READ-WRITE* file @f @= allocate fd
    f.@name = ...        // Translates to @f..point.*.@name = ...
    . . . . .
    av uns8 @n = f.@name // Translates to av uns8 @n = @f..point.*.@name
    . . . . .
```

### 11.2.1 Pointer Type Conversions

You can define an implicit conversion of one pointer type to another, after the manner of Section 11.1.4. An example is:

```
// This function enables implicit conversion of `dp ...' to
// `ap ...', where the latter has the constant 0 for a base
// and the dp value for its offset, provided the dp target is
// *GLOBAL*.
//
dp *GLOBAL* int std @zero:
    zero = 0
function std ap QU$1 *GLOBAL* PST$1 @r = std *IMPLICIT* *CONVERSION*
        ( std dp QU$1 *GLOBAL* PST$1 @p ):
    std data for dp ddp = std data for dp ( @p )
    std data for ap dap:
        dap.@base  = std @zero
        dap.offset = ddp.address
    @r = *UNCHECKED* ( dap )
```

As a convenience feature, we can define a `*POINTER* *IMPLICIT* *CONVERSION*` (non-reference) function instead. Suppose we are given:

```
pointer type P1 is type D1
pointer type P2 is type D2
```

Then this function has prototype:

```
function D2 d2 = ma? *POINTER* *IMPLICIT* *CONVERSION* ( D1 d1 )
```

where *ma* refers to any module in which both $D1$ and $D2$ are defined. This implicitly defines the function:

```
function P2 Q$1 PST$1 @p2 =
            ma? *IMPLICIT* *CONVERSION* ( P1 Q$1 PST$1 @p1 ):
```

```
D1 d1 = D1 ( @p1 )
D2 d2 = ma? *POINTER* *IMPLICIT* *CONVERSION* ( d1 )
@p2 = *UNCHECKED* ( d2 )
```

where *ma* is the same module as that of the `*POINTER* *IMPLICIT* *CONVERSION*` function, if any.

Similarly one can define an `*UNCHECKED*` function to convert *P*1 values to *P*2 values. As a convenience, this can be done by defining:

```
function D2 d2 = ma? *POINTER* *UNCHECKED* *CONVERSION* ( D1 d1 )
```

which implicitly defines the function:

```
function P2 Q$1 PST$1 @p2 = ma? *UNCHECKED* ( P1 Q$1 PST$1 @p1 ):
    D1 d1 = D1 ( @p1 )
    D2 d2 = ma? *POINTER* *UNCHECKED* *CONVERSION* ( d1 )
    @p2 = *UNCHECKED* ( d2 )
```

An example is the builtin:

```
// This function enables *UNCHECKED* conversion of `ap ...' to
// `dp ...' where the latter is the sum of the base and offset
// of the ap.
//
function std data for dp r = std *POINTER* *UNCHECKED* *CONVERSION*
        ( std data for ap dap ):
    std data for dp ddp:
        ddp.address = dap.base + dap.offset
    r = ddp
```

Lastly, one can define an explicit pointer conversion function with prototype:

```
function P2 Q$1 PST$1 @p2 = ma? P2 ( P1 Q$1 PST$1 @p1 )
```

## 11.3 Inline Function Declarations

The syntax of a function declaration is:

***function-declaration*** ::= *function-prototype* **:**
*statement*+
| *function-prototype* **: *DEFERRED***

***function-prototype*** ::=
*function-specializer*? **function** *prototype-result-list* **=**
*module-abbreviation*? *prototype-pattern*
| *function-specializer*? **function** *module-abbreviation*? *prototype-pattern*

***function-specializer*** ::= **`macro`** | **`constant`** | **`inclusion`** | **`allocator`**

***prototype-result-list*** ::=
*prototype-result-declaration* { **`,`** *prototype-result-declaration* }$^{\star}$

***prototype-result-declaration***
::= *prototype-result-variable-declaration*
| *prototype-next-variable-declaration*

*module-abbreviation* ::= see p36

***prototype-result-variable-declaration*** ::=
*initial-variable-declaration* that has no *allocation-call*

*initial-variable-declaration* ::= see p44

***prototype-next-variable-declaration*** ::=
*next-variable-declaration* [May have *allocation-call*.]

*next-variable-declaration* ::= see p45

***prototype-argument-declaration***
::= *prototype-result-variable-declaration* { **`?=`** *default-value* }$^{?}$
| **`macro`** *variable-name* { **`?=`** *default-value* }$^{?}$
| *prototype-result-variable-declaration* **`*=*`** *required-value*

***default-value*** ::= *expression*

***required-value*** ::= *constant-expression*

*variable-name* ::= see p36

*expression* ::= see p27

*constant-expression* ::= a **`const`** valued expression as defined on p93

***prototype-pattern*** ::= *parenthesized-pattern-argument-list*$^{\star}$ *pattern-term*$^{+}$

***pattern-term*** ::= *function-term-name* *pattern-argument-list*$^{\star}$

*function-term-name* ::= see p36

***pattern-argument-list***
::= **`(`** *prototype-argument-declaration* { **`,`** *prototype-argument-declaration* }$^{\star}$ **`)`**
| **`[`** *prototype-argument-declaration* { **`,`** *prototype-argument-declaration* }$^{\star}$ **`]`**

***parenthesized-pattern-argument-list*** ::=
*pattern-argument-list* with parentheses **`( )`** (and not square brackets **`[ ]`**)

1. A *prototype-pattern function-term-name* must not be an initial segment of any other *function-term-name* in the same *prototype-pattern*.

2. *Function-term-names* (in a *prototype-pattern*) may not be *member-names*$^{p36}$ (compare with *reference-function-declaration*$^{p102}$).

3. *Pattern-argument-lists* appearing before the first *function-term-name* may not use square **`[ ]`** brackets (compare with *reference-function-declaration*$^{p102}$).

4. There must be a *parenthesized-pattern-argument-list* that is not omittable (it must have a *prototype-argument-declaration* that has no *default-value*) either immediately before or immediately after the first *function-term-name* in a *prototype-pattern*. This distinguishes a *function-call* from a *reference-expression*: see Rule 4 on page 58.

5. Result and argument *variable-names* in a *function-prototype* must not begin with a *module-abbreviation*.

6. If a *prototype-result-declaration* contains $N > 0$ *pointer-type-names*, then the *pointer-variable* it declares typically begins with exactly $N$ '`@`'s. It may have fewer '`@`'s, including none, but may not have more.

   If the declaration contains a `PST$...` wildcard that is resolved into a list with *pointer-type-names*, the *prototype-result-declaration* after wildcard substitution will necessarily have fewer '`@`'s than *pointer-type-names*.

7. For a *prototype-result-declaration* of the form '**`next`** *v*', *v* must be the *variable-name* in a *prototype-argument-declaration* of the form '... *type-name v*'.

8. Result and argument *variable-names* in a *function-prototype* must be distinct, with an exception for the previous note.

9. In a *pattern-argument-list* a *prototype-argument-declaration* with no *default-value* cannot follow a *prototype-argument-declaration* with a *default-value*.

10. The first *prototype-argument-declaration* in a *parenthesized-pattern-argument-list* that is before the first *pattern-term* of a *prototype-pattern* must not have a *default-value*. I.e., these *parenthesized-pattern-argument-lists* cannot be omitted from *function-calls*.

11. If there are no *parenthesized-pattern-argument-lists* before the first *pattern-term* of a *prototype-pattern*, there must be a *parenthesized-pattern-argument-list* immediately after the *function-term-name* in that *pattern-term*, and the first *prototype-argument-declaration* in that *parenthesized-pattern-argument-list* must not have a *default-value* (i.e., this *parenthesized-pattern-argument-list* cannot be omitted from *function-calls*).

    This distinguishes *function-calls* from *reference-expressions*.

12. If the first *function-term-name* in the *prototype-pattern* is a *type-name* or *pointer-type-name*, the *function-declaration* must have a *prototype-result* with exactly one *prototype-result-variable-declaration* that has for its type this same *type-name* or has for its pointer type this same *pointer-type-name*, and the *prototype-pattern* may not have any *parenthesized-pattern-argument-lists* before the first *function-term*. The function declared is called a **explicit type conversion function**, and a *function-call* to the function is called an **explicit type conversion**.

13. A wild-card[p37] name of the form `T$...` can only match a *type-name*. A wild-card name of the form `P$...` can only match a *pointer-type-name*. A wild-card name of the form `Q...$...` can only match a possibly empty sequence of *qualifier-names*. A wild-card name of the form `PST$...` can only match an optional sequence of *pointer-type-names* and *qualifier-names* beginning with a *pointer-type-name*, all followed by a *type-name*.

    Syntactically `T$...` and `PST$...` behave as a *type-name*, `P$...` behaves as a *pointer-type-name*, and `Q...$...` behaves as a *qualifier-name*.

14. *Function-declarations* with *function-specifiers* other than `allocator` declare **special-functions**. These may not compile run-time code, and all their *prototype-result-variable-declarations* must be of the form '`const` *non-pointer-variable*. These functions are referred to as **macro functions**, **constant functions**, and **inclusion functions**, respectively.

15. A `*DEFERRED*` *function-declaration* may not be a special function.

16. An `allocator` function is just like any other inline function except that it can only be called in an *allocation-call*[p45] or from inside another `allocator` function.

An example of an inline function declaration and an inline function call is:

```
function F ( int x ) G ( int y ) H ( int z ?= 7 ) I ( int w ):
    . . . . . .
F ( 10 ) I ( 8 ) G ( 6 )
    // Equivalent to F ( 10 ) G ( 6 ) H ( 7 ) I ( 8 )
```

The *function-term-names* in the declaration are matched to those in the call, but need not have the same order in the call, except for the first *function-term-name* which must be the same in the declaration and the call. Thus the *call-terms* of the call are re-ordered to match the order of the *pattern-terms* of the declaration. If one of the *pattern-terms* is omitted in the call, but all its arguments have *default-values*, the *pattern-term* with its *default-values* will be inserted into the call (here `H ( 7 )` is inserted). However note that the first declaration argument list (here `( int x )`) may not have a default value.

For an argument list in the prototype to match an argument list in the call, both must be surrounded by the same brackets; either both have `( )` or both have `[ ]`, except that implied parentheses in the call are treated as `( )` during matching.

As an example of this last,

```
r = x + y
```

is treated for matching purposes is as

```
( r ) "=" ( ( x ) "+" ( y ) )
```

Note that *quoted-marks* and *quoted-separators* in *function-term-names* may appear with or without quotes in *call-term-names*.[p57] Thus we have the example:

```
function int z = ( int x ) "%%" ( int y ):
    . . . . . .
int v = 5 %% 6       // Legal
int w = 5 "%%" 6     // Legal
```

However, quoting an operator will cause it to be not recognized as an operator. For example:

`x + y * z` parses as `{ { "x" }, "+", { { "y" }, "*", { "z" } } }`

whereas

`x "+" y * z` parses as `{ { "x", "+", "y" }, "*", { "z" } }`

In the latter, `{ "x", "+", "y" }` cannot be recognized as a *function-call* because the arguments `"x"` and `"y"` are not bracketed (i.e., are not `{ "x" }` and `{ "y" }`), and will also not be recognized as a *reference-expression* (because `"+"` cannot be in a *variable-name* or begin a *member-name*) or *constant*.

A *pattern-term* with the syntax:

***boolean-pattern-term*** ::=
*function-term-name* **`( bool`** *variable-name* **`?=`** *default-value* **`)`**

triggers special syntax in a call that matches the prototype. In the call:

| | | |
|---|---|---|
| *function-term-name* | is equivalent to | *function-term-name* `( TRUE )` |
| **`no`** *function-term-name* | is equivalent to | *function-term-name* `( FALSE )` |
| **`not`** *function-term-name* | is equivalent to | *function-term-name* `( FALSE )` |
| omitted *function-term-name* | is equivalent to | *function-term-name* **`(`** *default-value* **`)`** |

Thus the example:

```
function F ( int x ) OPTION ( bool y ?= TRUE )
    . . . . . .
F ( 5 )            // Equivalent to F ( 5 ) OPTION ( TRUE )
F ( 5 ) OPTION     // Equivalent to F ( 5 ) OPTION ( TRUE )
F ( 5 ) no OPTION  // Equivalent to F ( 5 ) OPTION ( FALSE )
```

A function prototype result may be named '**`next`** *v*' if the prototype has an argument named *v*, in which case a *function-call* matching the prototype must match the prototype argument *v* to a *reference-expression e* that does not have the form '**`next`** *variable-name*'. Then the *assignment-result* corresponding to the prototype '**`next`** *v*' may be omitted, and will be taken to be '**`next`** *e*' if *e* is a *variable-name* naming a stack **`co`** variable for which '**`next`** *e*' is legal, and will otherwise be taken to be simply the *reference-expression e*.

For example:

```
function next x = inc ( int x ):
    next x = x + 1
. . . . . .
int y = ...
```

```
inc ( y )  // Equivalent to `next y = y + 1'.
ap *READ-WRITE* int @z:
    z = ...
inc ( z )  // Equivalent to `z = z + 1'.
```

If a *required-value* is given in a prototype, the call must have an equal `const` valued actual argument value in order for the call to match the prototype. Note that the argument variable type need not be `const`, as `const` values can be converted to run-time values: equality will be checked after conversion to the argument type. Matches to prototypes with more *required-values* are preferred over matches to prototypes with less *required-values*. Thus the example:

```
function F ( int x ) G ( int y *=* 5 ):  // First F declaration
    . . . . . .
function F ( int x ) G ( int y ?= 5 ):  // Second F declaration
    . . . . . .
int z = 5
F ( 8 )            // Matches only second F declaration
F ( 8 ) G ( 5 )    // Matches preferred first F declaration
F ( 8 ) G ( 6 )    // Matches only second F declaration
F ( 8 ) G ( z )    // Matches only second F declaration
                   // (z is not const valued).
```

A *prototype-argument-declaration* with the syntax:

***prototype-argument-declaration*** ::=
**`macro`** *variable-name* { **`?=`** *default-value* }$^{?}$

causes the argument value in a call to be the parse of the actual argument, which is a `const` value. If the *default-value* is used, it is evaluated as a *constant-expression*.

When a **macro function** is called it must return a `const` result that is a parsed expression which replaces the *function-call*. For example:

```
macro function const r = sum ( macro a1, macro a2 ?= { 1 } ):
    r = { a1, "+", a2 };

int x = ...
int y = sum ( x, 5 ) // Same as: int y = x "+" 5
                     // Parsed equivalent:
                     //     { "int" "y" } "=" { { x } "+" { 5 } }
                     // a1 is { x }, a2 is { 5 }
int y = sum ( x )    // Same as: int y = x "+" 1
                     // Parsed equivalent:
                     //     { "int" "y" } "=" { { x } "+" { 1 } }
                     // a1 is { x }, a2 is { 1 }
```

When a **constant function** is called its must return a `const` result that becomes the sole

result of the *function-call*. The difference between a `constant` function the same function without the `constant` *function-specifier* is that the former cannot compile run-time code whereas the latter can. Thus the `constant` *function-specializer* effectively does nothing but promise that the function will not compile run-time code. Examples:

```
constant function const r = sum ( const a1, const a2 ):
    r = a1 + a2;

const x = 10
const y = sum ( x, 5 )  // Sets y to 15
                        // a1 is 10, a2 is 5, r is 15
                        // sum ( x, 5) can be thought of as
                        //             being replaced by 15

constant function const r = list ( const a1, const a2 ):
    r = { a1, a2 }

const z = list ( 4, 5 )  // Sets z to { 4, 5 }
    // list ( 4, 5 ) can be thought of as being replaced by { 4, 5 }
```

A ***constant-expression*** is an *expression* that is either a *constant*[p39], a compile-time *reference-expression*[p54], or a *function-call* to a `constant` function.[9] *Constant-expressions* are used to compute parameters that must be known at compile-time.

Arguments to special functions (such as `constant` functions) need not have the `const` type. The values of such arguments cannot be accessed by the functions, but their types, which are `const` values, can be.

An **inclusion function** declaration can only be matched to a *function-call* at statement level (i.e., outside explicit or implicit parentheses or brackets) in a *call-assignment-statement*. When an **inclusion function** is called it must return a `const` result that is a parsed list of *statements* that replaces the *call-assignment-statement*. See Section 11.5[p107] for examples.

A '`*DEFERRED*`' *function-declaration* permits inline functions defined between it and a later non-`*DEFERRED*` companion *function-declaration* to call the function. An example is:

```
function F2 ( const i): *DEFERRED*
function F1 ( const i):
    if i != 0:
        <do F1 thing>
        F2 ( i - 1 )

F1 ( 1 )   // Compile Error: Call F2(0) cannot be expanded.
```

[9] A *rational-constant* is both a *constant* and a *function-call* to a `constant` function.

```
function F2 ( const i )
    if i != 0:
        <do F2 thing>
        F1 ( i - 1 )

F1 ( 5 )   // Legal, expands to:
           //    <do F1 thing>
           //    <do F2 thing>
           //    <do F1 thing>
           //    <do F2 thing>
           //    <do F1 thing>
           // Would not be legal if the deferred
           // function declaration were omitted,
           // as then no F2 declaration would be
           // visible to the statements of F1.
```

Here the statements of `F1` compile in the context of the declaration of `F1` and need the `*DEFERRED*` declaration of `F2` in that context to enable these statements to call `F2`. Given that a call is enabled, the situation where the statements of `F2` are provided later is permitted.

A `*DEFERRED*` declaration and its companion non-`*DEFERRED*` declaration must have identical prototypes, except:

- *Default-values* and *required-values* may appear only in the `*DEFERRED*` declaration and be omitted in the companion, but not vice-versa.
- If *default-values* and *required-values* appear in both the `*DEFERRED*` declaration and its companion, they must evaluate to the same *const* value. They need not have the same *constant-expressions*, and note that the two *constant-expressions* are each evaluated where their prototype is declared, and therefore are evaluated in two different contexts.

A `*DEFERRED*` inline *function-declaration* may have at most one companion.

The prototype of an inline *function-declaration* is visible to the *statements* of that same declaration, and therefore an inline function can call itself without having any `*DEFERRED*` companion.

Recursion in inline function calls must be limited by `const` variables such as the counter `i` in the above example, for if it is not, there will be a compile error when the compiler decides the inline nesting is too deep or the code generated by one statement is too much.

A special function cannot be `*DEFERRED*`.

### 11.3.1 Inline Call-Declaration Matching

Each *function-call* in a statement must be matched to a single *function-declaration*, else compilation of the statement fails with a compile error.

The first step in processing a *statement* is to visit in bottom-to-top left-to-right order all the *expressions* in the *statement* and apply the following algorithm which evaluates *constant-expressions*, analyzes *reference-expressions*, and processes macros and inclusions:

**Bottom-Up Expression Algorithm**

1. This algorithm is applied to an *expression* $E$, and terminates after any step that annotates $E$.

   When this algorithm searches for a non-reference declaration matching an *expression*, it uses the Function Declaration Search Algorithm below.

2. If the entire *expression* $E$ has the form:

   *module-abbreviation* `(` *subexpression* `)`

   where the parentheses may not be implied, move the *module-abbreviation* to the beginning of the parsed *subexpression*, and continue to the next step with the modified *expression*. See p59 for an example.

3. If $E$ is a *constant*[p39] other than a *rational-constant*, evaluate $E$ to a `const` value $C$, and annotate $E$ with $C$ and with the type `const` (*Rational-constants* are evaluated as *function-calls* to a `constant` function in Step 6 below.)

4. If $E$ is a compile-time *reference-expression*[p54], evaluate $E$ to a reference pointer. Then if $E$ is storing, annotate $E$ with the reference pointer, and if $E$ is loading, load the value $C$ pointed at and annotate $E$ with $C$ and the `const` type.

5. If $E$ is a run-time *reference-expression*[p50], evaluate the type of $E$'s reference pointer and annotate $E$ with this type. Also annotate each index subexpression with its target type.

6. If $E$ is a *function-call*, search for constant *function-declarations* matching $E$. If one is found, evaluate the declaration's function to get a `const` value $C$, and annotate $E$ with $C$ and the `const` type.

7. If $E$ is a *function-call*, Search for macro *function-declarations* matching $E$. If one is found, evaluate the declaration's function to get a `const` value $R$, replace $E$ by $R$, and apply the Bottom-Up Expression Algorithm to $R$ (re-analyze the subexpressions of $R$).

8. If $E$ is a *function-call* that is top level in a *call-assignment-statement* $S$, search for inclusion *function-declarations* matching $E$.

If one found, evaluate the declaration's function to obtain `const` return values. The first return value replaces $S$ (not $E$).

Subsequent return values are assigned to $S$'s *assignment-results* in left-to-right order. Thus if $S$ has $N$ *assignment-results* (all of which must be `const` variables), then the execution must return at least $N + 1$ results (all of which must be `const` values). If there are excess results, discard them. If there are too few, it is a compile error.

If $S$ has been replaced, terminate this algorithm (and begin processing the replacement for $S$).

The second step in processing a *statement* is to visit in top-down order all the *expressions* $E$ in the *statement* and locate for each $E$ that are *function-calls* (and not annotated with a `const` value by the Bottom-Up Algorithm above) a matching *function-declaration*. Or more specifically, a *function-declaration* whose *function-prototype* matches the *function-call*. In this step all non-special *function-declarations* that are in scope are included in the search (special *function-declarations* were processed by the above Bottom-Up Expression Algorithm).

Furthermore in this step each $E$ has a target type $T$ (or types), and *function-prototypes* that match the expression are required to have a result of exactly this type (or results of exactly these types).

After the Bottom-Up Expression Algorithm, *assignment-results* have a known type, which becomes the target type of expressions that set these results. If an *assignment-result* is a variable declaration, the declaration specifies the type of the *assignment-result*. If the *assignment-result* is a *reference-expression*, the Bottom-Up Expression Algorithm determines the type of the *assignment-result*.

For an *expression* that is a *function-call* argument, the declaration matched to the *function-call* specifies the target type of the argument *expression*. For example, in '`int z = x + 5`' the target type of `+` is `int`, the declaration matched to '`x + 5`' (the only one with an `int` result type) has `int` result and argument types, so `int` is the target type of '`x`' and '`5`'.

For an index of a *reference-expression*, the index *expression* is annotated with its target type when its containing *reference-expression* is analyzed by the Bottom-Up Expression Algorithm.

As a special case, in a relational expression such as '`x < y + 5`' all the operands are assigned the same target type by the following procedure. If one of the operands is an explicit type conversion [p89], the type of the explicit type conversion becomes the target type of all the operands (as in '`x < flt ( y + 5 )`'). Otherwise, if one of the operands is a *reference-expression*, its type is tested as the target type of all the operands, and if successful, its type is used (in '`x < y + 5`', the type of '`x`' becomes the target type of '`y + 5`'), If several operands are *reference-expressions*, this succeeds if one of these has a type into which the types of the others can be implicitly converted.

Consider the case of *expression* $C$ which is a *function-call* containing an argument *expression* $E$. The various *function-declarations* that might match $C$ specify various target types for

$E$. Some of these target types may be the same, so if we find matches for $E$ with a given target type, we should remember what we found so we do not re-compute the match for that target type if it is needed again. Therefore, when we find a *function-declaration* $D$ matching $E$ for target type $T$, we annotate $E$ with $T \mapsto D$. These annotations may then be used to compile the *statement* after this second step is completed.

Given all this, when the second step visits an *expression* $E$ with target type $T$ the second step executes:

**Top-Down Expression Algorithm**

1. This algorithm is applied to an *expression* $E$ and its target type $T$ found by top-to-bottom left-to-right transversal of the *expressions* of the current statement. It annotates *expressions* with maps from possible target types to declarations containing code that can execute the *expressions*, When this algorithm searches for a declaration matching an *expression*, it uses the Function Declaration Search Algorithm below.

2. If $E$ already has an annotation $T \mapsto \ldots$, terminate this algorithm. In what follows, processing of $E$ is terminated whenever $E$ is annotated with $T \mapsto \ldots$.

3. If $E$ is annotated with a `const` value (from the Bottom-Up Step), then if $T$ equals `const`, annotate $E$ with $T \mapsto *EXACT*$. Otherwise search the implicit conversion graph [p81] for some shortest path $D_1, D_2, \ldots, D_n$ from $T$ to `const`, and if one is found annotate $E$ with $T \mapsto D_1, D_2, \ldots, D_n$, or if not found, annotate $E$ with $T \mapsto *\texttt{FAIL}*$.

4. If $E$ is a *reference-expression* [p48] of type $T'$, then if $T$ equals $T'$ annotate $E$ with $T \mapsto *EXACT*$. Otherwise search the implicit conversion graph [p81] for some shortest path $D_1, D_2, \ldots, D_n$ from $T$ to $T'$, and if one is found annotate $E$ with $T \mapsto D_1, D_2, \ldots, D_n$, or if not found, annotate $E$ with $T \mapsto *\texttt{FAIL}*$.

5. If $E$ is a *function-call*, search for non-special *function-declarations* that match $E$. If none are found, annotate $E$ with $T \mapsto *\texttt{FAIL}*$. If more than one is found (of highest rank), annotate $E$ with $T \mapsto *\texttt{AMBIGUOUS}*$. Otherwise if just one *function-declaration* $D$ of highest rank was found, annotate $E$ with $T \mapsto D$.

   If $E$ is the top level *function-call* in a *call-assignment-statement*, and the latter has more than one *assignment-result*, then $T$ is replaced by a list $T_1, T_2, \ldots, T_m$ of target types (the types of the *assignment-results*), and the search for *function-declarations* is limited to those whose prototypes have at least $m$ results with the first $m$ of these being of exactly types $T_1, T_2, \ldots, T_m$ respectively.

The search algorithm used by the above is:

**Function Declaration Search Algorithm**

1. This algorithm examines all *function-declarations* in the current context [p116] that meet criteria specified by the search and applies to each the Function Declaration Matching Algorithm for *function-declarations* (see below). Search criteria can restrict the search to, for example, `macro` *function-declarations*, or to give another example, to all non-special *function-declarations* with a given target type (type of their first result).

2. After examining all the appropriate declarations, this algorithm has a set of declarations satisfying the Function Declaration Matching Algorithm. If the set is empty, this algorithm reports failure. Otherwise all but the highest ranking matches are discarded. If there is only one match left, this algorithm reports success along with the single highest ranking matching declaration. If there are more than one highest ranking match, this algorithm reports ambiguity failure.

The matching algorithm used by the above to match a *function-call* with a *function-declaration* is as follows.

**Function Declaration Matching Algorithm**

1. If the call begins with a *module-abbreviation* and the *prototype-pattern* in the *function-declaration* either does not have a preceding *module-abbreviation*, or is preceded by one that names a different module from that of the call, the call-declaration match fails.

   If the *function-call* begins with a *module-abbreviation* and *prototype-pattern* is preceded by a *module-abbreviation* identifying the same module, or if the *function-call* does not begin with a *module-abbreviation* and *prototype-pattern* is not preceded by a *module-abbreviation* (both are non-external), the match is marked as **module proficient**.

   In what follows, any *module-abbreviation* beginning the *function-call* is ignored, and the remainder of the *function-call* is matched to the *function-declaration*'s prototype.

2. The **key** of a *function-declaration* is the number of argument lists before the first *function-term-name* in its *prototype-pattern* followed by that first *function-term-name*. For example, the key of '`(...)  "+" (...)`' is '`() "+"`'.

   The *function-call* must begin with the key of the *function-declaration*, else the call-declaration match fails.

   This implies that argument lists before the first *call-term-name* in a *function-call* cannot be omitted and must be bracketed by actual or implied parentheses.

   When matching two *function-term-name* lexemes, if the one from the *prototype-pattern* is `"` quoted, the one in the *function-call* may be quoted or not quoted.

   The first *function-term-name* in the call may or may not be followed by *call-argument-lists*.

3. The remainder of the *function-call*, the part after any *call-argument-lists* following the first *function-term-name*, is scanned for *prototype-pattern function-term-names* other than the first one, and also for the words '`no`' and '`not`'. At each point in the scan one of the following must be found: (1) a *function-term-name* from the *prototype-pattern*, other than the first such, or (2) the word '`no`', or (3) the word '`not`', or (4) a bracketed argument list (with explicit or implicit brackets). If none of these is found at some point in the scan, the call-declaration match fails. If a *function-term-name* is found more than once, the call-declaration match fails. Every '`no`' and '`not`' must immediately precede a *function-term-name*, else the call-declaration match fails.

   If the scan succeeds, the *function-call* is divided into *call-terms*, each beginning with a *function-term-name* possibly preceded by '`no`' or '`not`', and ending with argument lists (that may be omitted). If there are argument lists before the first *function-term-name* (in the key), the situation is treated as if both prototype and call began with identical virtual *function-term-names*.

   Then each *call-term* in the *function-call* is matched to the *pattern-term* with the same name in the *function-declaration*.

4. If a *call-term* begins with '`no`' or '`not`', the *call-term* must have no argument lists and be matched to a *boolean-pattern-term*[p91], else the call-declaration match fails. In this case the beginning '`no`' or '`not`' is removed and the argument list '`( FALSE )`' is appended to the *call-term*.

   If a *call-term* that does not begin with '`no`' or '`not`' matches a *boolean-pattern-term* and has no argument lists, the argument list '`( TRUE )`' is appended to the *call-term*.

   (By the rules below, if a *boolean-pattern-term* is not matched to any *call-term*, it will be given its *default-value*.)

5. A *call-term* must match its corresponding prototype *pattern-term* according the rules that follow. Failure of any call-prototype term match causes the prototype-call match to fail.

6. For a *call-term* to match its corresponding *pattern-term*, both must have the same number of *argument-lists*, the same brackets, and the same number of arguments in corresponding *argument-lists*, after the *call-term* has been **adjusted**. The following are permitted adjustments.

   For every *pattern-term* that has no corresponding *call-term* (because its *function-term-name* was not found in the call), a *call-term* consisting of just the *pattern-term*'s *function-term-name* is appended to the *function-call*. After this the *call-terms* are re-ordered so their order matches that of their associated *pattern-terms*.

   A *call-term argument-list* with implied parentheses is treated as if it had `( )` parentheses.

If in a left-to-right scan of a *call-term*, a *call-argument-list* with `( )` is expected but instead a `[ ]` bracketed *call-argument-list* or the end of the *call-term* is found, an empty list `()` is inserted into the *call-term*.

If a *call-argument-list* is shorter than the corresponding *pattern-argument-list*, and all omitted arguments at the end of the *call-argument-list* have *default-values* in the *pattern-argument-list*, the *default-values* corresponding to the omitted arguments are added to the end of the *call-argument-list*. The *default-values* are compiled in the context of the *function-declaration* and not the context of the *function-call*: see p119.

At this point the *pattern-argument-lists* in the prototype *pattern-term* must match in order all the *call-argument-lists* in the *call-term* in type of bracket and number of arguments, else the call-declaration match fails.

7. If all the above is successful, then *actual-arguments* in the call are matched to corresponding *prototype-argument-declarations* in the *function-declaration* according to the rules that follow. Failure of any of these matches causes the call-declaration match to fail.

8. This section is skipped for special functions (functions whose *prototype-result* has the form '`macro` *variable-name*', '`constant` *variable-name*', or '`inclusion` *variable-name*').

   The types in the *prototype-result-variable-declarations* are matched left-to-right to the target types for these results. Any wildcards [p37] in a *prototype-result-variable-declaration* are filled in from the corresponding target type, left-to-right. If a wildcard gets more than one value from this process, the call-declaration match fails.

   Note that *prototype-next-variable-declarations* are ignored during this matching.

   Then if any prototype result type is not identical to its corresponding assignment target type, the call-declaration match fails (i.e., there is no implicit conversion of function result types).

9. *Prototype-argument-declarations* are matched to an *actual-arguments* and processed left to right.

   If a *prototype-argument-declaration* $PAD$ is matched to an *actual-argument* $AA$ and $PAD$ has a wildcard not assigned in Step 8, $AA$ must be *reference-expression* or a *constant-expression*, (i.e., annotated with a `const` value by the Bottom-Up Algorithm) else the call-declaration match fails. The wildcard is assigned from information in the *reference-expression*'s type (the type of value the reference pointer points at) or *constant-expression*'s type (which is just `const`).

   If different values are assigned to the same wildcard by this process (by different prototype-actual argument matches, or by assignments made in Step 8) the call-declaration match fails.

10. If any *prototype-argument-declaration* wildcards are unassigned at this point, the call-declaration match fails.

11. If a *prototype-argument-declaration* has a *required-value*, its matching *actual-argument* must be a *constant-expression* with a value equal to the *required-value* (which is itself the result of a *constant-expression* evaluated in the context of the *function-declaration*), after both are converted to the argument type specified by the *prototype-argument-declaration*, else the call-prototype match fails. Note that the argument type need not itself be `const`.

    If the argument type is `const` and the values being compared are maps, the maps must be identical (they are compared as pointers and not as lists of elements).

12. Matches between *prototype-argument-declaration* types and acceptable *actual-argument* subexpression target types are then checked, except for `macro` arguments (that require no run-time typing). Specifically, if a *prototype-argument-declaration* specifies a target type $T'$ for an *actual-argument expression* $E'$, the Top-Down Expression Algorithm is applied recursively to $E'$ with target type $T'$ to get an annotation $T' \mapsto \ldots$ on $E'$.

    Then if any argument $T'$ is mapped to `*FAIL*` or `*AMBIGUOUS*` the call-declaration match fails.

    If any is mapped to `*EXACT*`, the call-prototype map is marked as **conversion proficient**. Note that only a single argument needs to be `*EXACT*` in order for the entire call-declaration match to be marked conversion proficient.

13. Some *prototype-argument-declarations* may correspond to *prototype-next-variable-declarations* and for these the *actual-arguments* must be either a `co` *variable* $v$ for which '**`next`** $v$' is legal, or a readable and writable *reference-expression*. However no checks are done on these requirements during the Function Declaration Matching Algorithm. These requirements are checked a final call-declaration match is found, and any requirement failure becomes a compile error.

14. If the call-declaration match has been successful, it is assigned a **rank** that is the sum of:

    - the number of required arguments the prototype has
    - the negative of the number of `T$...` wildcards the prototype has; a wildcard name appearing more than once in the prototype is counted only once
    - a very large number if the match is conversion proficient
    - an even larger number if the match is module proficient

The following are examples using the builtin prototypes

```
function N r = std (N v1) "+" ( N v2 )
function bool r = std (N v1) "==" ( N v2 )
function flt64 r = std flt64 (N v1)
```

which exist for every builtin number type `N`.

```
int32 x = ...
flt64 r1 = x + 5
    // Target type flt64 selects N = flt64 for "+".
    // Implicit conversions flt64 <--- int32 and
    // flt64 <--- const exist.
bool b1 = ( x == 5 )
    // Target type of "==" is bool.  Reference expression x selects
    // N = int32 as target type for "==" operands, which works
    // because implicit conversion int32 <--- const exists.
bool b2 = ( x == 5.5 )
    // Ditto, but when evaluated in the compiler with target type int32,
    // std *IMPLICIT* *CONVERSION* ( 5.5 ) fails and gives a
    // compile error, as 5.5 cannot be represented as an int32.
flt64 y = 5.5
bool b3 = ( x == y )
    // Reference expression y selects N = flt64 as target type
    // for "==" operands, which works because implicit conversion
    // flt64 <--- int32 exists.  Reference expression x selects
    // N = int32, which fails because implicit conversion
    // int32 <--- flt64 does NOT exist.
bool b4 = ( flt64 ( x ) == 5 )
    // The explicit type conversion function flt64 selects
    // N = flt64 which works because flt64 <--- const exists.
```

## 11.4 Reference Function Declarations

A **reference function** is a function that can be called by an initial part of a *reference-expression*. This part is called a *reference-call*.

A *function-variable-name* appearing at the beginning of a *reference-expression* as its root is a *reference-call*.

A compound *reference-expression* can be a *reference-call*.

Note that if a *reference-expression* contains a *module-abbreviation*, it is at the very beginning and is part of the root (*function-*)*variable-name*.

The syntax of a *reference-function-declaration* is:

***reference-function-declaration*** ::=
*reference-function-prototype* **:**
*statement*$^{+}$
| *reference-function-prototype* **: *DEFERRED***

***reference-function-prototype***

::= **load reference function**
*prototype-result-variable-declaration* **=** *function-variable-name*
| **store reference function**
*function-variable-name* **=** *prototype-result-variable-declaration*
| **point reference function**
*prototype-result-variable-declaration* **=** *function-variable-name*
| **load reference function**
*prototype-result-variable-declaration* **=** *reference-prototype-pattern*
| **store reference function**
*reference-prototype-pattern* **=** *prototype-result-variable-declaration*
| **point reference function**
*prototype-result-variable-declaration* **=** *reference-prototype-pattern*

*reference-function-specifier* ::= **load** | **store** | **point**

*prototype-result-variable-declaration* ::= see p88

*function-variable-name* ::= see p36

***reference-prototype-pattern*** ::= *reference-pattern-base* *reference-pattern-offset*

***reference-pattern-offset*** ::= *member-name*
| *member-name* *reference-pattern-index-list*
| *reference-pattern-index-list*

***reference-pattern-base*** ::= **(** *prototype-result-variable-declaration* **)**

***reference-pattern-index-list*** ::=
**[** *reference-pattern-index-declaration* { **,** *reference-pattern-index-declaration* }$^\star$ **]**

***reference-pattern-index-declaration*** ::= *prototype-result-variable-declaration*

*member-name* ::= see p36

1. A *reference-function-declaration* is compile-time if its *prototype-result-variable-declaration* before or after the **=** sign in its *reference-function-prototype* has type `const`, and is run-time otherwise. *Reference-pattern-index-declarations* have type `const` if and only if the *reference-function-declaration* is compile-time.

2. A `point` *reference-function-declaration* cannot be compile-time.

3. A *function-variable-name* begins with a *module-abbreviation* if and only if it is external (just like other *variable-names*). In the external case the *reference-function-declarations* for the *function-variable-name* must appear in the module (file) named.

4. A *reference-function-declaration* containing a *reference-pattern-base* is external if and only if all the *pointer-type-names* and *type-names* in its *reference-pattern-base* are external or are wildcards, and at least one that is external is in the module containing the declaration.

5. Default values, required values, and `macro` arguments are not permitted.

6. At most two *reference-function-declarations* in a context may have the same *function-variable-name*, and if there are two, one must have the `load` *reference-function-specifier*, and the other must have the `store` *reference-function-specifier*.

7. At most two *reference-function-declarations* in a context may have the same *reference-pattern-base* argument type and *member-name*, and if there are two, one must have the `load` *reference-function-specifier*, and the other must have the `store` *reference-function-specifier*.

8. At most two *reference-function-declarations* in a context may have the same *reference-pattern-base* argument type and no *member-name*, and if there are two, one must have the `load` *reference-function-specifier*, and the other must have the `store` *reference-function-specifier*.

9. Wild-cards may appear in a *reference-pattern-base*. Wild-cards appearing elsewhere in the *reference-function-prototype* must also appear in its *reference-pattern-base*.

   Syntactically `T$...` and `PST$...` behave as a *type-name*, `P$...` behaves as a *pointer-type-name*, and `Q...$...` behaves as a qualifier.

### 11.4.1 Reference Call-Declaration Matching

In the case of a *function-variable-name* reference call, the *function-variable-name* is located in the current context just like any other *variable-name*. The *function-variable-name* begins the *reference-expression* and is the root of the *reference-expression*. The *function-variable-name* can have at most two *reference-function-declarations*, and if it has two, one must be `load` and one must be `store`. One of these is excluded by context. If the root is the entire *reference-expression* and that is storing, then `load` is excluded. Otherwise `store` is excluded. So the *function-variable-name* can have at most one non-excluded *reference-function-declaration*, and it is a compile error if it has none (e.g., if a storing *simple-reference-expression* has only a `load` *reference-function-declaration*).

In the case of a compound *reference-expression*, we have a *reference-base* and a following *reference-offset*, and the type of the *reference-base* has already been computed at compile-time. In some situations, the *reference-offset* can be either a *member-name* followed by an *index-list*, or it can be just the *member-name* by itself. Regardless, the following algorithm is used to find a *reference-function-declaration* matching the *reference-base* and one of the possible following *reference-offsets*:

**Reference Function Declaration Search Algorithm**

1. This algorithm examines all *reference-function-declarations* in the current context [p116] that meet criteria specified in the following, and applies applies the Reference Function Declaration Matching Algorithm (below) to each.

2. If the *reference-base* is followed by a *member-name* that is followed by an *index-list*, this algorithm is applied first to a *reference-offset* consisting of both the *member-name* and the *index-list*. If no declarations are found, this algorithm is reapplied to a *reference-offset* consisting of just the *member-name*.

3. If the compound *reference-expression* consisting of the *reference-base* followed by the *reference-offset* is loading, `store` *reference-function-declarations* are excluded from the search, and if instead it is storing, `load` *reference-function-declarations* are excluded.

4. After examining all the appropriate declarations, this algorithm has a set of declarations satisfying the above criteria. If the set is empty, this algorithm reports failure. Otherwise all but the highest ranking matches are discarded. If there is only one match left, this algorithm reports success along with the single highest ranking matching declaration. If there are more than one highest ranking match, this algorithm reports ambiguity failure.

### Reference Function Declaration Matching Algorithm

1. The *reference-offset* and the *reference-pattern-offset* must match in that

   (a) If one has a *member-name*, both must have the same *member-name*, else the call-declaration match fails.

   (b) If one has an *index-list*, both must have *index-lists* with the same number of elements else the call-declaration match fails (but the types of the elements are not involved in matching).

2. The type of the *reference-base* in the call is matched to the type of the *reference-pattern-base* in the declaration. If the latter has wildcards, the wildcards are assigned information from the *reference-bases*'s type.

   If one wildcard is assigned two or more different values, or zero values, the call-declaration match fails.

   If after this has been done, the type of the *reference-base* does not exactly match the type of the *reference-pattern-base*, the call-declaration match fails.

3. If the call-declaration match has been successful, it is assigned a **rank** that is:

   - the negative of the number of wildcards the prototype has; a wildcard name appearing more than once in the prototype is counted only once

Example:

```
// The *UNCHECKED* function used below is a builtin function
// that performs a variety of conversions which violate type
// checking.
//
// For one of these uses *UNCHECKED* has the prototype:
//
//    av Q$ PST$ @r = std *UNCHECKED*
//        ( ap Q$ PST$ @p, int offset, int lower, int upper )
//
// This takes the ap @p, adds the offset argument to the
// pointer offset, and returns this as an av with bounds
// lower and upper.
//
// For another of these uses *UNCHECKED* has the prototype:
//
//    ap Q$1 PST$1 @r = std *UNCHECKED* ( ap Q$1 PST$2 @p )
//
// and simply changes the type of value pointed at by an ap.

// In my vector X, a vector with flt64 elements is located
// at X.offset from the address of X and allows index range
// from 0 through X.length-1.
//
type my vector:
    int offset        // Offset of first element in bytes.
    int length        // Number of elements.
    align 64
    *LABEL* first     // Offset of first element in bits.

point reference function ap Q$1 flt64 @x =
        ( ap Q$1 my vector @v ) [ int index ]:
    av Q$1 flt64 @p = *UNCHECKED*
            ( @v, v.offset, 0, v.length )
        // See *UNCHECKED* above.
    @x = @p[index]

type my data:
    *INCLUDE* my vector
    flt64[2]

point reference function ap Q$1 my vector @x =
    ( ap Q$1 my data @y ) ..point:
```

```
    @x = *UNCHECKED* ( @y )

ap *READ-WRITE* my data @D:    // `@= allocate' is implied
    D.offset = D.first / 8     // 8 converts bits to bytes.
    D.length = 2

D[0] = 5.5      // Computes and uses pointer @D[0]
D[1] = -7.33    // Computes and uses pointer @D[1]
flt64 x = D[0]  // Now x == 5.5
flt64 y = D[1]  // Now y == -7.33
flt64 z = D[2]  // Run time error: 2 >= upper bound of av
                // that equals D.length.

D[2] = 1.0      // Run-time error: bounds limit exceeded.
```

## 11.5 Inclusions

An example of an inclusion is:

```
// Create int V and iterate for V = 0, 1, 2, ..., LIMIT-1.
//
//     for ( V, LIMIT ):
//         STATEMENTS*
//
// is equivalent to:
//
//     int V = 0
//     while V < LIMIT:
//         STATEMENTS*
//         next V += 1
//
inclusion function const r = for ( macro V, macro LIMIT ):
    r = { { { "int" } + V, "=", { 0 } },
          { { "while", { V, "<", LIMIT } },
            com *SUBBLOCK* +
                { { "next" } + V, "+=", { 1 } } } }

// Compute sum = 1 + 2 + ... + 15
//
int sum = 0
for ( x, 15 ):
    next sum += x + 1
```

```
// The above for ... statement is expanded to:
//
//     int x = 0
//     while x < 15:
//         next sum += x + 1
//         next x += 1
//
// because:
//
//   com *SUBBLOCK* equals:
//     { { { "next", "sum" }, "+=", { { "x" }, "+", { 1 } } },
//       ".initiator" = ":", ".terminator" = *INDENTED-PARAGRAPH* }
```

Although the value of an inclusion is to be a sequence of parsed *statements*, it is not necessary to include some of the annotations in these. Specifically, it is not necessary to include the `.position` or the following `.initiator`s or `.terminator`s:

**Unnecessary Annotations**

| `.initiator` | `.terminator` |
|---|---|
| `*LOGICAL-LINE*` | `"<LF>"` |
| `"("` | `")"` |

because the `.position` of the function call will be added if no `.position` is given, logical lines can be identified from context, and `()` bracketed subexpressions are equivalent to subexpressions with implied brackets (i.e., with no `.initiator` or `.terminator`).

However, any `.separator` and the following must be included:

**Necessary Annotations**

| `.initiator` | `.terminator` |
|---|---|
| `":"` | `*INDENTED-PARAGRAPH*` |
| `"["` | `"]"` |
| `"{"` | `"}"` |

In the example at the beginning of this section, `com *SUBBLOCK*` evaluates to the subblock in the '`for ...`' statement, and this subblock has `".initiator" = ":"` and `".terminator" = *INDENTED-PARAGRAPH*`, which are the necessary annotations that are carried over into the concatenation `com *SUBBLOCK* + {...}`.

The *include-assignment-statement* can be used to return a list of parsed *statements*:

***include-assignment-statement*** ::=
`const` *non-pointer-variable* `= *INCLUDE*` *include-argument-list*$^{?}$ `:` *statement*$^{\star}$
| `const` *non-pointer-variable* `=` `*INCLUDE*` *include-argument-list*$^{?}$ `:` *restricted-statement*

***include-argument-list*** ::= `( )` | `(` *include-argument* { `,` *include-argument* }$^{\star}$ `)`

***include-argument*** ::= *word* beginning with a capital *letter*

***restricted-statement*** ::= see p70

*non-pointer-variable* ::= see p36

An *include-statement* is executed at compile time; its *statements* are parsed and the parser output is returned.

Each *include-argument* must be a `const` variable assigned a value before the *include-statement* compiles. Everywhere this variable appears in the parsed *statements* of the *include-statement*, the value of the variable is substituted for the variable name.

Substitution for *include-arguments* obeys the following rules:

- If the *include-argument* value is a list, and if the instance being substituted is an element of a list, the *include-argument* list is spliced into the instance containing list. However any *include-argument* dictionary entries are discarded.

  For example, let `X` be an *include-argument*.

  If `X = {"A", "B", ".separator" = ","}` then
  `"Y", "=", { "X", "C" }` becomes `"Y", "=", { "A", "B", "C" }`
  and
  `"Y", "=", { "X", "C", ".separator" = "," }`
  becomes
  `"Y", "=", { "A", "B", "C", ".separator" = "," }`
  The *include-argument* value list may be empty. If `X = {}` then
  `"Y", "=", { "foo", "X" }` becomes `"Y", "=", { "foo" }`
  If you do not want a list valued *include-argument* to be spliced in, use `( )` parentheses around the instance being substituted. Thus if `X = { { "A" }, "*", { "B" } }` then
  `"Y", "=", { ("X"), "+", { 1 } }`
  becomes
  `"Y", "=", { ( { "A" }, "*", { "B" } ), "+", { 1 } }`

- Otherwise the non-list value of the *include-argument* replaces the instance in the parsed *statement*. Thus if `X = "A"` then
  `{ "Y", "X" }, "=", { 2 }` becomes `{ "Y", "A" }, "=", { 2 }`

So in the example at the beginning of this section, `for` inclusion could have written as:

```
inclusion function const r = for ( macro V, macro LIMIT ):
    const P = com *SUBBLOCK*
    r = *INCLUDE* ( V, LIMIT, P ):
        int V = 0
        while V < LIMIT:
            P
            next V += 1
```

The following is a second example of an inclusion function:

```
inclusion function const r =
        my vector ( macro TYPE-NAME, macro ELEMENT-TYPE ):
    const ALIGNMENT = ELEMENT-TYPE..alignment
    r = *INCLUDE* ( TYPE-NAME, ALIGNMENT ):
        type TYPE-NAME:
            int offset     // Offset of first element in bytes.
            int length     // Number of elements.
            align ALIGNMENT
            *LABEL* first  // Offset of first element in bits.

        point reference function ap Q$1 ELEMENT-TYPE @x =
                ( ap Q$1 TYPE-NAME @v ) [ int index ]:
            av Q$1 ELEMENT-TYPE @p = *UNCHECKED*
                    ( @v, v.offset, 0, v.length )
                // See *UNCHECKED* above.
            @x = @p[index]

        allocator function ap QW$1 TYPE-NAME @r =
                (uns length, uns alignment )
                allocate [uns count]:
            @r = std ( length + count * ELEMENT-TYPE..size,
                       alignment ) allocate
            r.offset = r.first/8
            r.length = count

my vector ( my data, flt )
ap my data @D @= allocate [2]
    D[0] = 100
    D[1] = 200
flt x0 = D[0]
flt x1 = D[1]
flt x2 = D[2]  // Run-time error: index too large
```

## 11.6 Out-of-Line Function Declarations

An out-of-line function prototype is a limited subset of an inline function prototype which ensures that there is a single ordered list of arguments. To obtain a more flexible interface, an out-of-line function call should be embedded in an inline function that pre-processes the arguments.

The syntax of an out-of-line function declaration is:

***out-of-line-function-declaration*** ::=
*out-of-line-function-prototype* `:`
*statement*$^{+}$
| *out-of-line-function-prototype* `: *DEFERRED*` *foreign-function-name*$^{?}$

***out-of-line-function-prototype*** ::=
`out-of-line function` { *prototype-result-list* `=` }$^{?}$
*out-of-line-function-name* *parenthesized-pattern-argument-list*$^{+}$

***out-of-line-function-name*** ::= *module-abbreviation*$^{?}$ *basic-name*

***foreign-function-name*** ::= *quoted-string*

*prototype-result-list* ::= see p88

*module-abbreviation* ::= see p36

*basic-name* ::= see p36

*parenthesized-pattern-argument-list* ::= see p88

1. The rules for inline *function-declarations* on p87 must be followed where applicable.
2. *Out-of-line-function-declarations* must be top level; they cannot be inside sub-blocks.
3. `macro` and required arguments are not allowed.
4. Wild-cards are not allowed.
5. Functions with *foreign-function-names* are called **foreign** and are `*DEFERRED*` but have no companions. Foreign functions are called using their associated *out-of-line-function-name*.
6. Arguments and results of foreign C and C++ functions cannot have user defined types or user defined pointer types, but may have builtin pointer types (`dp`, `ap`, `bsp`, etc.) pointing at builtin or user defined target types.
7. The rules for inline '`*DEFERRED*`' *function-declarations* and their companions on p94 must be followed for `*DEFERRED*` non-foreign `out-of-line-function-declarations` and their companions.

An out-of-line function can be called with a normal *function-call*$^{p57}$.

Unlike inline functions, an out-of-line function can be called from a statement for which only a `*DEFERRED*` declaration of the out-of-line function is visible. A missing companion declaration is not a compile-time error, but will be a run-time error if the function is actually called at run-time.

Like inline functions, a `*DEFERRED*` non-foreign *out-of-line-function-declaration* can have only one companion. If a `*DEFERRED*` non-foreign *out-of-line-function-declaration* is external,[p118] its companion may be anywhere in the scope of the declaration, including in another module or another module's body that imports the declaration's module. This allows a module to call out-of-line functions defined by a companion in another module that imports the first module.

### 11.6.1 Function Type Declarations

A function type whose values are pointers to out-of-line functions may be declared by:

***function-type-declaration*** ::=
    `type` `function-type-name` `is` *function-type-prototype*
***function-type-name*** ::= *type-name*
*type-name* ::= see p36
***function-type-prototype*** ::=
    `function` { *prototype-result-list* `=` }? `()` *parenthesized-pattern-argument-list*+

1. *Function-type-declarations* must be top level; they cannot be inside subblocks.

Here the *function-type-prototype* is just like an *out-of-line-function-prototype* except that the *out-of-line-function-name* is replaced by `()` and the word '`out-of-line`' is omitted as being superfluous.

The only operations defined on function type values are copying them, comparing them with `==` and `!=`, and calling them. A call to such a value must be a *call-expression*[p113] and must be the right side (after the `=`) of a *call-assignment-statement.*[p63].

A function constant can be declared by:

***function-constant-declaration*** ::=
        *function-type-name function-constant-name* `:`
            *statement*+
    | *function-type-name*
            *function-constant-name* `:` `*DEFERRED*` *foreign-function-name*?
***function-constant-name*** ::= see p36
*foreign-function-name* ::= see p111

1. *Function-constant-declarations* must be top level; they cannot be inside subblocks.

2. The first line of a *function-constant-declaration* behaves like an *out-of-line-function-declaration* first line made by replacing '*function-type-name function-constant-name*' by the *function-type-prototype* specified by the *function-type-name* and:

   (a) adding '`out-of-line`' to the beginning

   (b) replacing the `()` by the *function-constant-name*

   However, a *function-constant-name* is not an *out-of-line-function-name* and cannot be used in a *function-call* to call the function. Instead it is the *variable-name* of a `co` variable that must be used in a *call-expression* to call the function.

The *function-constant-name* is declared as a run-time `co` variable of type named by the *function-type-name* with a value that is a run-time pointer to the declared out-of-line function.

Functions accessed by function type values must be called by:

***call-expression*** ::=
    `call` *function-expression* *parenthesized-call-argument-list*$^{+}$

***function-expression*** ::= *reference-expression*

*reference-expression* ::= see p49

*parenthesized-call-argument-list* ::= see p57

1. The *function-expression* must evaluate to a function-type value.
2. The *parenthesized-call-argument-list* and the function type's *parenthesized-pattern-argument-list* must match as if they were part of an out-of-line *function-call* and *out-of-line-function-prototype*.

## 11.7 Module and Body Declarations

A **module** is a file whose first statement is a *module-declaration*:

***module-declaration*** ::= *simple-module-declaration*
| *simple-module-declaration*`:`
*import-clause*$^{\star}$

***simple-module-declaration*** ::= `module` *module-name* `as` *module-abbreviation*

***module-name*** ::= *quoted-string*

*module-abbreviation* ::= see p36

***import-clause*** ::= `import` *module-name* `as` *module-abbreviation*

1. A *module-declaration* may only appear as the first statement of a module file.
2. In a *module-declaration*, all *module-abbreviations* must be distinct (but the *module-names* need not be).

The compiler maps *module-names* to POSIX file names in an implementation dependent manner. The file that contains the module cannot contain anything else.

The *module-abbreviation* associated with a *module-name* may differ in different files. Specifically, the *module-abbreviation* for a module used in the module's own module file need not be the same as the *module-abbreviations* used for the module in files that import the module.

The module **"standard"** with module abbreviation **std** is builtin and contains the builtin types and functions. The *import-clause*

```
import "standard" as std
```

is implied in every *module-declaration* and *body-declaration*.

A **body** is a file whose first statement is a *body-declaration*:

***body-declaration*** ::= **body** *body-name* **of** *module-name*:
*body-clause*$^\star$

***body-name*** ::= *quoted-string*

*module-name* ::= see p113

***body-clause*** ::= *import-clause* | *after-clause*

*import-clause* ::= see p113

***after-clause*** ::= **initialize after** *body-name*

1. A *body-declaration* may only appear as the first statement of a body file.
2. In a *body-declaration*, the *module-abbreviations* of imported modules must be distinct (but the imported *module-names* need not be). However no imported module may have the *module-abbreviation* or *module-name* of the body's module.

The compiler maps *body-names* to POSIX file names in an implementation dependent manner. The file that contains the body cannot contain anything else.

A **body** is an extension of the module named in the first line of the *body-declaration*.

A body implicitly imports the module it extends. Within the body that module has the same *module-abbreviation* that it had in the module's own file.

The *after-clauses* name other bodies, not necessarily in the same module, and determine the order in which bodies are initialized: see 11.7.1 p114.

If a module or body imports another module, it also implicitly imports all modules imported (explicitly or implicitly) by that other module.

### 11.7.1 Program Initialization

A module is initialized by executing its top level *statements* in the order in which they appear in the module. Similarly a body is initialized by executing its top level *statements* in the

order in which they appear in the body.

The order in which modules and bodies are initialized is determined by the following rules.

1. If a module or body imports another module, the imported module is initialized before the importing module or body.
2. A module is initialized before any of its bodies.
3. If a body contains an *after-clause*, the body is initialized after the body named in the *after-clause*.

The conceptual directed graph whose nodes are modules and bodies and whose arrows connect each module or body to the modules and bodies it must be initialized after is called the '**initialization graph**' and must be acyclic.

# 12 Parser Output

The output produced by the parser when it parses code is as follows. In the following `*LOGICAL-LINE*` and `*INDENTED-PARAGRAPH*` name special constants.

Recall that the input is a sequence of logical lines. Also, *rational-constants*[p40] are not produced by the parser: they are parsed as an operator name *word* followed by a *quoted-string*.

For a logical line, the parser produces a list with the annotations:

```
".initiator" => *LOGICAL-LINE*, ".terminator" => "<LF>"
```

The list elements are strings and numbers representing non-*quoted-string* lexemes, and lists representing subexpressions and *quoted-string* lexemes.

Quoted strings are represented by a list with the string as a single element and a `.type` annotation equal to `"<Q>"` (recall that `<Q>` in a quoted string represents the double quote `"`). Thus the lexeme `"Hello"` becomes:

```
{ "Hello", ".type" => "<Q>" }
```

For an explicitly bracketed subexpression the parser produces a list which has the annotations:

```
".initiator" => "(", ".terminator" => ")"
```

or

```
".initiator" => "[", ".terminator" => "]"
```

The list elements are strings and numbers representing non-*quoted-string* lexemes, and lists representing subexpressions and *quoted-string* lexemes.

For an implicitly bracketed subexpression the parser produces a similar list which has no `.initiator` or `.terminator` annotations.

Recall that an indented paragraph may appear at the end of a logical line.

For an indented paragraph the parser produces a list which has the annotations:
`".initiator" => ":", ".terminator" => *INDENTED-PARAGRAPH*`
The list elements are logical lines.

The parser does not introduce implied brackets where there are explicit or implied brackets, but will introduce implied brackets surrounding quoted strings. For example,

```
x + y parses as { { "x" }, "+", { "y" } }
( x ) + y parses as
          { { "x", ".initiator" => "(", ".terminator" => ")" },
            "+", { "y" } }
( x, "y" ) parses as
           { { "x" }, { { "y", ".type" => "<Q>" } },
             ".initiator" => "(", ".terminator" => ")",
             ".separator" => "," }
```

Operators that are separators, such as '`,`', are not included as elements of a list, but become a **`.separator`** annotation of the list. Thus,

```
( x, y ) parses as { "x", "y",
                     ".separator" => ",",
                     ".initiator" => "(",
                     ".terminator" => ")" }
```

Operator operands become lists in parser output; for example, the statement '`X = Y`' outputs '`{ { "X" }, "=", { "Y" } }`'.

An example is given in Figure 7 p117.

# 13 Scope

A *declaration* has a **scope** that is the set of statements in which any names or prototypes defined by the *declaration* are recognized.

Generally the scope of a *declaration* includes the *statements* in any *block* at the end of the *statement* containing the *declaration* (a *statement* is a logical line which can end in a *block*), and all *statements* following the *statement* containing the *declaration* up to the end of the smallest *block* containing the *declaration*.

The **context** of a statement is the set of declarations whose scope the statement is in.

A **top-level** *declaration* is a *declaration* that is not in a *statement* inside any *block*. The scope of a top-level *declaration* lasts to the end of the file containing the *declaration*. The scope of a top-level *declaration* in a module file is extended to include each body file of the module.

Parser Input:

```
if X < Y:
    X = Y
    Y = Y + 5 * Z
    A 1, B = B, A 1
    const P = "HOHO"
    const Q = 5 + P
    const R = ( 5 + Q )
```

Parser Output:

```
{ "if",
  { { "X" }, "<", { "Y" } },
  { { { "X" }, "=", { "Y" },
      ".initiator" => *LOGICAL-LINE*, ".terminator" => "<LF>" },
    { { "Y" }, "=",
      { { "Y" }, "+", { { 5 }, "*", { "Z" } } },
      ".initiator" => *LOGICAL-LINE*, ".terminator" => "<LF>" },
    { { { "A", 1 }, { "B" }, ".separator" => "," },
      "=",
      { { "B" }, { "A", 1 }, ".separator" => "," },
      ".initiator" => *LOGICAL-LINE*, ".terminator" => "<LF>" },
    { { "const", "P" }, "=", { { "HOHO", ".type" => "<Q>" } },
      ".initiator" => *LOGICAL-LINE*, ".terminator" => "<LF>" },
    { { "const", "Q" }, "=",
      { { 5 }, "+", { "P" } },
      ".initiator" => *LOGICAL-LINE*, ".terminator" => "<LF>" },
    { { "const", "R" }, "="
      { { 5 }, "+", { "Q" },
        ".initiator" => "(", ".terminator" => ")" },
      ".initiator" => *LOGICAL-LINE*, ".terminator" => "<LF>" },
   ".initiator" => ":", ".terminator" => *INDENTED-PARAGRAPH*
  },
  ".initiator" => "*LOGICAL-LINE*", ".terminator" => "<LF>"
}
```

Figure 7: Parser Output Example

Some *declarations* are **external**. These must be top-level declarations in a module (and not in a body), and must contain a *module-abbreviation*. This *module-abbreviation* must name the module containing the top-level *declaration*, except for companions of external deferred *out-of-line-function-declarations* (see below). The scope of an external *declaration* is extended to include all modules and bodies that import, explicitly or implicitly [p114], the module containing the *declaration*.

A *initial-variable-declaration* is external if the variable-name declared begins with a *module-abbreviation*.

A *next-variable-declaration* is external if the variable-name declared begins with a *module-abbreviation*. However, the *next-variable-declaration* must be in the same module file as the *initial-variable-declaration* whose variable name it shares.

A *type-declaration*, *pointer-type-declaration*, or *function-type-declaration* is external if the *type-name*, *pointer-type-name*, or *function-type-name* declared begins with a *module-abbreviation*, except that *type-declarations* or not external if they are expansions of a *type-declaration* appearing in another module.

A body may import two different definitions of a defined type, one from the module containing the root of the type, and one from the body's own module which is an expansion of the first definition. The unexpanded definition is ignored and the expanded definition is used.

Components of a *type-declaration* (fields, subfields, labels) are external if they are compiled when their defined type's external/private switch [p75] is set to external. The *field-type-names* of an external field must be external.

A *function-constant-declaration* is external if the *function-constant-name* declared begins with a *module-abbreviation*.

A *function-declaration* or *out-of-line-function-declaration* is external if the *prototype-pattern* in the *declaration* is immediately preceded by a *module-abbreviation*.

A *reference-function-declaration* containing a *function-variable-name* is external if the *function-variable-name* begins with a *module-abbreviation*. A *reference-function-declaration* containing a *reference-pattern-base* is external if and only if all the *pointer-type-names* and *type-names* in its *reference-pattern-base* are external or are wildcards, and at least one that is external is in the module containing the declaration.

*Prototype-result-declaration* and *prototype-argument-declaration variable-names* cannot begin with a *module-abbreviation*, and therefore these *declarations* can never be external.

However, the prototype argument and result types of an external function or reference function must be external.

The companion of a deferred *declaration* must be in the same file as the deferred *declaration*, except that if the deferred *declaration* is in a module, its companion may be in a body of the module (but not in modules or bodies that merely import the module). If the deferred *declaration* is external and its companion is in a body of its module, files that import the

module are within the scope of the deferred *declaration* but not the scope of its companion. As an exception to the rules just stated, a deferred external *out-of-line-function-declaration* may have its companion anywhere within the scope of the original declaration.

A *type-declaration* that ends with the `***` may have expansions anywhere within the scope of the *type-declaration*.

If two different *declarations* of a *variable-name*, *function-variable-name*, or *function-constant-name* have overlapping scope, one of these must necessarily include the other, and the declaration with the smaller scope is said to '**hide**' the other declaration, unless one *declaration* is deferred and the other is its companion. It is a compile error if one of these declarations hides another.

The same applies to declarations of *defined-type-names*, *pointer-type-names*, and *function-type-names*, except that hiding does not apply to *type-declarations* that expand previous *type-declarations*.

All these names have the property that a name with a *module-abbreviation* is not the same name as that name with the *module-abbreviation* dropped. E.g., '`mom X`' is not the same name as '`X`' if `mom` is a *module-abbreviation*.

A *next-variable-declaration* is allowed within the scope of a previous declaration of its *variable-name*, including a *next-variable-declaration* of this *variable-name*, if it is not within a smaller block than the previous declaration. Note that *next-variable-declarations* can be implied in *block-assignment-statements*[p46] and *loop-assignment-statements*[p69].

Function prototypes cannot hide each other. If the current context contains two function declarations whose prototypes both match a call, the call is ambiguous and in error, even if the scope of one declaration is within the scope of the other.

*Statement-labels* (*block-labels*, *exit-labels*, and *loop-labels*), have as their scope the block in which they are defined. It is a compile error if *statement-labels* hide each other.

When a *function-call* to an inline function is expanded, the context of the expansion is not the current context but is rather the context of the inline *function-declaration* that provided the *statements* executed by the call. Also the context in which any *default-value* expression provided by a *function-declaration* is compiled is the not the current context but is rather the context of that *function-declaration*.

Similarly when a *reference-call* to a reference function is expanded, the context of the expansion is not the current context but is rather the context of the *reference-function-declaration* that provided the *statements* executed by the call.

Code produced by inclusions during *statement* compilation is compiled in the context of the *statement* replaced by the code. See Inclusions (p107).

An example is:

```
module "my_own_module" as mom:
    // `import "standard" as std' is implied
```

```
    import "George's_own_module" as gom
    // gom contains:
    //    function int32 z = gom ( int32 x ) "+" ( int32 y )

int32 mom my constant = 44
int32 my constant = 55

int32 x1 = mom my constant // == 44
int32 x2 = my constant     // == 55

function int32 y = mom my function ( int32 x ): // [A]
    ... function body omitted ...
function int32 y = my function ( int32 x ):     // [B]
    ... function body omitted ...

int32 y1 = mom my function ( 100 )  // Uses [A]
int32 y2 = my function ( 100 )      // Compile error:
                                    // ambiguity between [A] and [B]

function int32 z = my inline function ( int32 x, int32 y ):
    int32 z1 = gom ( x + y )
        // Uses gom's + operator.
        // Compiles as as `gom (x) "+" (y)'.
    int32 z2 = std ( x + y )
        // Uses builtin std's + operator.
        // Compiles as as `std (x) "+" (y)'.
    z  = z1 + z2
        // Compiles as `z = ( (x) "+" (y) )'.
        // Compile error, ambiguous: both std + operator
        // and gom's + operator match the call to "+".
```

More specifically, when a function declaration is used, the *module-abbreviation* beginning the function call may be omitted if the function declaration is the only function declaration within scope that matches the usage, according to Step 1 of the Function Declaration Matching Algorithm, p98. Thus given:

```
function int32 y = mom my external function ( int32 x ):
    ... function body omitted ...
```

the lines:

```
int32 y = mom my external function ( x )
int32 y = my external function ( x )
```

are equivalent if no *function-declarations*

```
function int32 r = ma my external function ( int32 v )
function int32 r = my external function ( int32 v )
```

are in the current context, where *ma* is a module abbreviation for a module other than `"my_own_module"`.

# 14 Lifetimes

The **lifetime** of a variable, i.e., a piece of memory, is the time interval from the time that the variable is allocated to the time that the variable is deallocated. The compiler tracks lifetimes by assigning each variable a lifetime type and for variables in the local stack a separate lifetime depth.

The possible variable lifetimes are:

**global** The variable is stored in the global stack [p133]. The variable lifetime starts when the variable is created and stops when the program terminates.

Global variables are external variables created by a *initial-variable-declaration* or *next-variable-declaration* of an *assignment-statement*, or by an *allocation-call* to the `allocate` allocator function to set a `*GLOBAL*` pointer.

**local** The variable is stored in the local stack [p133]. The variable lifetime starts when the variable is created and stops when the scope of the declaration creating the variable ends.

Local variables are non-external variables created by a *initial-variable-declaration* or *next-variable-declaration* of an *assignment-statement*, or by a *prototype-result-variable-declaration* during a function call, or by an *allocation-call* to the `allocate` allocator function to set an undeclared pointer.

**heap** The variable is stored in the heap (i.e., garbage collectible memory). The variable lifetime starts when the variable is allocated to the heap, and stops when the variable can no longer be referenced by following a chain of pointer values the root of which is a global variable or is a local variable with a lifetime that has not yet terminated.

A pointer value has a pointer type that may have a **lifetime qualifier** which tells something about the lifetimes of the target variables in the datum the pointer points at. The possible lifetime qualifiers are:

| | |
|---|---|
| `*GLOBAL*` | The target variables are in the global stack. |
| `*HEAP*` | The target variables are either in the heap or they are in the global stack.<br><br>Note: allowing `*HEAP*` pointers to point into the global stack requires the garbage collector to be able to distinguish global stack and heap addresses and treat them differently. |
| **(undeclared)** | The pointer type has no lifetime qualifiers. There are two cases: |
| **(local)** | The compiler knows that the target variables are in the local stack, and also knows the lexical depth of the statement that created the variables. |
| **(anywhere)** | The compiler does not know whether the target variables are in the global stack, the local stack, or the heap. However, the compiler does know that the lifetime of the target variables contains the lifetime of the variable holding the pointer value. |

A pointer value whose type has the `*GLOBAL*` lifetime qualifier is called a **global pointer**. A pointer value whose type has the `*HEAP*` lifetime qualifier is called a **heap pointer**. A pointer value with undeclared pointer type that is known to point at target variables in the local stack is called a **local pointer**. A pointer value with undeclared pointer type that is not known to be a local pointer is called an **anywhere pointer**.

Somewhat similarly for variables. A **pointer variable** is a variable storing a pointer. A variable whose type is a pointer type with the `*GLOBAL*` lifetime qualifier is called a **global pointer variable**. A variable whose type is a pointer type with the `*HEAP*` lifetime qualifier is called a **heap pointer variable**. A variable whose type is a pointer type with undeclared lifetime qualifier is called an **undeclared pointer variable**.

We also divide variables into the following three classes:

| | |
|---|---|
| **target variables** | The variable is located in the target datum of a pointer. |
| **local non-target variables** | The variable is stored in the local stack and is not a target variable. |
| **global non-target variables** | The variable is stored in the global stack and is not a target variable. |

The `null` allocator function [p163] is special in that it can set a pointer of any lifetime type to a particular address that addresses a block of virtual memory that is inaccessible, and causes a memory fault when accessed. A pointer allocated by `null` is always a global, heap, or local pointer, depending on whether the pointer's type has a `*GLOBAL*` or `*HEAP*` qualifier or is undeclared, and is never an anywhere pointer. The lifetime assigned to target variables of a pointer allocated by `null` is a convenient fiction, since such variables can never be referenced.

The following rules prevent a pointer to a target with a given lifetime from being stored in a

variable with a longer lifetime. The first of these rules govern global and heap pointers and pointer variables and are:

**(L1)** A global pointer value can be stored in any global pointer variable.

**(L2)** A heap pointer value can be stored in any heap pointer variable.

**(L3)** A global pointer value can be stored in any heap pointer variable.

Note: This requires the garbage collector to be able to distinguish global stack and heap addresses and treat them differently.

**(L4)** A value read from a global pointer variable is a global pointer.

**(L5)** A value read from a heap pointer variable is a heap pointer (has pointer type with `*HEAP*` qualifier), even if in fact it points at the global stack.

**(L6)** Undeclared pointer variables that are in the target of a global pointer are global pointer variables.

Undeclared pointer variables that are in the target of a heap pointer are heap pointer variables.

In other words, a lifetime qualifier of a pointer $P$ is added to the type of any undeclared pointer variables in the target of $P$.

An example is:

```
type list element:
    ap list element @next
        // When in the global stack, @next has *GLOBAL*
        // lifetime qualifer implied by (L6).

// In the following, `@= allocate' allocating the global stack
// is implied because the pointers are *GLOBAL*.

// FIFO implemented as a circular list:
//
ap *GLOBAL* list element @last:  // `@= allocate' is implied
    last.@next = @last
    // Last element is pointed at by last.@next
    // First element is pointed at by last.next.@next
    // Empty list has last.@next == @last

// Put two elements in list.
//
ap *GLOBAL* list element @X  // `@= allocate = *DEFERRED*' is implied
ap *GLOBAL* list element @Y  // `@= allocate = *DEFERRED*' is implied
```

```
ap *GLOBAL* list element @X:
    // Put X on empty list
    X.@next = @X
    last.@next = @X
    // List now consists of (X)
ap *GLOBAL* list element @Y:
    // Add Y to end of list
    Y.@next = last.next.@next
    last.next.@next = @Y
    last.@next = @Y
    // List now consists of (X, Y)

. . . . . .

ap *READ-WRITE* ap *GLOBAL* list element @@Z:
                        // `@= allocate' is implied.
    // Set @Z to beginning of list and delete it.
    // If list empty, set @Z == @last.
    //
    @Z = last.next.@next    // Also works if last.@next == @last
    if last@next == @last:  // List is empty
        // Do nothing
    else if Z.@next == @Z:  // List has only one element
        last.@next = @last
    else:
        last.next.@next = Z.@next
```

A local pointer is tagged automatically by the compiler with the lexical depth of the *statement* that created the pointer. Top level *statements* have lexical depth 1. *Statements* in a subblock of a top level *statement* have lexical depth 2, and so forth. Details, including how to handle inline function calls, are below.

**(L7)** A pointer into the local stack created by an *allocation-call* in an *assignment-statement* of depth $D$ is a local pointer of depth $D$.

**(L8)** A local pointer of depth $D$ may be stored in an undeclared local non-target pointer variable that is not an `out-of-line` function argument variable (it may be an inline function argument variable). The compiler then annotates the pointer variable as holding a local pointer of depth $D$.

**(L9)** A pointer value read from an an undeclared non-target pointer variable that is annotated as holding a local pointer of depth $D$ is a local pointer of depth $D$.

**(L10)** A local pointer of depth $D$ may be stored in an undeclared target pointer variable that is in the target of a local pointer of depth $D$.

**(L11)** A pointer value read from an an undeclared target pointer variable in the target of a local pointer of depth $D$ is a local pointer of depth $D$.

Note that local pointers cannot be stored in global non-target variables or in pointer variables in the target of a global or heap pointer. If a local non-target variable holds a local pointer $P$ of depth $D$, all the undeclared target pointer variables reachable by tracing undeclared target pointer variable values starting at $P$ necessarily hold local pointers of depth $D$. So in a network of data completely connected by local pointers, all these local pointers have the same depth.

An example is:

```
type list element:
    ap list element @next

// In the following, `@= allocate' to the local stack is implied
// because the pointers are (lifetime) undeclared.

// FIFO implemented as a circular list:
//
ap list element @last:  // `@= allocate' is implied
    last.@next = @last
    // Last element is pointed at by last.@next
    // First element is pointed at by last.next.@next
    // Empty list has last.@next == @last

// Put two elements in list.
//
ap list element @X  // `@= allocate = *DEFERRED*' is implied
ap list element @Y  // `@= allocate = *DEFERRED*' is implied

// Note: @last, @X, @Y, last.@next, X.@next, etc.
// all have depth 1

ap list element @X:
    // Put X on empty list
    X.@next = @X
    last.@next = @X
    // List now consists of (X)
ap list element @Y:
    // Add Y to end of list
    Y.@next = last.next.@next
```

```
    last.next.@next = @Y
    last.@next = @Y
    // List now consists of (X, Y)

. . . . . .

ap *READ-WRITE* ap list element @@Z:
                        // `@= allocate' is implied.
    // Set @Z to beginning of list and delete it.
    // If list empty, set @Z == @last.
    //
    @Z = last.next.@next    // Also works if last.@next == @last
    if last@next == @last:  // List is empty
        // Do nothing
    else if Z.@next == @Z:  // List has only one element
        last.@next = @last
    else:
        last.next.@next = Z.@next
```

The last kind of pointer is the **anywhere pointer**. *Out-of-line* undeclared pointer arguments must be anywhere pointers. The rules are:

**(L12)** If a global or heap pointer is stored in an undeclared local non-target pointer variable, the value stored becomes an anywhere pointer, and the compiler annotates the pointer variable as holding an anywhere pointer.

**(L13)** If any pointer is stored in an undeclared *out-of-line* function argument variable, the value stored becomes an anywhere pointer, and the compiler annotates the argument variable as holding an anywhere pointer.

**(L14)** If an anywhere pointer is stored in an undeclared local non-target pointer variable, the value stored becomes an anywhere pointer, and the compiler annotates the pointer variable as holding an anywhere pointer.

**(L15)** A pointer value read from an an undeclared non-target target pointer variable that is annotated as holding an anywhere pointer is an anywhere pointer.

**(L16)** A pointer value read from an an undeclared target pointer variable in the target of an anywhere pointer is an anywhere pointer.

Note that because of (L6) the pointer value may actually point at the global stack or the heap.

Note that because of (L11) the pointer value may actually point at the local stack.

Note that an anywhere pointer can only be stored in local non-target variables, and any undeclared target pointer variable value that is reachable from an anywhere pointer becomes an anywhere pointer when it is read.

The rules for assigning depth to *statements* are as follows:

**(S1)** A *statement* that is not inside any other statement (i.e., is not in a sub-block; i.e., is top-level) is assigned depth 1.

**(S2)** A *statement* that is inside a sub-block $S$ at the end of an *assignment-statement* of depth $D$, and is not also inside a sub-block of $S$, is assigned depth $D + 1$.

**(S3)** An inline *function-call* is treated as creating a sub-block immediately inside the *statement* containing the *function-call*. The argument and result variables created by the inline *function-prototype* are treated as having been created by this *statement*.

Thus if an inline function is called from a *statement* of depth $D$, the *statements* in the sub-block $S$ at the end of the inline function declaration that are not inside a sub-block of $S$ will have depth $D + 1$ and the argument and result variables of the inline function declaration will have depth $D$.

Note: If an inline function is called from a *statement* of depth $D$, the depth $D'$ of the function's declaration must such that $D' \leq D$, since the calling *statement* must be in the scope of the declaration.

Note that because of (S1) and (S2) and the fact that an `out-of-line` function declaration must be top level, the sub-block $S$ at the end of an `out-of-line` declaration is assigned depth 2.

The last example modified to use inline functions is:

```
type list element:
    ap *READ-WRITE* list element @next
        // List elements must be *READ-WRITE*

// In the following, `@= allocate' to the local stack is implied
// because the pointers are (lifetime) undeclared.
```

```
// FIFO implemented as circular list:
//
ap *READ-WRITE* list element @last:  // `@= allocate' is implied
    last.@next = @last
    // Last element is pointed at by last.@next
    // First element is pointed at by last.next.@next
    // Empty list has last.@next == @last

// Put two elements in list.
//
ap *READ-WRITE* list element @X
    // `@= allocate = *DEFERRED*' is implied
ap *READ-WRITE* list element @Y
    // `@= allocate = *DEFERRED*' is implied

// Note: @last, @X, @Y, last.@next, X.@next, etc.
// all have depth 1

// Add element to end of circular list:
//
function add ( ap *READ-WRITE* list element @element ):
    // element depth is taken from actual argument
    if last.@next == @last:
        element.@next = @element
    else:
        element.@next = last.next.@next
        last.next.@next = @element
    last.@next = @element

// Put elements in list.
//
add ( @X )           // X is first element
add ( @Y )           // Y is second element

. . . . . .

// Remove an element from the beginning of the circular
// list and return it
//
function ap *READ-WRITE* list element @r =
        remove ( ap *READ-WRITE* list element @last ):
    @r = last.next.@next    // Also works if last.@next == @last
```

```
        if last@next == @last:  // List is empty
            // Do nothing
        else if r.@next == @r:  // List has only one element
            last.@next = @last
        else:
            last.next.@next = r.@next

    // Take element off list.
    //
    ap *READ-WRITE* list element @Z = remove ( @last )
```

A *reference-expression* with no *reference-calls* that computes a pointer assigns to that pointer the lifetime qualifiers given for the root of the *reference-expression*. E.g., a pointer to a member of the target of a global pointer is a global pointer. When the *reference-expression* does not contain a *reference-call*, this makes sense because a member of the target of a global pointer is contained in the target of the global pointer.

A `point reference function` assigns lifetime qualifiers according to instructions in its *reference-function-prototype*. By using wildcards the prototype may copy root lifetime qualifiers to the result or change these qualifiers.

The garbage collector must be able to find all heap pointer variables in the local and global stacks. Since anywhere pointer variables in the local stack may also point at heap data, the garbage collector must also be able to find them.

# 15 Memory Management

Space for local variables allocated by *variable-declarations* is allocated to the currently executing out-of-line function frame or by calls to ‘`allocate` to the stack after this frame: see p60 and p62 for details.

Pointers to the heap (**heap pointers**) normally use the stub-body concept: the heap pointer points not at the body of a heap datum, but instead at a stub which begins with a body pointer at the body. This can be used in a variety of garbage collection schemes. All these schemes require that some special action be taken when a heap pointer is read into a stack variable, or when a heap pointer is written into a heap datum, or when any location in the stack or heap that stores a heap pointer has its value changed.

However instead of using stub-body, heap pointers can optionally be interpreted as pointing directly at the heap datum.

Specifically, the compiler recognizes the following options:

**`copying-gc`**, **`marking-gc`**, **`counting-gc`**, or **`no-gc`**

These options specify the kind of garbage collection (**GC**) algorithm being used.

With `no-gc`, there is no heap and no garbage collection. The garbage collection algorithms are described below.

**`separate-stub`** or **`integrated-stub`**

Heap pointers always point at a stub. With the `separate-stub` option, heap pointers point at a stub that is separate from the heap datum body, and the first word of the stub points at the body. With the `integrated-stub` option, the body is immediately after the stub, though the stub may begin with a forwarding pointer that points at the true location of both the stub and the body (this is done in `copying-gc`).

**`read-gc`** or **`write-gc`**

These are sub-options of GC, indicating whether the GC is read-oriented or write-oriented. See the descriptions of GC below.

**`deallocation-allowed`** or **`deallocation-forbidden`**

These are sub-options of GC, indicating whether a datum body can be deallocated without the datum being garbage collected. With the `separate-stub` option, the pointer to a deallocated body merely points at inaccessible memory. With the `integrated-stub` option, the `copying-gc` option allows the datum to consist of just a forwarding pointer pointing at inaccessible memory, but other `...-gc` options do not support deallocation without the datum being garbage collected.

GC runs interleaved with non-GC execution. The above compiler options control default inline functions that do the following during non-GC execution:

- Read a non-pointer from a heap datum.
- Read a heap pointer from a heap datum.
- Write a heap pointer to a heap datum.
- Write a non-pointer to a heap datum.
- Write a heap pointer to a non-loop iteration stack variable.
- Write a heap pointer to a loop iteration stack variable.
- Deallocate a location that contains a heap pointer.

The three kinds of GC are discussed in detail in the following sub-sections.

## 15.1 Copying Garbage Collection

In copying garbage collection the stub of a datum is the datum's first word. When the datum is first allocated, this stub points at itself. Then during GC the body will be copied and for

a time have two stubs, the stub in the old body that was the source of the copy, and the stub of the new body that was the destination of the copy. Both will point at the new body, and the new body will hold the datum itself, while the old body will no longer be accessed except for its stub.

GC works in cycles. At the start of each cycle, all heap data are in a contiguous virtual memory space called the old space. A new large contiguous virtual memory space is allocated called the new space. The boundary address between these can be use to tell if a body is in old space or new space: just compare the body pointer with the boundary address. The object of the GC cycle is to copy all active data from old space to new space, update all active heap pointers to point at new space stubs, and then discard old space completely.

GC performs a basic operation on heap pointers which we will call **pointer-update**. In this a heap pointer is checked to see if it points at new space, and if not, is replaced by a pointer that points at new space. If the heap pointer points at a body pointer in old space that itself points at old space, the body pointed at is moved to new space, and the heap pointer is replaced by a pointer to the body in new space. If the heap pointer points at a body pointer in old space that points at new space, the heap pointer is simply replaced by that body pointer.

When a new body is created, it is allocated to new space. Places for new or copied bodies in new space are allocated at the 'end' of new space.

GC goes through new space from beginning to end updating all the pointers in bodies it encounters. This is called 'scavenging'. At any time there is an address that is the boundary between the scavenged bodies at the beginning of new space and the non-scavenged bodies at the end of new space. This address can be use to tell if a new space body has been scavenged.

The root pointers of GC are in the global memory (memory allocated at load time) and the stack.

The GC can be either read-oriented or write-oriented.

A read-oriented GC updates heap pointers when they are read from bodies by non-GC execution. The GC begins by updating all heap pointers in global memory and the stack, and from this point on, all pointers in global memory and the stack point to new space and no special action is required when a pointer is written to a body (which itself will be in new space) by non-GC execution.

A write-oriented GC updates heap pointers when they are written into scavenged bodies by non-GC execution. After all objects in new space have been scavenged, the GC updates all pointers in global memory and the stack, and if this does not move any bodies to new space, GC is done; otherwise GC resumes scavenging. Heap pointers in global memory and the stack need not be updated during non-GC execution, and no special action is taken when reading a heap pointer from a body.

The advantage of write-oriented over read-oriented is that write operations are less frequent

than read operations and therefore write-oriented may be more efficient. The disadvantage is that in order to finish, the GC must update all pointers in the local stack while non-GC execution is changing the local stack. This could be done by having GC update pointers from the bottom to the top of the local stack, and terminate GC if the non-GC execution pops the local stack so it no longer includes non-updated pointers.

It is possible to implement a deallocate operation which deallocates a body, except for its body pointer. A deallocated body has a body pointer pointing at a large area of inaccessible virtual memory, so a memory fault will occur if the body is accessed. To copy a deallocated body one just makes a copy of just the body pointer without changing this body pointer, which is left pointing at inaccessible memory. The update operation must do extra work to detect deallocated bodies if they are permitted.

The `no-stub` option may not be used with copying GC.

## 15.2 Marking Garbage Collection

In marking GC, stubs are allocated to a separate space from bodies, and bodies are not copied during GC. Bodies are copied by a separate activity called compaction that is independent of GC. The advantages are that there is less body copying and also that less memory space is required. Also deallocated bodies do not require the update operation to do extra work.

The simplest marking GC uses stubs that begin with a body pointer followed by two list pointers (for doubly linked lists), a marked flag, and a scavenged flag (the flags can generally be put in the same words as the list pointers). There are two lists of stubs: an old space list and a new space list. To move a stub from old space to new space, it is unlinked from the old space list and linked onto the end of the new space list, and its marked flag is set. When a datum is scavenged, its scavenged flag is set.

Otherwise marking GC is just like copying GC.

A variant has only one list pointer associated with the stub and there is just one stub list. The marked and scavenged flags are used as before. At its end, GC goes through the list of all stubs and frees unmarked stubs along with their bodies. However, the list of stubs to be scavenged must be maintained separately, typically as a list of vectors whose elements point at stubs to be scavenged. When a stub is first marked it is put on the list of stubs to be scavenged.

## 15.3 Counting Garbage Collection

In counting GC each stub has a reference count. The fundamental non-GC operation is storing a pointer P in a location. The required steps are:

(1) save the location's previous value S
(2) add one to the reference count of the stub pointed at by P
(3) subtract one from the reference count of the stub pointed at by S; if that reference count is now zero, collect the stub and its body
(4) store P in the location

This operation must be used when a pointer location is updated in the stack or in a body. There is also an operation for deallocating a location containing a pointer, which omits steps (2) and (4).

Bodies may or may not be allocated separately from stubs. If separate (the `stub` option), deallocated bodies may be implemented and bodies may be compacted separately from GC. Or the `no-stub` option may be used with counting GC.

Of course reference counting GC cannot collect data containing pointer loops, such as circular lists.

## 15.4 Locating Global and Local Heap Pointers

A list of global and local variables that contain heap pointers needs to be made available to the garbage collector. As per the Lifetimes section $^{p121}$, these are pointers of declared `*HEAP*` type plus anywhere pointers.

From the point of view of the garbage collector, variables are held in two stacks: the **global stack** and the **local stack**. These stacks are each a sequence of blocks. Each block has at its beginning a **block type** ID which references load time data that tells the length of the block and contains a map specifying which words of the block contain heap pointers. Some types of blocks contain a vector of similar elements, and for these there is a separate count of the number of vector elements at the beginning of the block and the block type data only specifies that length of an element and which words of the element contain heap pointers.

An out-of-line function call begins by allocating a block in the local stack to hold its local `co` variables. All the variables are allocated at the beginning of the call execution, so this block does not vary in size or layout during the function call execution. Executions of '`@= allocate`' to an undeclared pointer during the function call will add a block to the local stack if the allocated memory size cannot be determined at compile time. At the end of the out-of-line function call, all the blocks it allocated to the local stack are deallocated.

In the case of a statement containing '@= allocate' to an undeclared pointer in a loop iteration, execution of the statement will re-use the block allocated by the last execution of that statement, unless that block is too small, in which case a new block of at least twice the size of the last block will be allocated. This scheme allocates at most 4 times as much memory as is actually used by the statement during any of its executions.

When a block is allocated it is zeroed. This allows the garbage collector to avoid variables containing heap pointers that have not yet been set.

The execution of each module and body file initialization is treated the same as an execution of an out-of-line function call, except that at load time, before initialization, a block holding the external variables of each module is allocated to the global stack. This allows addresses of external variables to be computed at load time.

Statements containing `@= allocate` to a `*GLOBAL*` pointer can be executed anytime and allocate blocks to the global stack.

Blocks in the global stack are never deallocated.

# 16 Non-Function Operators

Many operators map onto *function-calls*. For example,

`x + y` maps onto `x "+" y`

However the following operators do not map onto functions:

prefix **`type`**
prefix **`pointer type`**
right **`function`**
infix **`reference function`**
prefix **`out-of-line function`**
afix infix **`is type`**
afix infix **`is function`**
See Declarations p71.

prefix **`call`**
See Call-Expression p113

prefix **`if`**
prefix **`else if`**
initial **`else`**
afix right **`:`** (the colon operator)
See Conditional Statements p70.

afix subblock
postfix subblock
See Assignment Statements p59 and Conditional Statements p70.

prefix **`loop`**
prefix **`while`**
prefix **`until`**
prefix **`exactly`**
prefix **`at most`**

afix **times**
See *iteration-control*[p67].

infix **=**
See Assignment Statements[p59].

infix **+=**
infix **-=**
infix ***=**
infix **/=**
infix **|=**
infix **&=**
infix **^=**
infix **<<=**
infix **>>=**
The statement '`x += y`' is syntactic sugar for '`next x = x + y`', if `x` is `co` and '`next x`' is defined, or for '`x = x + y`' if `x` is `*READ-WRITE*`.

Similarly for the other operators of the form '*B*`=`' where *B* is a binary operator, the statement '`x` *B*`= y`' is syntactic sugar for '`next x = x` *B* `y`' or '`x = x` *B* `y`'.

infix **@=**
See Allocation Call[p45].

infix **--->**
See Abbreviation Statements[p38].

nofix **,** (comma)
Becomes a `.separator` annotation on a list. See the operator separator format[p32].

infix **if**
infix afix **else**
Must be used in an *else-expression* with target type *T* which has the syntax:

***else-expression*** ::= *if-expression* { **else** *if-expression* }$^\star$ **else** *T-expression*
***if-expression*** ::= *T-expression* **if** *bool-expression*
*bool-expression* ::= *expression* with target type `bool` if *T* is not `const` and otherwise target type `const`
*T-expression* ::= *expression* with target type *T*

The *bool-expressions* are evaluated left to right until one evaluates to `true` or `TRUE`. Then the corresponding *T-expression* (the one in the same *if-expression* as the *bool-expression*) is evaluated and returned. If all *bool-expressions* evaluate to `false` or `FALSE`, the *T-expression* after the last `else` is evaluated and returned.

infix **BUT NOT**
'`x BUT NOT y`' is syntactic sugar for '`x AND ( NOT ( y ) )`'.

infix **AND**
: ‘`x AND y`’ is syntactic sugar for ‘`y if x else FALSE`’.

infix **OR**
: ‘`x OR y`’ is syntactic sugar for ‘`TRUE if x else y`’.

prefix **NOT**
: ‘`NOT x`’ is syntactic sugar for ‘`FALSE if x else TRUE`’.

# 17 Builtin Abbreviations

*Module-abbreviations* are automatically deduced for function calls, but not for data types, pointer types, or variables used as global constants, such as **`true`** or **`false`**. In order to avoid having to input a *module-abbreviation* with every use of these names, *abbreviation-statements* are used (p38). The following are the builtin abbreviations:

```
                int  --->  std int
               int8  --->  std int8
              int16  --->  std int16
              int32  --->  std int32
              int64  --->  std int64
             int128  --->  std int128

                uns  --->  std uns
               uns8  --->  std uns8
              uns16  --->  std uns16
              uns32  --->  std uns32
              uns64  --->  std uns64
             uns128  --->  std uns128

               bool  --->  std bool
               true  --->  std true
              false  --->  std false

               TRUE  --->  std TRUE
              FALSE  --->  std FALSE
               NONE  --->  std NONE
              UNDEF  --->  std UNDEF
      LOGICAL-LINE*  --->  std *LOGICAL-LINE*
INDENTED-PARAGRAPH*  --->  std *INDENTED-PARAGRAPH*
```

```
  flt  --->  std flt
 flt8  --->  std flt8
flt16  --->  std flt16
flt32  --->  std flt32
flt64  --->  std flt64

   dp  --->  std dp
   ap  --->  std ap
  bsp  --->  std bsp
   av  --->  std av
  bsv  --->  std bsv
```

# 18 Constant Functions and Compiler Constants

Constant functions have `const` results and do not produce run-time code. [p92] The functions described in the following sections are builtin constant functions.

Some compilation related functions and constants are in the `"compiler"` module, which is abbreviated here as '`com`'.

Unless stated otherwise, builtin constant functions obey the following rules:

1. Boolean values are represented by the special constants `TRUE` and `FALSE`.
2. Errors in arguments, such as passing a `string` when a `number` or `rational` is required, result in the function doing nothing but returning the `UNDEF` special constant and producing a compiler error message.
3. If a result or argument is said to be an integer, it may be either a `number` with an integral value, or a `rational` with denominator `1`.
4. A **`number`** is a `const` value representable by a *number-constant.* [p40].

   A function (e.g., `"+"` or `"=="`) with at least one non-integer `number` argument will convert all `rational` arguments to `numbers` before using them, will do all internal calculations with `numbers` and not `rationals`, and will return any numeric results as `numbers`. If there is an error `NaN` will be returned and there will be a compile error message.

   A function with at least one `rational` argument and no non-integer *number* arguments will convert all integer `numbers` to `rationals` before using them, will do all internal calculations with `rationals`, and will return any numeric results as `rationals`. If there is an error `UNDEF` will be returned and there will be a compile error message.
5. `Number` values too positive or negative to store are converted to `+Inf` or `-Inf`. `Number` values too small to store are converted to `+0` or `-0`, preserving the sign of the value. There will be no compile error message if these conversions are necessary.

6. Arguments out of range (e.g., a non-integer where an integer is required) produce an `UNDEF` result of `rationals` or a `NaN` result for `numbers` plus a compile error message.

## 18.1 Compile Time General Functions

```
constant function const r = std (const v1) "==" ( const v2 )
constant function const r = std (const v1) "!=" ( const v2 )
```

Comparisons of `const` values of different types treat the values as unequal, except in the case that one argument is a `number` and the other a `rational`.

```
constant function const r = std type ( const v1 )
```

Returns the type of `v1` as one of the strings:

```
"special"   "number"   "rational"   "string"   "map"
```

## 18.2 Compile Time Numeric Functions

```
constant function const r = std number ( const v1 )
constant function const r = std rational ( const v1 )
```

These convert their argument to a number or rational. If the argument is a string, it must have the format of a number or rational constant (rational operator followed by quoted string). If the argument is a map, the map must have two elements that are strings, with the first being a rational operator and the second being the quoted string it operates on. Map annotations are ignored, so the map may be ` ' quoted. E.g.,

```
`B# "1.1"'    { "B#", "1.1" }
```

A conversion error produces an `UNDEF` result and an error message. Note that finite numbers can always be converted to rationals, and rationals can always be converted to numbers, though these may be `+Inf` or `-Inf`.

```
constant function const r = std "+" ( const n1 )
constant function const r = std "-" ( const n1 )
constant function const r = std (const n1) "+" ( const n2 )
constant function const r = std (const n1) "-" ( const n2 )
constant function const r = std (const n1) "*" ( const n2 )
constant function const r = std (const n1) "/" ( const n2 )
```

Standard arithmetic operators on numbers or rationals `n1` and `n2`, done using IEEE number or rational arithmetic. For numbers, dividing by `0`, adding `+Inf` to `-Inf`, a `NaN` argument, etc. return `NaN` and no compiler error message. For rationals, dividing by `0` returns `UNDEF` and outputs a compiler error message.

```
constant function const r = std (const i1) "&" ( const i2 )
constant function const r = std (const i1) "|" ( const i2 )
```

```
constant function const r = std (const i1) "^" ( const i2 )
constant function const r = std (const i1) "<<" ( const i2 )
constant function const r = std (const i1) ">>" ( const i2 )
```

Standard bitwise operators on integers `i1` and `i2` that are treated as two's complement. For the shift operators `"<<"` and `">>"`, `i2`, the amount of the shift, must not be negative.

```
constant function const r = std (const n1) "==" ( const n2 )
constant function const r = std (const n1) "!=" ( const n2 )
constant function const r = std (const n1) "<" ( const n2 )
constant function const r = std (const n1) "<=" ( const n2 )
constant function const r = std (const n1) ">" ( const n2 )
constant function const r = std (const n1) ">=" ( const n2 )
```

Standard comparison operators on numbers or rationals `n1` and `n2`. Infinities are treated as actual numbers with absolute value larger than any real number: e.g., if `x` is not a `NaN`, '`x <= +Inf`' is always `TRUE` and '`x == +Inf`' is `TRUE` iff `x` is `+Inf`. If an argument is a `NaN`, all comparisons return `FALSE` except `!=` which returns `TRUE`, and there is a compiler error message.

```
constant function const r1, const r2 = std floor
      (const n1, const n2 ?= 1 )
constant function const r1, const r2 = std ceiling
      (const n1, const n2 ?= 1 )
constant function const r1, const r2 = std truncate
      (const n1, const n2 ?= 1 )
constant function const r1, const r2 = std round
      (const n1, const n2 ?= 1 )
```

These divide `n1` by `n2` and return `r1` as the result rounded to an integer and `r2` as the remainder. Here `floor` rounds toward negative infinity, `ceiling` rounds towards positive infinity, `truncate` rounds toward zero, and `round` rounds to the nearest integer, or to the even integer if there are two nearest integers.

Both `number` and `rational` computation is supported. Dividing by zero is produces `NaN`s or `UNDEF`s and an error message.

```
constant function const r = std numerator ( const r1 )
constant function const r = std denominator ( const r1 )
```

These functions return the numerator and denominator of `r1`, which must be a rational. Both numerator and denominator are integer `rationals`.

```
constant function const r = std is nan ( const v1 )
constant function const r = std is infinite ( const v1 )
constant function const r = std is finite ( const v1 )
constant function const r = std is rational ( const v1 )
constant function const r = std is numeric ( const v1 )
```

Return `TRUE` if `v1` is a NaN `number` (`is nan`), is a `+Inf` or `-Inf` `number` (`is infinite`), is a `number` that is neither of these (`is finite`), is a `rational` (`is rational`), or is either a `number` or a `rational` (`is numeric`), and `FALSE` otherwise. Returns `FALSE` if `v1` is neither a `number` or a `rational`.

## 18.3 Compile Time String Functions

```
constant function const r = std "#" ( const s )
```

Returns the length of string `s` as a non-negative integer `number`.[10] Note that `#` is a prefix operator.

```
constant function const r = std ( const s1 ) "+" ( const s2 )
```

Returns the concatenation of string `s1` and string `s2`.

```
constant function const r = std sprintf ( const format, const a1 = "",
                                          const a2 = "", const a3 = "",
                                          const a4 = "", const a5 = "" )
```

Returns the string made by calling the UNIX `sprintf` function as per:

```
sprintf ( format, a1, a2, a3, a4, a5 )
```

where `format` is a string. Not all of the data arguments `a1`, `a2`, `a3`, `a4`, and `a5` need be used by the `format`. Data arguments may be numbers or strings. Rational data arguments are converted to numbers first. Map data arguments are not allowed.

```
constant function const r = std (const s1) "==" ( const s2 )
constant function const r = std (const s1) "!=" ( const s2 )
constant function const r = std (const s1) "<" ( const s2 )
constant function const r = std (const s1) "<=" ( const s2 )
constant function const r = std (const s1) ">" ( const s2 )
constant function const r = std (const s1) ">=" ( const s2 )
```

Standard lexigraphic comparison operators on strings `s1` and `s2`. Characters are compared by comparing their `UTF-8` representations as strings of unsigned 8-bit bytes. With exceptions for unnormalized `UTF-8` encodings,[11] this is equivalent to comparing the characters by comparing their UNICODE codes as unsigned 32-bit integers.

```
constant function const r = std explode ( const s )
```

Returns a map that is a vector whose elements are unsigned integer `numbers` equal to the UNICODE codes of the characters of string `s`.

```
constant function const r = std implode ( const m )
```

Given a map `m` that is a vector whose elements are unsigned integer `numbers` that are

[10] The length of a `string` cannot be above $2^{48}$ while `numbers` can precisely store integers up to $2^{53}$.

[11] An unnormalized `UTF-8` encoding for a character is one taking more bytes than necessary. For example, `NUL` with UNICODE code `0`, can be encoded in 1-byte if normalized, or in 2-, 3-, or 4- bytes unnormalized.

UNICODE codes of characters, return the string whose characters are those specified by the map elements in the order specified by the map.

`constant function const r = std` **`compile re`** ( const s )

Compile the regular expression represented by the string `s` and return an integer that references the compiled expression.

Regular expressions are those recognized by the `pcre32` subroutine library for linux: see `pcrepattern[3]` in the linux documentation. The only line ends recognized by `\R` and `$` are LF, CR, and CRLF (no other `pcre32` options are used). By default `^` matches the beginning of the string being matched and `$` matches the end. This can be changed by the `(?i)` option setter in the regular expression.

`function` **`free re`** ( const i )

Free the memory used by the compiled regular expression referenced by the integer `i`. Does nothing if `i` does not reference a compiled regular expression.

`constant function const r = std` **`match re`** ( const i, const s )

Matches the string `s` to the compiled regular expression referenced by the integer `i`. Returns a map `r` that is a vector of substrings matched. If there is no match this is an empty list. If there is a match, `r[0]` is the string matched. If there are subpattern matches, `r[i]` is the string matched by the `i`'th subpattern.

During matching `s` is stored as an exploded vector of unsigned 32-bit unicode values. Matched strings are subvectors which are imploded to make `const` string values.

`constant function const r = std` **`scan`** ( const s )

Scan the string `s` and return a map that is a vector containing the list of lexemes in `s`. Brackets and operators are not specially recognized and are returned as strings. Quoted strings inside `s` are returned as vector elements that are maps of the form:

`{ "`*represented-string*`", ".type" => "<Q>" }`

Syntax errors produce compiler error messages and return `UNDEF`.

`constant function const r = std` **`parse brackets`** ( const s )

Equivalent to '*value-of-*`s`'. See *phrase-constants*: p43.

Specifically, parse the string `s` recognizing brackets but not recognizing operators and return a map. Syntax errors produce compiler error messages and return `UNDEF`.

`constant function const r = std` **`unparse brackets`** ( const v, const f ?= "%0.16g" )

Inverse of `parse brackets`. Return a string $S$ such that '$S$' equals `v` with numeric rounding differences. Numbers are printed in $S$ using `f` as a printf format.

`constant function const r = std` **`parse`** ( const s )

Equivalent to `{*` *value-of-*`s *}`. See *expression-constants*: p43.

Specifically, parse the string `s` and return a map. Syntax errors produce compiler error messages and return `UNDEF`.

`constant function const r = std` **`unparse`** ( const v, const f ?= "%0.16g" )

Inverse of `parse`. Return a string $S$ such that `{* "`$S$`" *}` equals `v` with numeric rounding differences. Numbers are printed in $S$ using `f` as a printf format.

## 18.4 Compile Time Map Functions

A `const` map value is actually a pointer to the map, and not the whole map itself. A *map-constant* creates a new map, distinct from every other map (so you can have multiple different empty maps).

A map may be read-write or read-only. Read-only maps cannot be modified. Each *map-constant* makes a separate read-only map that can be made read-write permanently or temporarily by the following:

`constant function const r = std` **`read-write`** `( const m )`
`constant function const r = std` **`read-only`** `( const m )`

Makes the map `m` read-write or read-only and returns `m`.

When the map is created by a *map-constant*, it is made read-only.

Each map *dictionary-entry* can be separately made read-write or read-only. When created, the entry is read-only. This can be changed by the following:

`constant function const r = std` **`read-write`** `( const m, const s )`
`constant function const r = std` **`read-only`** `( const m, const s )`

Makes the map label `s` (a string) of the map `m` read-write or read-only, and returns `m`. The dictionary entry labeled `s` in the map `m` can be written if it is read-write and the whole map is separately read-write.

When a value is first written at a label, the label is created for the map and set to read-only.

In addition, the compiler will mark some map labels as **protected**. Such entries are read-only to the code being compiled, and may either be permanently read-only or may be written only by the compiler during compilation.

`constant function const r = std "#" ( const m )`

Returns the length of the vector part of the map `m` as a non-negative integer `number`.[12] Note that `#` is a prefix operator.

`constant function const r = std` **`labels`** `( const m )`

Returns a list of the *dictionary-labels* (see p42) of map `m`. The *dictionary-labels* are

[12] The length of a `map` cannot be above $2^{48}$ while `numbers` can precisely store integers up to $2^{53}$.

strings. The list may be empty.

```
load reference function const r = ( const m ) [ const s ?= NONE ]
store reference function ( const m ) [ const s ?= NONE ] = const v
```

Here `m` is map and `m[s]` is used to reference a vector element or dictionary entry of `m` as follows:

1. If the value of `s` is a non-negative non-rational integer number, the `s`+1'st element of the vector of `m` is referenced. If this is being read and it does not exist, `NONE` is returned and there is no compile error. If the element is being written but does not exist, or the map is read-only, a compile error results; otherwise the element value is changed.
2. If the value of `s` is a negative non-rational integer, `s` is replaced by `# m + s`, and things are as in the last paragraph (note `# m + s < # m`). The element does not exist if `# m + s < 0`.
3. If the value of `s` is a string, the dictionary entry of `m` with label `s` is referenced. If this is being read and it does not exist, `NONE` is returned. If it is being written and it does not exist, and if the map is read-write, the entry is created and made read-only. Else if the entry exists and is read-write, and the map is read-write, the entry value is changed. Else if the entry exists and is read-only, or the map is read-only, a compile error results.
4. If the value of `s` is `NONE`, reading pops a value from the end of the vector of `m` and returns the value popped (as in '`v = m[]`'), or just returns `NONE` if the vector is empty, and writing pushes the value written to the end of the vector of `m` (as in '`m[] = v`').
5. Otherwise if `s` is neither a non-rational integer or a string, a compile-error results.

```
constant function const r = std copy ( const m )
constant function const r = std copy top ( const m )
```

Returns a new map whose contents is a copy of the contents of map `m`. The new map is read-only if and only if `m` is, and the labels in the new map are read-only if and only if the corresponding labels in `m` are.

If any vector or dictionary element values are maps, they are copied recursively by `copy`, but not by `copy top`, which only copies the element values of `m` and does not copy recursively.

Changing the values of elements of the new map will not change the contents of `m`. For `copy` modifying the element values that are maps will not change `m`, but for `copy top`, modifying these element values will change the elements of `m`.

```
constant function const r = std duplicate ( const m )
constant function const r = std duplicate top ( const m )
```

Ditto, but if any read-only map is to be copied, the pointer to the map is copied and no new map is made.

`constant function const r = std slice ( const m, const i, const n )`

`i` and `n` must be non-rational integers.

If `i < 0` then `i` is replaced by `# m + i` before the following is done.

Then if `n > 0`, returns a new map consisting of just a vector of the elements `m[i]`, `m[i+1]`, ..., `m[i+n-1]`. If any of these elements do not exist, they are omitted (e.g., if `i >= # m` or `i + n <= 0` the empty map is returned).

If `n == 0` the empty map is returned.

If `n < 0` the elements returned are `m[i+n]`, `m[i+n+1]`, ..., `m[i-1]`.

`constant function const r = std splice ( const m, const i, const n, const v )`

`i` and `n` must be non-rational integers.

If `i < 0` then `i` is replaced by `# m + i` before the following is done.

Then if `n > 0`, edits `m` by replacing the vector element sequence `m[i]`, `m[i+1]`, ..., `m[i+n-1]` by the vector elements of `v`. Dictionary elements of `v` are ignored; dictionary elements of `m` are unchanged.

If `n == 0` the elements of `v` are pushed into `m` just after `m[i]`.

If `n < 0` the elements replaced are `m[i+n]`, `m[i+n+1]`, ..., `m[i-1]`.

The elements being replaced need not (all) exist. `m` is thought of as an infinite sequence of existing and non-existing elements indexed by integers, including negative integers. The members of `m` to be removed are made non-existent, then the elements of `v` are pushed into the sequence, and then non-existent elements are deleted.

So if `i >= # m` and `n >= 0` then the elements of `v` are appended to `m`, and if `i = 0` and `n < 0`, the elements of `v` are prepended to `m`.

The edited map `m` is returned.

`constant function const r = std truncate ( const m, const i )`

An optimized version of `splice` that removes the elements `m[i]`, `m[i+1]`, ..., from the end of the vector of `v`. The edited map `m` is returned. It is not an error if no elements are removed.

If `i` $< 0$ it is replaced by `# m + i`. If `# m + i < 0`, all elements of the vector are removed.

`constant function const r = std push ( const m, const v )`

Appends `v` to the vector of `m` and returns `m`.

`constant function const r = std push ( const m, const v, const i )`

Executes `push(m,v)` `i` times. `i` must be a non-rational, non-negative integer.

`constant function const r = std append ( const m1, const m2 )`

Appends the vector elements of the map `m2` to the vector of `m1` and returns `m1`.

`constant function const r = std pop ( const m )`

Deletes the last vector element of `m` and returns it. Returns `NONE` if `m` is empty.

`constant function const r = std pop ( const m, const i )`

Deletes the last `i` vector elements of `m` and returns a map containing them in the same order. If there are fewer than `i` vector elements in `m`, the returned vector will have only `# m` elements. `i` must be a non-rational, non-negative integer.

`constant function const r = std ( const m1 ) "+" ( const m2 )`

Concatenates the maps `m1` and `m2`. Specifically, `r` has the vector elements of `m1` followed by the vector elements of `m2`, and the dictionary elements of both `m1` and `m2`. It is a compile error if `m1` and `m2` have conflicting dictionary elements, that is, there is a label such that each map has a dictionary element with that label but the values of these two elements are not `==`.

## 18.5 Type, Field, Subfield, and Pointer Type Maps

Types, fields, subfields, and pointer types are described at compile-time by `const` map values which user code can access. These are read-write as a whole, but some of their dictionary entry labels are protected. The following sections describe protected labels provided by the compiler. Unless otherwise specified, these have values that do not change during compilation.

Compiled code may add its own labels to these maps. To prevent conflict, the labels provided by the compiler begin with '.', so that to use them to access a map dictionary entry you must use double dots: '..'. E.g., `int..size`.

In the following a **name string** is a string consisting of a sequence of one or more *words* and *natural-numbers*, separated by single spaces, and beginning with a *word*. *Natural-numbers* are represented by strings of 1 to 15 decimal digits with no high-order zeros (zero is represented by '0'). Lexical *names*$^{p35}$ not containing quoted words, marks, or separators can be represented naturally by name strings. Name strings are used in this section to represent type, field, and subfield names and origin labels.

*Module abbreviations* in a name string are replaced by **compiler module abbreviations** which are words of the form `M$`$n$, where $n$ is a natural number. `M$0` is always the abbreviation for the `std` module. These compiler module abbreviations are specific to the entire compilation and are not dependent on which module or body a definition appears in. Two code *module-abbreviations* designating the same module will be replaced by the same compiler module abbreviation.

Abbrevations (e.g., `int ---> std int`) do not apply to name strings. Thus `"int"` is not used; `"M$0 int"` is used instead.

`com module dictionary`

A dictionary mapping compiler module abbreviations to strings that are *module-names*. Specifically: if `M$`$n$ maps to $N$ then both `com module dictionary[`$n$`] == "`$N$`"` and `com module dictionary["M$`$n$`"] == "`$N$`"`. For example,

```
com module dictionary[0] == "standard"
com module dictionary["M$0"] == "standard"
```

### 18.5.1 Number Type Maps

At compile-time a *number-type-name* can be used as a *variable-name* that names a `co const` variable with a map value called a **number type map**.

The compiler defined attributes of a number type map are:

**`.kind => "number"`**

**`.name`**

The name of the type as a name string.

**`.size`**
**`.alignment`**

The `.size` is the number of bits taken by a value of the given type at run-time. The `.alignment` is the alignment in bits of an aligned value of the given type at run-time. E.g., `int64..size == 64, int64..alignment == 64`.

**`.min`**
**`.max`**

The minimum and maximum values of a number type $N$. These are `const` values which must be converted to the run-time type $N$ before being used (e.g., by the expression '$N$($N$..max)'). These values are exact; for some $N$, these values can be *number-constants*, but for others they must be *rational-constants* (e.g., `int64..max` is `X#"7FFFFFFFFFFFFFFF"`).

**`.indefinite`**

The value of the indefinite integer for a signed integer type $I$. Not defined for non-integer and unsigned integer types.

The **indefinite integer** is returned by some operations, e.g. `round`, when they cannot return a correct integer value.[13]

The value is typically $-2^{S-1}$ where $S$ is the size of $I$ in bits.

[13] See documentation of the intel IA-64 FIST instruction.

This is a `const` value which must be converted to the run-time type $I$ before being used (e.g., by the expression '$I$($I$..indefinite)'). This value is exact; for some $I$, the values can be a *number-constant*, but for others it must be a *rational-constant* (e.g., `int64..indefinite` is `-B#"1e63"`).

Example:

```
int64 = { .kind => "number",
          .name => "M$0 int64",
          .size => 64,
          .alignment => 64,
          .min => -X#"8000000000000000",
          .max => X#"7FFFFFFFFFFFFFFF",
          .indefinite => -X#"8000000000000000" }
```

### 18.5.2 Pointer Type Maps

At compile-time a *pointer-type-name* can be used as a *variable-name* that names a `co const` variable with a map value called a **pointer type map**.

The compiler defined attributes of a pointer type map are:

**`.kind => "pointer"`**

**`.name`**

The name of the pointer type as a name string.

**`.data type`**

The type map of the data type of the pointer type.

Example:

```
ap = { .kind => "pointer",
       .name => "M$0 ap",
       .data type => M$0 ap data }

M$0 ap data = { .kind => "defined type",
                .name => "M$0 ap data",
                .size => 128,   // Assumes int..size == 64
                .alignment => 64,
                .expandable => FALSE,
                .final => TRUE,
                .components => {
                    base => { .kind => "field",
                              .name => "base",
                              .parent => M$0 ap data,
```

```
                                    // Note absence of quotes
                                .offset => 0,
                                .field type => int,
                                    // Note absence of quotes
                                .qualifiers => {},
                                .dimensions => {},
                                .subfields => {} },
                        offset => { .kind => "field",
                                    .name => "offset",
                                    .parent => M$0 ap data,
                                        // Note absence of quotes
                                    .offset => 64,
                                    .field type => int,
                                        // Note absence of quotes
                                    .qualifiers => {},
                                    .dimensions => {},
                                    .subfields => {} } } }
```

### 18.5.3 Function Type Maps

At compile-time a *function-type-name* can be used as a *variable-name* that names a `co const` variable with a map value called a **function type map**.

The compiler defined attributes of a function type map are:

**`.kind => "function"`**

**`.name`**

The name of the function type as a name string.

**`.size`**
**`.alignment`**

Same as for number type maps: see p146

### 18.5.4 Pointer Value Type Maps

A **pointer value type** is a type of a pointer value. Textually it is:

{ *pointer-type-name qualifier-name*$^\star$ }$^+$
{ *number-type-name* | *function-type-name* | *defined-type-name* | *pointer-type-name* }

**`.kind => "pointer value"`**

**`.pointer type`**

The pointer type map of the first *pointer-type-name* of the pointer value type as.

**.target qualifiers**

The first list of *qualifiers* of the pointer value type. List of strings, each a *word* naming a qualifier. May be empty list.

**.target type**

The remainder of the pointer value type, after the initial

*pointer-type-name qualifier-name*$^\star$

This is a type map (for number, function, pointer, defined type, pointer value).

### 18.5.5 Defined Type Maps

At compile-time a *defined-type-name* can be used as a *variable-name* that names a `co const` variable with a map value called a **defined type map**.

The compiler defined attributes of a defined type map are:

**.kind => "defined type"**

**.name**

The *defined-type-name* of the defined type as a name string.

**.size**
**.alignment**

Same as for number type maps: see p146

**.expandable**

The `.expandable` attribute is `TRUE` if the defined type definition allows expansions, and is `FALSE` otherwise. An expansion not ending in `***` will set this `FALSE`.

**.final**

The `.final` attribute is set `TRUE` by a *type-declaration* for this defined type that has no ending `***` and that is in the same module or body as the root of the defined type. Otherwise the attribute is `FALSE`.

This must be `TRUE` if the *defined-type-name* is to be used as a field name or `*INCLUDE*` name.

**.components**

A dictionary map mapping the *data-labels* of the defined type components (fields, subfields, labels) to their definitions (field maps, subfield maps, and label maps).

Components may be added during compilation of type expansions. This map be empty if a type has no components, or if the type is currently `*DEFERRED*` and not yet defined by the compilation.

Private components are omitted in the version of this map that is imported.

### 18.5.6 Field Maps

Each field of a type has a `const` map value called a **field map** which is in the `.fields` dictionary of a type map. The compiler defined attributes of a field map are:

**`.kind => "field"`**

**`.name`**

The name of the field (*field-label*) as a name string. May be `NONE` for a field with subfields.

**`.external`**

The `.external` attribute is `TRUE` if the field is external and `FALSE` if it is private.

**`.parent`**

The defined type map of the type of containing this field.

**`.offset`**

Offset in bits of the field within a value of its parent type.

**`.field type`**

The type of the field. May be a number type map, function type map, pointer type map, defined type map, or pointer value type map.

**`.qualifiers`**

List of strings, each a *word* naming a qualifier of the field. May be empty list.

**`.dimensions`**

List of strictly positive integers, the dimensions of the field, or `NONE` if no dimensions.

**`.subfields`**

Dictionary of subfield maps for the subfields of the type. The labels of the dictionary entries are the names of the subfields as name strings. Empty if there are no subfields.

Subfields are also entries in the parent's `.components`.

Example:

```
type my type:
    int32 X[4,3]
    av *READ-WRITE* flt @Y

// The value of the global const variable `my type' is:
{ .kind => "defined type",
  .name => "my type",
```

```
.size => 640,
.alignment => 64,
.expandable => FALSE,
.final => TRUE,
.components => {
    "X" => { .kind => "field",
             .name => "X",
             .parent => my type, // Note absence of quotes
             .offset => 0,
             .qualifiers => {},
             .field type => int32, // Note absence of quotes
             .dimensions => {4, 3},
             .subfields => {}
           },
    "@Y" => { .kind => "field",
              .name => "@Y",
              .parent => my type,  // Note absence of quotes
              .offset => 384,      // 12 * 32
              .qualifiers => {},
              .field type =>
                  { .kind => "pointer value",
                    .pointer type => av
                        // Note absence of quotes
                    .target qualifiers => { "*READ-WRITE*" },
                    .target type => flt
                        // Note absence of quotes
                  },
              .dimensions => {},
              .subfields => {}
            } } }
```

### 18.5.7 Subfield Maps

Each subfield of a field has a `const` map value called a **subfield map** which is in the `.subfields` dictionary of a field map. The compiler defined attributes of a subfield map are:

**`.kind`** `=> "subfield"`

**`.name`**

The name of the subfield (*non-pointer-label*) as a name string.

**.external**

The .external attribute is TRUE if the subfield is external and FALSE if it is private.

**.parent**

Field map of the field of containing this subfield.

**.bits**

A list of two integers: {*highbit*,*lowbit*}.

**.subfield type**

Type map for the type of the subfield. This is always a std number type or std bool.

**.dimensions**

List of strictly positive integers, the dimensions of the subfield, or NONE if no dimensions.

### 18.5.8 Label Map

Each *LABEL* *type-subdeclaration* has a const map value called a **label map** which is in the .components directory of its parent defined type map. The compiler attributes of a label map are:

**.kind => "label"**

**.name**

The name of the label (*origin-label*) as a name string.

**.external**

The .external attribute is TRUE if the label is external and FALSE if it is private.

**.parent**

The defined type map of the type of containing this label.

**.offset**

Offset in bits assigned to this label.

## 18.6 Compile Time Compilation Functions

The following functions can be called by any inline function:

```
constant function com compile error ( const message )
constant function com compile errorf ( const format, const a1 = "",
                                       const a2 = "", const a3 = "",
                                       const a4 = "", const a5 = "" )
```

This function outputs a compiler error message. The expression calling the inline

function that executes a call to this function is also identified in the message.

In the first form `message` is a string. The second form calls `sprintf` [p140] on the arguments to produce the message string.

`load reference function const r = com new variable name`

This function returns a *variable-name* of the form `"V$..."` that is unique and distinct from any other variable name in the current compilation. In particular, the function does not return the same *variable-name* twice.

The following functions are can be called by `macro` and/or `inclusion` functions:

`load reference function const r = com *STATEMENT*`

This *function-variable-name* returns the parsed *statement* containing the `inclusion` function call currently executing at compile-time.

`load reference function const r = com *SUBBLOCK*`

This *function-variable-name* returns the subblock at the end of the parsed *statement* containing the `inclusion` function call currently executing at compile-time. This is annotated with `".initiator" = ":"` and `".terminator" = *INDENTED-PARAGRAPH*`.

If instead of ending with a subblock, the *statement* ends with the `":"` operator followed by a *restricted-statement* $S$, this function returns:

`{` $S$`, ".initiator" = ":", ".terminator" = *INDENTED-PARAGRAPH* }`

In other words, this function packages $S$ in the form of a 1-*statement* subblock.

`constant function const r = com name of ( const variable name )`

Here '`variable name`' is a *variable name* declared in a non-macro *prototype-argument-declaration* of a `macro` or `inclusion` *function-declaration* $D$, and the `name of` function is called by $D$. The type of the variable in this *prototype-argument-declaration* cannot be `const`.

This function returns a *variable-name* of the form `V$...` created by calling `com new variable name` that can be used inside run-time code produced by the macro or inclusion function to reference the value of the actual argument associated with the *prototype-argument-declaration* of the '`variable name`'. More explicitly, if the *prototype-argument-declaration* has the form '$TTT\ V$' where $TTT$ consists of run-time types and qualifiers and $V$ is a *variable-name*, and if

$W$ `= name of ( "`$V$`" )`,

is executed in the inline function, and if `"V$..."` is returned as the value of $W$, then code of the form:

$TTT$ `V$...=` *`actual-argument`*

is executed at run-time just before the *statement* containing the executing macro or inclusion.

`constant function const r = com type of ( const variable name )`

Here `variable name` is the same as in `com name of` and the result returned is the type of the named variable.

For example:

```
function macro r = inc ( macro v, int w ):
    const W = name of ( "w" )
    return { { v }, "+=", { W } }
ap *READ-WRITE* int @x:
    x = 5
inc ( x, 10 ) // Same as:
              //     int V$1 = 10
              //     x += V$1
    // Now x == 15.
```

## 18.7 Compile Time Exception and Trace Support

An L-Language **ID** is a `const` integer storable at run time in an `uns` variable that identifies something. *Throw-exception-statements* may have **exception IDs** and *trace-statements* may have **trace IDs**.

The `std` module defines the following `const` maps and functions:

```
const std exception ID map = { "NONE" }
const std trace ID map = { }
```

```
constant function const r = std new exception ID ( const name )
```

A call to this function pushes `name` to the end of `std exception ID map` and returns the new length of the map minus 1.

```
constant function const r = std new trace ID ( const name )
```

A call to this function pushes `name` to the end of `std trace ID map` and returns the new length of the map minus 1.

There must be no more than 64 trace IDs.

## 18.8 Compile Time Machine Parameters

```
const com atomic types = { "int", "uns", ...  }
```

These are the types for which atomic operations [p165] are defined.

```
const com hardware overflow = ...     [TRUE or FALSE]
```

`TRUE` iff hardware computes the overflow `bool` for integer addition and subtraction. [p162] `FALSE` if this is computed when needed by software (much more slowly).

# 19 Builtin Run-Time Functions and Constants

Run-time functions execute at run-time, and but may have parts that execute at compile-time, and may even return `const` results.

The L-Language built-in run-time functions are very basic and provide only functionality that cannot be efficiently provided by library functions.

## 19.1 Builtin Run-Time Constants

```
bool std true = 1
bool std false = 0
```

## 19.2 Builtin Implicit Conversions

See 11.1.4 p80 for non-builtin conversions.

### 19.2.1 Numeric Implicit Conversions

Any value of number or `bool` type `N1` can be implicitly converted to a value of number type `N2` if every value of type `N1` can be precisely represented by a value of type `N2`, or if `N1` is an integer type and `N2` is a floating point type.

Note that a larger floating point type cannot be implicitly converted to a smaller floating point type, as such would create a cyclic loop in the implicit conversion graph p81. Also note that implicit conversions are the shortest path in this graph, e.g., the exact path `int32` → `flt64` will be used and the inexact path `int32` → `flt32` → `flt64` will not be.

More specifically, the implicit conversions are defined by:

```
function N2 r = std *IMPLICIT* *CONVERSION* ( N1 v )
```

in the following cases:

| | N1 | | |
|---|---|---|---|
| N2 | flt64 | flt32 | flt16 |
| flt64 | no | yes | yes |
| flt32 | no | no | yes |
| flt16 | no | no | no |
| int... | no | no | no |
| uns... | no | no | no |

| | N1 | | | |
|---|---|---|---|---|
| N2 | int64 | int32 | int16 | int8 |
| flt64 | yes | yes | yes | yes |
| flt32 | yes | yes | yes | yes |
| flt16 | yes | yes | yes | yes |
| int64 | no | yes | yes | yes |
| int32 | no | no | yes | yes |
| int16 | no | no | no | yes |
| int8 | no | no | no | no |
| uns... | no | no | no | no |

| | N1 | | | | |
|---|---|---|---|---|---|
| N2 | uns64 | uns32 | uns16 | uns8 | bool |
| flt64 | yes | yes | yes | yes | yes |
| flt32 | yes | yes | yes | yes | yes |
| flt16 | yes | yes | yes | yes | yes |
| int64 | no | yes | yes | yes | yes |
| int32 | no | no | yes | yes | yes |
| int16 | no | no | no | yes | yes |
| int8 | no | no | no | no | yes |
| uns64 | no | yes | yes | yes | yes |
| uns32 | no | no | yes | yes | yes |
| uns16 | no | no | no | yes | yes |
| uns8 | no | no | no | no | yes |

In addition an implicit conversion from a `const` value to each run-time numeric type `N` is defined:

```
function N r = std *IMPLICIT* *CONVERSION* ( const v )
```

This will result in a compile error if `v` is not a number or rational or has a value that cannot be represented exactly in a value of an integer type `N`. For example, converting `5.5` to an `int` will be a compile error, while converting `5.0` to an `int` or `1e1000` to a `flt` will not be (in the latter case the result is `+inf`).

### 19.2.2 Pointer Implicit Conversions

An aligned pointer may be implicitly converted to a direct pointer:

```
function dp Q$ PST$ @r = std *IMPLICIT* *CONVERSION ( ap Q$ PST$ @p )
```

The base address of `@p` is added to the offset of `@p` to produce the value of `@r`.

### 19.2.3 Qualifier Implicit Conversions

Implicit conversions of qualifiers may occur whenever a pointer value is copied, unlike other implicit conversions. The following are qualifier conversions:

1. `co` may be replaced by `ro`
2. `*READ-WRITE*` may be replaced by `ro`
3. `*READ-WRITE*` may be replaced by `*WRITE-ONLY*`
4. `*GLOBAL*` may be replaced by `*HEAP*` (See [L3] in Section 14 $^{p121}$)
5. `*GLOBAL*` and `*HEAP*` qualifiers may be deleted when storing a value in certain locations: see [L12] and [L13] of Section 14 $^{p121}$.

## 19.3 Builtin Explicit Conversions

See 11.1.4 $^{p80}$ for non-builtin conversions.

### 19.3.1 Numeric Explicit Conversions

```
function F r = std F (N v)
```

Where `F` is any builtin floating point type and `N` is any builtin number or `bool` type.

Converts `v` to the type `F`. The result may be `+Inf` or `-Inf`, or precision may be lost.

```
function I r = std floor (F v1, F v2 ?= 1.0 )
function I r = std ceiling (F v1, F v2 ?= 1.0 )
function I r = std truncate (F v1, F v2 ?= 1.0 )
function I r = std round (F v1, F v2 ?= 1.0 )
```

Where `I` is any builtin signed integer type and `F` is one of `flt`, `flt64`, or `flt32`.

These divide `v1` by `v2` and return `r` as the result rounded to an integer. Here `floor` rounds toward negative infinity, `ceiling` rounds towards positive infinity, `truncate` rounds toward zero, and `round` rounds to the nearest integer, or to the even integer if there are two nearest integers.

The floating point flags set are those set by division (see below), plus the inexact flag may be set if the division quotient is not a integer, plus the invalid flag is set if the result is outside the range storable in `I`. In the last case, and the result is an indefinite integer. $^{p146}$

```
function I1 r = std *UNCHECKED* ( I2 v )
```

Where `I1` and `I2` are any builtin integer types such that at least one of the following is true:

1. `I1` is shorter than `I2`
2. `I1` and `I2` are of equal length and one is `int...` while the other is `uns...`
3. `I2` is `int...` while `I1` is `uns...`.

These in effect convert `v` to a bit-string of unbounded length (e.g., by two's complement sign extension) and then truncate it to the length of `I1`.

```
function N r, std FP flags f = std D# ( ap QR$1 C s )
function N r, std FP flags f = std B# ( ap QR$1 C s )
function N r, std FP flags f = std X# ( ap QR$1 C s )
```

Where `N` is any number type and `C` is `uns8`, `uns16`, or `uns32` (for UTF8, UTF16, or UTF32 strings).

These convert the string `s` to the number `r`, where the string is formatted as in Section 7.2.1 [p40].

Error flags are set in `f` (see p161). If there are no errors, no flags are set. If `s` does not have the proper format, or `N` is an integer type and the value cannot be stored in that type, the `Invalid` flag is set, `NaN` is returned for floating point types, and the indefinite integer [p146] is returned for integer types. If the value is to large to store and `N` is a floating point type, the `Overflow` flag is set and ±`Inf` is returned. If the value must be rounded when it is stored, the `Inexact` flag is set.

### 19.3.2 Pointer Explicit Conversions

```
function P$ Q$ PST$1 @r = std *UNCHECKED* ( P$ Q$ PST$2 @p )
```

This just copies the pointer, in effect changing the type of its target.

```
function ap Q$ PST$ @r = std *UNCHECKED* ( dp Q$ PST$ @p )
```

The base pointer of `r` is set to point at a location that is always `0` and the offset of `r` is set to point to the address stored as the value of `p`.

```
function av Q$ PST$ r = std *UNCHECKED*
  ( ap Q$ PST$ p, int offset, int lower, int upper )
```

The base pointer of `@r` is set to the base pointer of `@p`, the offset of `@r` is set to the sum of the offset of `@p` and the `offset` argument, and the bounds of `@r` are set from the `lower` and `upper` arguments.

## 19.4 Builtin Floating Point Operations

A floating point operation produces a finite number, infinity, or NaN. One particular NaN is special: the **floating point indefinite**:[14]

1. A floating point operation result is a finite number, an infinity, or a floating point indefinite, provided that all operands to the operation are finite numbers, infinities, or floating point indefinites. If such an operation has a floating point indefinite operand, its results will be floating point indefinites.

2. *Numeric-word* lexemes that represent NaNs all represent a floating point indefinite. In particular `nan`, `+nan`, and `-nan` all represent the same value.

3. There is one floating point indefinite value for each floating point number size, and when floating point values are converted to a different size, a floating point indefinite converts to the floating point indefinite of the target size.

```
function F r = std "+" ( F v1 )
function F r = std "-" ( F v1 )
function F r = std (F v1) "+" ( F v2 )
function F r = std (F v1) "-" ( F v2 )
function F r = std (F v1) "*" ( F v2 )
function F r = std (F v1) "/" ( F v2 )
```

Where `F` is one of `flt`, `flt64`, or `flt32`.

Standard arithmetic operators on numbers `v1` and `v2`, done using IEEE floating point arithmetic.

Floating point operations may set the following floating point flags:

Invalid — Set in the following cases. Returns `NaN`.

```
+Inf + -Inf          -Inf + +Inf
+Inf - +Inf          -Inf - -Inf
+Inf * 0             0 * +Inf
-Inf * 0             0 * -Inf
+Inf / +Inf          +Inf / -Inf
-Inf / +Inf          -Inf / -Inf
+0 / +0              +0 / -0
-0 / +0              -0 / -0
```

Also set when an operand is a signaling NaN, in which case the

[14]For INTEL X64 processors, the floating point indefinite NaN has all significand bits zero but the high order bit which is one, and has the sign bit set.

results are quiet NaNs that may not be floating point indefinite NaNs.[15]

Divide by Zero — Set when a non-zero value is divided by a zero value. Returns `+Inf` or `-Inf` with sign determined by the signs of the zero and non-zero values in the usual way.

Overflow — Set when the computed value is a number outside the range that can be stored because its absolute value is too large. Returns `+Inf` or `-Inf`.

Underflow — Set when the computed value is a number outside the range that can be stored because its absolute value is too small. Returns `+0` or `-0`.

Inexact — Set when the computed value cannot be precisely stored but is inside the range of absolute values that can be stored. Returns the nearest value that can be stored, with ties going to the value whose least significant bit is zero.

```
function F r = std floor (F v1, F v2 ?= 1.0 )
function F r = std ceiling (F v1, F v2 ?= 1.0 )
function F r = std truncate (F v1, F v2 ?= 1.0 )
function F r = std round (F v1, F v2 ?= 1.0 )
```

Where `F` is one of `flt`, `flt64`, or `flt32`.

These divide `v1` by `v2` and return `r` as the result rounded to an integer. Here `floor` rounds toward negative infinity, `ceiling` rounds towards positive infinity, `truncate` rounds toward zero, and `round` rounds to the nearest integer, or to the even integer if there are two nearest integers.

The floating point flags set are those set by division (see above), plus the inexact flag may be set if the division quotient is not a integer. If the division quotient is an infinity or `NaN`, `r` is set to this quotient.

```
function bool r = std (F v1) "==" ( F v2 )
function bool r = std (F v1) "!=" ( F v2 )
function bool r = std (F v1) "<" ( F v2 )
function bool r = std (F v1) "<=" ( F v2 )
function bool r = std (F v1) ">" ( F v2 )
function bool r = std (F v1) ">=" ( F v2 )
```

Where `F` is one of `flt`, `flt64`, or `flt32`.

Standard comparison operators on floating point numbers `v1` and `v2`. Infinities are treated as actual numbers with absolute value larger than any real number: e.g., if `x`

[15] See Table 4-7: Rules for Handling NaNs, of section 4.8.3.5, of *Intel 64 and IA-32 Architectures Software Developer's Manual*.

is not a NaN, 'x <= +Inf' is always true and 'x == +Inf' is true iff x is +Inf. If an argument is a NaN, the result is undefined and an invalid flag is set.

```
function bool r = std is nan ( F v1 )
function bool r = std is infinity ( F v1 )
function bool r = std is finite ( F v1 )
```

Where F is one of flt, flt64, flt32, or flt16.

| | |
|---|---|
| is nan | Returns true if v1 is any NaN number (not just the quiet NaN with zero significand bits except for the highest order one), and false otherwise. |
| is infinity | Returns true if v1 is +Inf or -Inf, and false otherwise. |
| is finite | Returns true if is nan and is infinity both return false, and returns false otherwise. |

```
type std FP flags data:
     uns flags
     [...]  bool invalid (operand for a given operation)
     [...]  bool divide by zero
     [...]  bool overflow
     [...]  bool underflow
     [...]  bool inexact
load reference function std FP flags data r = std FP flags register
store reference function std FP flags register = std FP flags data v
```

The hardware floating point flags can be read into a datum of type FP flags and an FP flags datum can be written to the hardware floating point flags by using the FP flags register reference functions. The exact bits in the data that contain the flags are implementation dependent, and are here represented by '...'.

If a memory space has more than one process (a.k.a., thread - i.e., local stack [p133]), each process has its own std FP flags register.

## 19.5 Builtin Integer Operations

```
function I r, bool cout, bool ovfl = std "+" ( I v1, bool cin = 0 )
function I r, bool cout, bool ovfl = std "-" ( I v1, bool cin = 1 )
function I r, bool cout, bool ovfl = std (I v1) "+" ( I v2, bool cin = 0 )
function I r, bool cout, bool ovfl = std (I v1) "-" ( I v2, bool cin = 1 )
```

Where I is one of: int int128 int64 int32 int16 int8
uns int128 uns64 uns32 uns16 uns8

Standard arithmetic operators on v1 and v2 treated as binary unsigned integers. When

values are interpreted as two's complement signed integers, these operations also give valid results.

`cin` is added to the result; `cout` is the carry from the result. Operands are made negative by bitwise complementing them and adding 1 by setting `cin = 1`.

`ovfl` is set to `1` if and only if the operation overflows when values are interpreted as signed two's complement integers. If S0, S1, and Sr are the signs of `v1`, `v2`, and `r`, for two operand "+" this would equal `S0 = S1 ≠ Sr`. `ovfl` is computed by some hardware, but is expensive to compute if not supported by hardware. See `com hardware overflow`. p154

```
function I r = std (I v1) "*" ( I v2 )
function U r, U cout = std (U v1) "*" ( U v2, U cin1 = 0, U cin2 = 0 )
```

Where `I` is one of: `int int128 int64 int32 int16`
and `U` is one of: `uns uns64 uns32`

The version without carries is the standard arithmetic multiply which truncates results that are outside the range of `I`.

The version with carries is integer multiply of N-bit unsigned integers to produce a 2N-bit product to which both `cin1` and `cin2` are added. Of the result `r` is the low order N bits and `cout` is the high order N bits.

```
function I r = std (I v1) "/" ( I v2 )
function U r, U cout = std (U v1, U cin = 0) "/" ( U v2 )
```

Where `I` is one of: `int int128 int64 int32 int16`
and `U` is one of: `uns uns64 uns32`

The version without carries is the standard arithmetic divide. An exception trap occurs if `v2 = 0`.

The version with carries is integer divide of a 2N-bit unsigned dividend made by concatenating `cin` (high order) and `v1` (low order) by an N-bit unsigned divisor `v2`. The result is an N-bit quotient `r` and an N-bit remainder `cout`. An exception trap occurs if `v2 ≤ cin` (this includes that case where `v2 = 0`).

```
function I r = std "~" ( I v1 )
function I r = std (I v1) "&" ( I v2 )
function I r = std (I v1) "|" ( I v2 )
function I r = std (I v1) "^" ( I v2 )
```

Where `I` is one of: `int int128 int64 int32 int16 int8`
`uns uns128 uns64 uns32 uns16 uns8 bool`

Standard bitwise operators, complement (`~`), and (`&`), or (`|`), and exclusive or (`^`), on integers `v1` and `v2` that are treated as two's complement if signed.

```
function I r = std (I v1) "<<" ( int v2 )
```

```
function I r = std (I v1) ">>" ( int v2 )
```

Where I is one of: `int int128 int64 int32 int16 int8`
`uns uns128 uns64 uns32 uns16 uns8`

Standard bitwise shifts of integer `v1` by the amount `v2`. If signed, `v1` is treated as two's complement. The amount of shift, `v2`, must be in the range [0,`I..size`); values out of range produce undefined results. Bits shifted out at the left or right side are discarded.

```
function bool r = std (I v1) "==" ( I v2 )
function bool r = std (I v1) "!=" ( I v2 )
function bool r = std (I v1) "<" ( I v2 )
function bool r = std (I v1) "<=" ( I v2 )
function bool r = std (I v1) ">" ( I v2 )
function bool r = std (I v1) ">=" ( I v2 )
```

Where I is one of: `int int128 int64 int32 int16 int8`
`uns uns128 uns64 uns32 uns16 uns8 bool`

Standard comparison operators on integers `v1` and `v2`.

## 19.6 Builtin Pointer Operations

```
allocator function ap Q$1 PST$1 r =
            std ( uns length, uns alignment ) null
allocator function av Q$1 PST$1 r =
            std ( uns length, uns alignment ) null [ uns count ]
```

These functions allocate a value of type `PST$1` (or `count` values of type `PST$1`) to null memory (memory in inaccessible virtual pages that memory fault if accessed) and return a pointer to the value(s).

```
allocator function ap QU$1 PST$1 r =
            std ( uns length, uns alignment ) allocate
allocator function av QU$1 PST$1 r =
            std ( uns length, uns alignment ) allocate [ uns count ]
```

These functions allocate a value of type `PST$1` (or `count` values of type `PST$1`) to local memory (the current out-of-line function execution's frame in the local stack: see p133) and return a pointer to the value(s).

```
allocator function ap QU$1 *GLOBAL* PST$1 r =
            std ( uns length, uns alignment ) allocate
allocator function av QU$1 *GLOBAL* PST$1 r =
            std ( uns length, uns alignment ) allocate [ uns count ]
```

These functions allocate a value of type `PST$1` (or `count` values of type `PST$1`) to global memory (the global stack: see p134) and return a pointer to the value(s).

## 19.7 Builtin Exception and Trace Support Operations

The following functions can be used in an *exception-subblock* to determine which variables of its containing *block-assignment-statement* or *loop-assignment-statement* have been set.

```
function bool r = std is set ( N value )
function bool r = std is set ( P$1 Q$1 PST$1 pointer )
```

where `N` is any number type.

Returns `true` if the `value` is not zero or the `pointer` is not `null`, and `false` otherwise. These functions should only be used inside *exception-subblocks*[p64].

Information about the last exception is recorded in the following:

```
type std exception data:
     uns ID          // exception ID from throw statement
     dp uns8 @pc     // program counter of throw statement
load reference function std exception data r = std exception register
store reference function std exception register = std exception data v
```

Read and write the current exception data which is stored in the `std exception register`. The exception data contain the exception ID and program counter of the last exception. These are set by a *throw-statement*[p65] that gives an exception ID. A *throw-statement* without an exception ID does not change the exception register.

If a memory space has more than one process (a.k.a., thread - i.e., local stack[p133]), each process has its own `std exception register`.

A program fault executes the equivalent of a *throw-statement* with one of the following exception IDs:

```
const std divide exception = std new exception ID ( "DIVIDE" )
           // Integer divide by zero.
const std inaccessible exception = std new exception ID ( "INACCESSIBLE" )
           // Read or write of inaccessible memory location.
const std write exception = std new exception ID ( "WRITE" )
           // Write of accessible but write-protected memory location.
```

See Compile Time Exception and Trace Support 18.7[p154] for the `std new exception ID` function.

A mask determining which *trace-statements* are active is recorded in the following:

```
load reference function uns64 r = std trace mask register
store reference function std trace mask register = uns64 v
```

Read and write the current trace mask which is stored in the `trace mask register`. A *trace-statement*[p71] with trace ID X will be executed if the current trace mask has bit `1 << X` on, and will be a no-operation otherwise.

If a memory space has more than one process (a.k.a., thread - i.e., local stack [p133]), each process has its own `std trace mask register`.

The following is an example trace mask:

```
uns64 trace file errors = 1 << std new trace ID ( "FILE ERRORS" )
    // Note: 1 is converted to target type uns64 before being shifted.
```

See Compile Time Exception and Trace Support 18.7 [p154] for the `std new trace ID` function.

# 20 Atomic Operations

**Atomic operations** read and write some RAM memory locations that are shared between multiple CPUs or processes or a process and a device.

When a process executes a program, the process may move memory reads and writes around so they are no longer done in the order that they would be done were the program statements executed in strict sequential order. The process does this if the effect of the process is not changed, under the assumption that the process is the only user of the RAM memory. But if there are multiple CPUs, or a multi-tasking system asynchronously switching one CPU between processes, this assumption is not correct.

There is also the possibility that if a memory location is being read by CPU 1 and at the same time written by CPU 2, what CPU 1 reads will consist partly of the location value before CPU 2's write and partly of the location value after CPU 2's write.

Input/output devices may be inactive, meaning that they cannot read or write RAM (device RAM or regular RAM), or active, meaning that they may read or write RAM. The registers of an inactive device may be treated as ordinary RAM memory, as long as only one CPU tries to access these registers.

A device may have a GO bit in one of its registers which if turned on by a write to that register will activate the device. It is assumed that if the value a CPU writes to the register has the same size as the register, the register write will complete before the device activates.

Similarly a device may have a READY bit in one of its registers which is turned on when the device deactivates itself. It is assumed that if a CPU reads from the register a value of the same size as the register, and the READY bit is on in that value, then the device has become inactive before the register read (so the rest of the value read will be that of an inactive device).

An operation is **atomic** if:

(A1) All operations on memory (reads, writes, atomic or not-atomic) executed by the current process that would be before this atomic operation in strict sequential program execution of the current process are executed before this atomic opera-

tion.

(A2) All operations on memory (reads, writes, atomic or not-atomic) executed by the current process that would be after this atomic operation in strict sequential program execution of the current process are executed after this atomic operation.

(A3) This atomic operation's action on a regular RAM location cannot be interrupted by another atomic operation's action on that location, where the other atomic operation is executed by a different CPU or process, unless both operations just read memory. The action may include both a read and a write, with the other CPUs operation not allowed to interrupt the interval between the read and write.

Note that a single atomic operation may both read and write a location. Also we do not consider atomic operations that operate on more than one location, though such are possible given supporting hardware, or can be emulated using our atomic operations as in Example 3 below.

Atomic operations can only be performed on locations of **atomic type**. Atomic types are listed in the `com atomic types` $^{p154}$ compile-time variable. The word-length integer types, `int` and `uns`, are always atomic types.

`function A r = std atomic read ( ap QR$1 A p )`

For `A` an atomic type, perform an atomic operation that reads the location pointed at by `p` and return the value read.

`function std atomic write ( ap QW$1 A p, A v )`

For `A` an atomic type, perform an atomic operation that writes the value `v` into the location pointed at by `p`.

`function std bool r = atomic compare and set ( ap QRW$1 A p, A vr, A vw )`

For `A` an atomic type, perform an atomic operation that first reads a value from the location pointed at by `p`, then compares the value to `vr`, and lastly, if and only if the compared values are equal, writes `wv` into the location. The value `r` returned is `true` is `wv` was written and `false` otherwise.

`load reference function std bool r = atomic nop`

This atomic function does nothing but force rules (A1) and (A2) to be obeyed. It always returns '`true`'.

Example 1:

A device has registers in global memory which are shared between the device and a process, and has GO and READY bits as described above. The registers are either owned by the process and the device is inactive, or the registers are owned by the device which is actively executing an operation.

When the device is inactive, parameters for the next device operation are written into device registers by the process. Then the GO bit is set by an atomic write of a value

whose size matches the size of the register containing the GO bit. This activates the device which starts performing the device operation. The atomic nature of this write guarantees that all process reads from and writes to device registers that are in the code before the GO bit is set are actually executed before the GO bit is set.

When the device is active, the register containing the READY bit is read by an atomic read of a value whose size matches that of the register containing the READY bit. The process may do this more than once, until a read reads an on READY bit indicating the device has become inactive. Then device operation results are read from device registers. The atomic nature of the READY bit read guarantees the all process reads from and and writes to device registers that are in the code after the READY bit was read as on are actually executed after the READY bit turned on.

Example 2:

A device has registers as in Example 1. The device also has a large memory accessible when the device is inactive via two registers: A and D. A holds the address of a location in device memory, and D holds the contents of that location: reading D reads the location contents, and writing D writes the location contents.

We assume the device is inactive and the process writes A and then reads or writes D. If either the write to A or the operation on D is atomic, it is guaranteed that A will be written before D is read or written. It may happen that A has an atomic type but D does not, in which case A should be written atomically. If neither A nor D have atomic types, an `atomic nop` can be put between the write of A and the operation on D.

Example 3:

A data structure which is shared among processes perhaps executing on different CPUs is guarded by a lock consisting of two atomic integer memory locations: B (before) and A (after). When the structure is not being written, B == A. A writer first increments B, then updates the structure, and then increments A.

To get a write lock, the writer reads A atomically and saves the value V, and then does an atomic compare-and-set on B that checks that B == V and if so writes V+1 (= B+1) to B and acquires the lock. Because the compare-and-set is atomic, A will be read before the compare-and-set, and all reads and writes after the compare-and-set in the code will execute after the compare-and-set. Because the reads of A and B are atomic, data read will not be corrupted by simultaneous atomic writes by another CPU. Because the write to B is atomic, another CPU using an atomic read of B will not see data corrupted by the atomic write to B.

If two CPUs attempt to get the write lock simultaneously, both read the same value V from A, and the first to execute the compare-and-set gets the lock.

To release the lock, the writing process just atomically writes the V+1 (= B) into A. Because the write is atomic, all reads and writes before it in the code are executed before it.

To read, the reader reads A atomically and saves the value V, then reads data from the structure, then reads B atomically and checks that the value read equals V. If the check passes, the data is uncorrupted by writing that is simultaneous with the reading. If the check fails, the data may be corrupted. The reader must be sure corrupted data does not destroy the integrity of the reader's execution, but can read B atomically and check the value against V at any time to see if the data read so far is uncorrupted. Because the reads of A and B are atomic, reads of the structure that are between the reads of A and B in the code will execute between the atomic reads of A and B, and also the values read from A and B will not be corrupted by atomic writes done by another CPU.